

8

The Structure of Hadrons

In Chapters 3 through 7, we have learned how to perform quantitative calculations for the electromagnetic interactions of leptons and quarks. The same techniques will allow us to compute the color interactions of quarks and gluons. There is, however, an immediate problem: experiments to study color (strong) interactions are performed with hadrons (e.g., proton beams or secondary π -beams interacting with nuclear targets), not with the quarks and gluons that are described by quantum field theory. This situation is similar to that encountered in atomic physics where experiments involving complex atoms have to be interpreted through the electromagnetic interactions of the constituent electrons. This analogy reveals the problem: we need to find the “wavefunctions” that describe, for example, a proton in terms of its constituent quarks and gluons. In this chapter, we present an experimental technique that allows us to determine the quark and gluon structure of hadrons; namely the deep inelastic scattering of leptons off hadronic targets. The structure functions so obtained will be presented in Chapter 9. Finally, in Chapters 10 and 11 we reach our goal and translate these quark–gluon QCD calculations into predictions for the results of experiments involving leptons and hadrons.

The discussion will soon reveal that these structure functions are not the static quark wavefunctions introduced in Chapter 2, although they are indirectly related to them.

8.1 Probing a Charge Distribution with Electrons. Form Factors

“Photographing” an object by scattering an electron beam off it is a well-proved technique in physics. Suppose we want to determine the charge distribution shown in Fig. 8.1, which could, for example, be the electron cloud of an atom. The procedure is to measure the angular distribution of the scattered electrons and compare it to the (known) cross section for scattering electrons from a point charge, in the form

$$\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{point}} |F(q)|^2, \quad (8.1)$$

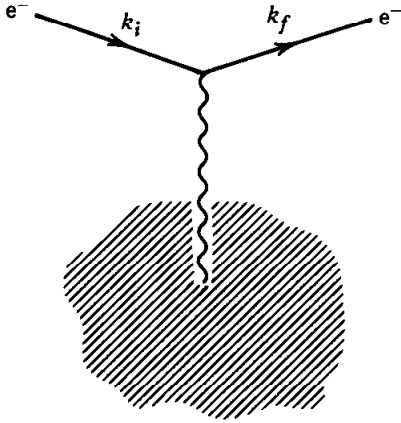


Fig. 8.1 Lowest-order electron scattering by a charge cloud.

where q is the momentum transfer between the incident electron and the target, $q = k_i - k_f$, see Fig. 8.1. We then attempt to deduce the structure of the target from the form factor $F(q)$ so determined.

We can gain insight into this technique by first looking at the scattering of unpolarized electrons of energy E from a static, spinless charge distribution $Ze\rho(\mathbf{x})$, normalized so that

$$\int \rho(\mathbf{x}) d^3x = 1. \quad (8.2)$$

For a static target, it is found that the form factor in (8.1) is just the Fourier transform of the charge distribution

$$F(\mathbf{q}) = \int \rho(\mathbf{x}) e^{i\mathbf{q}\cdot\mathbf{x}} d^3x, \quad (8.3)$$

while the reference cross section for a structureless target is

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{point}} \equiv \left(\frac{d\sigma}{d\Omega} \right)_{\text{Mott}} = \frac{(Z\alpha)^2 E^2}{4k^4 \sin^4 \frac{\theta}{2}} \left(1 - v^2 \sin^2 \frac{\theta}{2} \right), \quad (8.4)$$

where $k = |\mathbf{k}_i| = |\mathbf{k}_f|$, $v = k/E$, and θ is the angle through which the electron is scattered.

EXERCISE 8.1 It is useful practice of the techniques developed in the previous chapters to derive (8.3) and (8.4). We outline the various steps below. The electromagnetic field due to $Ze\rho(\mathbf{x})$ is $A^\mu = (\phi, \mathbf{0})$ where, using (6.59),

$$\nabla^2 \phi = -Ze\rho(\mathbf{x}).$$

Use (6.4) and (6.6) to show that the scattering amplitude is (see also Section 7.1)

$$T_{fi} = -i2\pi \delta(E_f - E_i) (-e\bar{u}_f \gamma_0 u_i) \int e^{i\mathbf{q} \cdot \mathbf{x}} \phi(\mathbf{x}) d^3x. \quad (8.5)$$

Justify

$$\int e^{i\mathbf{q} \cdot \mathbf{x}} \nabla^2 \phi d^3x = -|\mathbf{q}|^2 \int e^{i\mathbf{q} \cdot \mathbf{x}} \phi d^3x$$

and hence show that the integral in (8.5) is $Ze F(\mathbf{q})/|\mathbf{q}|^2$, see (7.9). Following the arguments of Section 4.3, verify that the differential cross section from a fixed target is

$$d\sigma = \frac{|T_{fi}|^2}{T} \frac{d^3k_f}{(2\pi)^3 2E_f} \left(\frac{1}{v 2E_i} \right), \quad (8.6)$$

with

$$d^3k_f \delta(E_f - E_i) = kE d\Omega.$$

Summing final, and averaging initial, electron spins give

$$\frac{1}{2} \sum_{s_f, s_i} |\bar{u}_f \gamma_0 u_i|^2 = 4E^2 \left(1 - v^2 \sin^2 \frac{\theta}{2} \right), \quad (8.7)$$

where θ is the angle introduced in Section 7.1. Check this answer with (6.25). Putting all this together yields the advertised result

$$\frac{d\sigma}{d\Omega} = \left(\frac{d\sigma}{d\Omega} \right)_{\text{Mott}} |F(\mathbf{q})|^2,$$

with the form factor given by (8.3).

EXERCISE 8.2 Show that if the electron beam is replaced by a beam of “point” spinless particles, the only change is that factor (8.7) is replaced by $4E^2$. This raises a question: why does the electron spin make no difference in the nonrelativistic limit, $v \rightarrow 0$? The remarks following (6.13) are the clue.

EXERCISE 8.3 By considering the electron helicity, explain why you would anticipate the $\cos^2(\theta/2)$ behavior of factor (8.7) in the extreme relativistic limit, see Section 6.6.

By virtue of the normalization condition, (8.2),

$$F(0) \equiv 1. \quad (8.8)$$

If $|\mathbf{q}|$ is not too large, we can expand the exponential in (8.3), giving

$$\begin{aligned} F(\mathbf{q}) &= \int \left(1 + i\mathbf{q} \cdot \mathbf{x} - \frac{(\mathbf{q} \cdot \mathbf{x})^2}{2} + \cdots \right) \rho(\mathbf{x}) d^3x \\ &= 1 - \frac{1}{6} |\mathbf{q}|^2 \langle r^2 \rangle + \cdots, \end{aligned} \quad (8.9)$$

where we have assumed that ρ is spherically symmetric, that is, a function of $r \equiv |\mathbf{x}|$ alone. The small-angle scattering therefore just measures the mean square radius $\langle r^2 \rangle$ of the charge cloud. This is because in the small $|\mathbf{q}|$ limit the photon in Fig. 8.1 is soft and with its large wavelength can resolve only the size of the charge distribution $\rho(r)$ and is not sensitive to its detailed structure.

EXERCISE 8.4 If the charge distribution $\rho(r)$ has an exponential form, e^{-mr} , show, using (8.3), that the form factor

$$F(|\mathbf{q}|) \propto \left(1 - \frac{q^2}{m^2} \right)^{-2}$$

with $q^2 = -|\mathbf{q}|^2$.

8.2 Electron-Proton Scattering. Proton Form Factors

The above discussion cannot be applied directly to yield the structure of the proton. First, the proton's magnetic moment is involved in the scattering of the electron, not just its charge. Second, the proton is not static, but will recoil under the electron's bombardment. If, however, the proton were a point charge e with a Dirac magnetic moment $e/2M$, then we already know the answer. We can take over the result for electron-muon scattering, (6.50), and simply replace the mass of the muon by that of the proton:

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{lab}} = \left(\frac{\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}} \right) \frac{E'}{E} \left\{ \cos^2 \frac{\theta}{2} - \frac{q^2}{2M^2} \sin^2 \frac{\theta}{2} \right\}, \quad (8.10)$$

where the factor

$$\frac{E'}{E} = \frac{1}{1 + \frac{2E}{M} \sin^2 \frac{\theta}{2}},$$

given by (6.48), arises from the recoil of the target.

Copying the calculation of the electron-muon cross section, the lowest-order amplitude for electron-proton elastic scattering, Fig. 8.2, is given by [see (6.8)]

$$T_{fi} = -i \int j_\mu \left(-\frac{1}{q^2} \right) J^\mu d^4x,$$

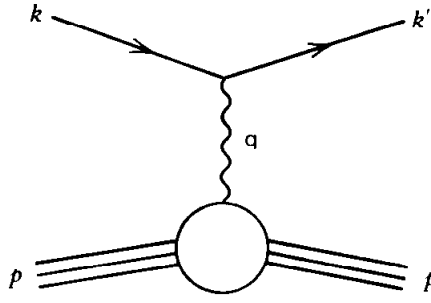


Fig. 8.2 Lowest-order electron-proton elastic scattering.

where $q = p' - p$ and the electron and proton transition currents are, respectively,

$$j^\mu = -e \bar{u}(k') \gamma^\mu u(k) e^{i(k' - k) \cdot x} \quad (8.11)$$

$$J^\mu = e \bar{u}(p') [\quad] u(p) e^{i(p' - p) \cdot x}, \quad (8.12)$$

see (6.6). Since the proton is an extended structure, we cannot replace the square brackets in (8.12) by γ^μ , as for point spin- $\frac{1}{2}$ particles in (8.11). But we know that J^μ must be a Lorentz four-vector, and so we must use the most general four-vector form that can be constructed from p , p' , q and the Dirac γ -matrices sandwiched between $\bar{u}$ and u . There are only two independent terms, γ^μ and $i\sigma^{\mu\nu}q_\nu$, and their coefficients are functions of q^2 (q^2 is the only independent scalar variable at the proton vertex). Terms involving γ^5 are ruled out by the conservation of parity. Therefore, quite generally, we may write the square bracket of (8.12) in the form

$$[\quad] = \left[F_1(q^2) \gamma^\mu + \frac{\kappa}{2M} F_2(q^2) i\sigma^{\mu\nu} q_\nu \right] \quad (8.13)$$

where F_1 and F_2 are two independent form factors and κ is the anomalous magnetic moment (see Exercise 6.2).

EXERCISE 8.5 Show that current conservation, $\partial_\mu J^\mu = 0$, rules out $(p - p')^\mu$ as a possible four-vector. Why do we not show a term involving $(p + p')^\mu$ in (8.13)?

EXERCISE 8.6 Show that $p \cdot q$ is not an independent scalar variable by expressing it in terms of the variable q^2 .

For $q^2 \rightarrow 0$, that is, when we probe with long-wavelength photons, it does not make any difference that the proton has structure at the order of 1 fermi. We effectively see a particle of charge e and magnetic moment $(1 + \kappa)e/2M$, where κ , the anomalous moment, is measured to be 1.79. The factors in (8.13) must therefore be chosen so that in this limit

$$F_1(0) = 1, \quad F_2(0) = 1. \quad (8.14)$$

The corresponding values for the neutron are $F_1(0) = 0$, $F_2(0) = 1$, and experimentally $\kappa_n = -1.91$.

If we use (8.13) to calculate the differential cross section for electron-proton elastic scattering, we find an expression similar to (8.10):

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{lab}} = \left(\frac{\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}} \right) \frac{E'}{E} \left\{ \left(F_1^2 - \frac{\kappa^2 q^2}{4M^2} F_2^2 \right) \cos^2 \frac{\theta}{2} - \frac{q^2}{2M^2} (F_1 + \kappa F_2)^2 \sin^2 \frac{\theta}{2} \right\}, \quad (8.15)$$

see (6.50). This is known as the Rosenbluth formula. The two form factors, $F_{1,2}(q^2)$, parametrize our ignorance of the detailed structure of the proton represented by the blob in Fig. 8.2. These form factors can be determined experimentally by measuring $d\sigma/d\Omega$ as a function of θ and q^2 . Note that if the proton were a point particle like the muon, then $\kappa = 0$ and $F_1(q^2) = 1$ for all q^2 , and (8.15) would revert to (8.10).

In practice, it is better to use linear combinations of $F_{1,2}$,

$$\begin{aligned} G_E &\equiv F_1 + \frac{\kappa q^2}{4M^2} F_2 \\ G_M &\equiv F_1 + \kappa F_2, \end{aligned} \quad (8.16)$$

defined so that no interference terms, $G_E G_M$, occur in the cross section. Equation (8.15) becomes

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{lab}} = \frac{\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}} \frac{E'}{E} \left(\frac{G_E^2 + \tau G_M^2}{1 + \tau} \cos^2 \frac{\theta}{2} + 2\tau G_M^2 \sin^2 \frac{\theta}{2} \right), \quad (8.17)$$

with $\tau \equiv -q^2/4M^2$.

Now that interference terms have disappeared, these proton form factors may be regarded as generalizations of the nonrelativistic form factor introduced in Section 8.1, and so it would be nice if we could interpret their Fourier transforms as the charge and magnetic moment distributions of the proton. Unfortunately, the recoil of the proton makes this impossible. However, it is possible to show that the form factors $G_E(q^2)$ and $G_M(q^2)$ are closely related to the proton charge and magnetic moment distributions, respectively, in a particular Lorentz frame, called the Breit (or brick wall) frame, defined by $\mathbf{p}' = -\mathbf{p}$.

EXERCISE 8.7 Show that the proton transition current, $J^\mu(x)$ of (8.12), can be rewritten in the form

$$J^\mu(0) = e \bar{u}(p') \left[\gamma^\mu (F_1 + \kappa F_2) - \frac{(p^\mu + p'^\mu)}{2M} \kappa F_2 \right] u(p). \quad (8.18)$$

Evaluate $J^\mu(0) \equiv (\rho, \mathbf{J})$ in the Breit frame ($\mathbf{p}' = -\mathbf{p}$). There is no energy

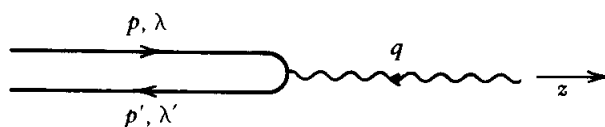


Fig. 8.3 The Breit or brick-wall frame, $\mathbf{p}' = -\mathbf{p}$.

transferred to the proton in this frame, and it behaves as if it had bounced off a brick wall, see Fig. 8.3. If the z axis is chosen along $\mathbf{p}$ and helicity spinors are used, show that

$$\begin{aligned} \rho &= 2Me G_E(q^2) & \text{for } \lambda = -\lambda', \\ J_1 \pm iJ_2 &= \mp 2|\mathbf{q}|e G_M(q^2) & \text{for } \lambda = \lambda' = \mp \frac{1}{2}, \end{aligned} \quad (8.19)$$

and that all other matrix elements are zero; λ and λ' denote the initial and final proton helicities, respectively. Determine the corresponding values of the helicity of the virtual photon.

In generalizing the form factor of Section 8.1, we have replaced $F(|\mathbf{q}|)$ by $F(q^2)$. However, as long as $|\mathbf{q}|^2 \ll M^2$, we can take over the Fourier transform interpretation of Section 8.1.

EXERCISE 8.8 Show that for $|\mathbf{q}|^2 \ll M^2$, the form factors G_E and G_M are the Fourier transforms of the proton's charge and magnetic moment distributions, respectively.

G_E and G_M are referred to as the electric and magnetic form factors, respectively. The data on the angular dependence of $ep \rightarrow ep$ scattering can be used to separate G_E, G_M at different values of q^2 , see (8.17). The result for $G_E(q^2)$ is

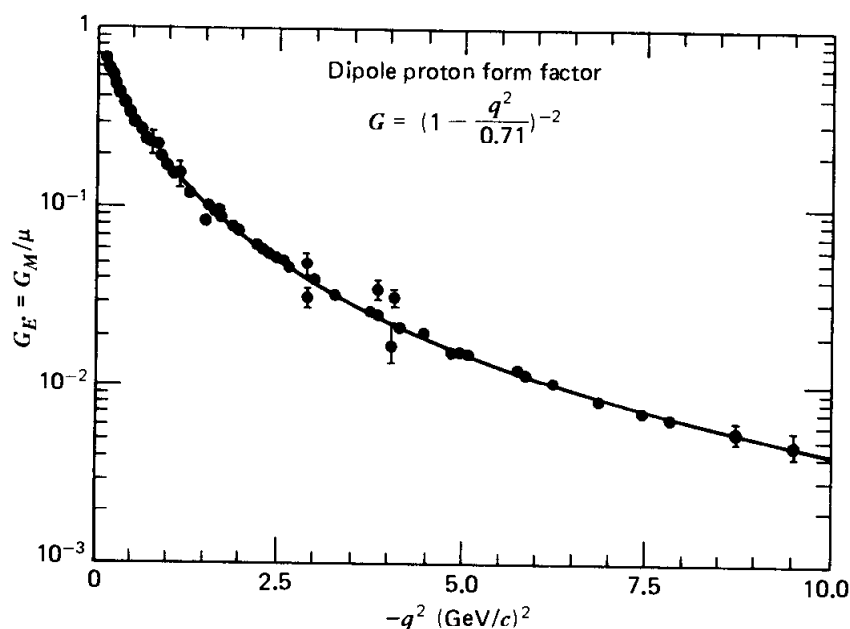


Fig. 8.4 The proton form factors as a function of q^2 .

shown in Fig. 8.4. $G_M(q^2)$ has the same q^2 dependence. A closer look at Fig. 8.4 reveals that

$$G_E(q^2) \approx \left(1 - \frac{q^2}{0.71}\right)^{-2} \quad (\text{in units of GeV}^2). \quad (8.20)$$

The behavior for small $-q^2$ can be used to determine the residual terms in the expansion of (8.9). In particular, the mean square proton charge radius is

$$\langle r^2 \rangle = 6 \left(\frac{dG_E(q^2)}{dq^2} \right)_{q^2=0} = (0.81 \times 10^{-13} \text{ cm})^2. \quad (8.21)$$

The same radius of about 0.8 fm is obtained for the magnetic moment distribution. Using the result of Exercise 8.4, we conclude that the charge distribution of the nucleon has an exponential shape in configuration space.

8.3 Inelastic Electron-Proton Scattering $ep \rightarrow eX$

Having measured the size of the proton, one might like to take a more detailed look at its structure by increasing the $-q^2$ of the photon to give better spatial resolution. This can be done simply by requiring a large energy loss of the bombarding electron. There is, however, a catch: because of the large transfer of energy, the proton will often break up, and the picture of Fig. 8.2 has to be generalized to Fig. 8.5. For modest $-q^2$, one might just excite the proton into a Δ -state and hence produce an extra π -meson, that is, $ep \rightarrow e\Delta^+ \rightarrow ep\pi^0$. In these events, the invariant mass (see Fig. 8.5) is $W^2 \approx M_\Delta^2$. When $-q^2$ is very large, however, the debris becomes so messy that the initial state proton loses its identity completely and a new formalism must be devised to extract information from the measurements. Figure 8.6 shows the invariant mass distribution. One notices the peak when the proton does not break up ($W \approx M$) and broader peaks when the target is excited to resonant baryon states. Beyond the resonances, the complicated multiparticle states with large invariant mass result in a smooth distribution in missing mass W .

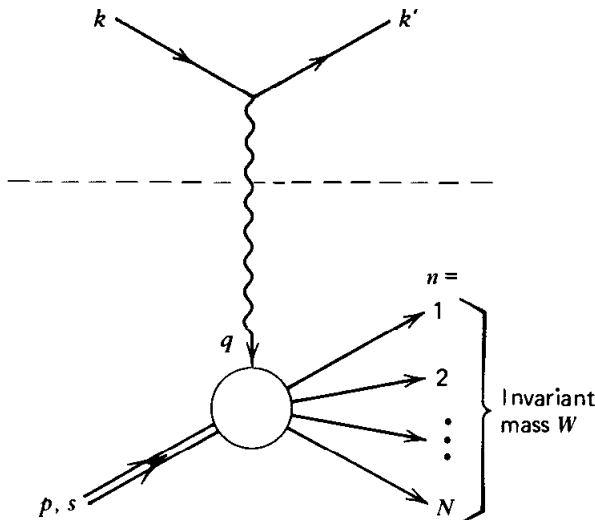


Fig. 8.5 Lowest-order diagram for $ep \rightarrow eX$.

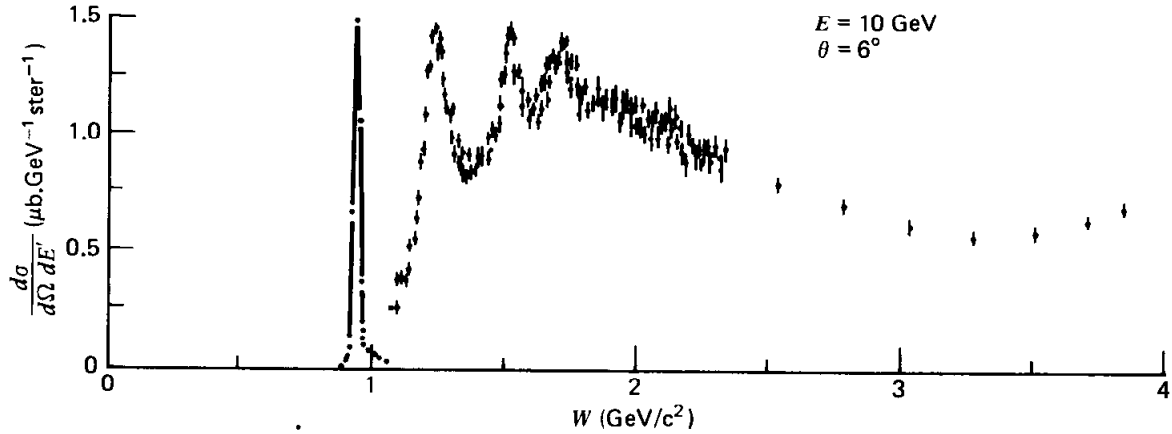


Fig. 8.6 The $ep \rightarrow eX$ cross section as a function of the missing mass W . Data are from the Stanford Linear Accelerator. The elastic peak at $W = M$ has been reduced by a factor of 8.5.

The problem now facing us is illustrated by recalling (8.11), (8.12), and Fig. 8.2. The switch from a muon to a proton target was made by replacing the lepton current j^μ ($\sim \bar{u}\gamma^\mu u$) by a proton current J^μ ($\sim \bar{u}\Gamma^\mu u$), and the most general form of Γ^μ was constructed. This is inadequate to describe the inelastic events of Fig. 8.5. Although everything above the dashed line in Fig. 8.5 remains unchanged (a fact which we shall exploit), the final state below it is not a single fermion described by a Dirac “ $\bar{u}$ ” entry in the matrix element or current. Therefore, J^μ must have a more complex structure than (8.12). Instead, the expression for the cross section [see (6.18)]

$$d\sigma \sim L_{\mu\nu}^e (L^p)^{\mu\nu} \quad (8.22)$$

is directly generalized to

$$d\sigma \sim L_{\mu\nu}^e W^{\mu\nu}, \quad (8.23)$$

where $L_{\mu\nu}^e$ represents the lepton tensor of (6.20), since everything in the leptonic part of the diagram above the photon propagator in Fig. 8.5 is left unchanged. The hadronic tensor $W^{\mu\nu}$ serves to parametrize our total ignorance of the form of the current at the other end of the propagator. The most general form of the tensor $W^{\mu\nu}$ must now be constructed out of $g^{\mu\nu}$ and the independent momenta p and q ($p' = p + q$). γ^μ is not included, as we are parametrizing the cross section which is already summed and averaged over spins. We write

$$W^{\mu\nu} = -W_1 g^{\mu\nu} + \frac{W_2}{M^2} p^\mu p^\nu + \frac{W_4}{M^2} q^\mu q^\nu + \frac{W_5}{M^2} (p^\mu q^\nu + q^\mu p^\nu). \quad (8.24)$$

We have omitted antisymmetric contributions to $W^{\mu\nu}$, since their contribution to the cross section vanishes after insertion into (8.23) because the tensor $L_{\mu\nu}^e$ is symmetric. Note the omission of W_3 in our notation; this spot is reserved for a parity-violating structure function when a neutrino beam is substituted for the electron beam, so that the virtual photon probe is replaced by a weak boson; see Perl (1974), Close (1979), or Llewellyn Smith (1972).

EXERCISE 8.9 Show that indeed $L_{\mu\nu}^e = L_{\nu\mu}^e$ and that

$$q^\mu L_{\mu\nu}^e = q^\nu L_{\mu\nu}^e = 0. \quad (8.25)$$

EXERCISE 8.10 Show that current conservation at the hadronic vertex requires

$$q_\mu W^{\mu\nu} = q_\nu W^{\mu\nu} = 0. \quad (8.26)$$

The proof may be left until after (8.39); it follows from $\partial_\mu \tilde{J}^\mu = 0$. As a result of (8.26), verify that

$$W_5 = -\frac{p \cdot q}{q^2} W_2,$$

$$W_4 = \left(\frac{p \cdot q}{q^2} \right)^2 W_2 + \frac{M^2}{q^2} W_1.$$

Thus, only two of the four inelastic structure functions of (8.24) are independent; so we may write

$$W^{\mu\nu} = W_1 \left(-g^{\mu\nu} + \frac{q^\mu q^\nu}{q^2} \right) + W_2 \frac{1}{M^2} \left(p^\mu - \frac{p \cdot q}{q^2} q^\mu \right) \left(p^\nu - \frac{p \cdot q}{q^2} q^\nu \right), \quad (8.27)$$

where the W_i 's are functions of the Lorentz scalar variables that can be constructed from the four-momenta at the hadronic vertex. Unlike elastic scattering, there are two independent variables, and we choose

$$q^2 \quad \text{and} \quad \nu \equiv \frac{p \cdot q}{M}. \quad (8.28)$$

The invariant mass W of the final hadronic system is related to ν and q^2 by

$$W^2 = (p + q)^2 = M^2 + 2M\nu + q^2. \quad (8.29)$$

EXERCISE 8.11 It is common to replace ν and q^2 by the dimensionless variables

$$x = \frac{-q^2}{2p \cdot q} = \frac{-q^2}{2M\nu}, \quad y = \frac{p \cdot q}{p \cdot k}, \quad (8.30)$$

where the four-momenta are shown on Fig. 8.5. Show that the allowed kinematic region for $ep \rightarrow eX$ is $0 \leq x \leq 1$ and $0 \leq y \leq 1$. Sketch this physical region in the ν, q^2 plane and check your answer with Fig. 9.3.

EXERCISE 8.12 Show that in the rest frame of the target proton,

$$\nu = E - E', \quad y = \frac{E - E'}{E},$$

where E and E' are the initial and final electron energies, respectively.

Evaluation of the cross section for $ep \rightarrow eX$ is a straightforward repetition of the same calculation for $e^- \mu^- \rightarrow e^- \mu^-$ (or $ep \rightarrow ep$) scattering with the substitution of $W_{\mu\nu}$, given by (8.27), for $L_{\mu\nu}^{\text{muon}}$ (or $L_{\mu\nu}^p$). Using the expression (6.25) for $(L^e)^{\mu\nu}$ and noting (8.25), we find

$$(L^e)^{\mu\nu} W_{\mu\nu} = 4W_1(k \cdot k') + \frac{2W_2}{M^2} [2(p \cdot k)(p \cdot k') - M^2 k \cdot k']. \quad (8.31)$$

In the laboratory frame, this becomes

$$(L^e)^{\mu\nu} W_{\mu\nu} = 4EE' \left\{ \cos^2 \frac{\theta}{2} W_2(\nu, q^2) + \sin^2 \frac{\theta}{2} 2W_1(\nu, q^2) \right\}, \quad (8.32)$$

see (6.44). By including the flux factor, (4.32), and the phase space factor for the outgoing electron, (4.24), we can obtain the inclusive differential cross section for inelastic electron-proton scattering, $ep \rightarrow eX$,

$$d\sigma = \frac{1}{4((k \cdot p)^2 - m^2 M^2)^{1/2}} \left\{ \frac{e^4}{q^4} (L^e)^{\mu\nu} W_{\mu\nu} 4\pi M \right\} \frac{d^3 k'}{2E'(2\pi)^3}, \quad (8.33)$$

where $|\overline{\mathcal{M}}|^2$ is given by the expression in the braces [recall (6.18)]. The extra factor of $4\pi M$ arises because we have adopted the standard convention for the normalization of $W^{\mu\nu}$. Inserting (8.32) in (8.33) yields

$$\left. \frac{d\sigma}{dE' d\Omega} \right|_{\text{lab}} = \frac{\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}} \left\{ W_2(\nu, q^2) \cos^2 \frac{\theta}{2} + 2W_1(\nu, q^2) \sin^2 \frac{\theta}{2} \right\},$$

(8.34)

where, as usual, we neglect the mass of the electron.

8.4 Summary of the Formalism for Analyzing ep Scattering

Although we have achieved our objective, it is informative to take a second look at the formalism. The final result, (8.34), may be reexpressed in the form

$$\frac{d\sigma}{dE' d\Omega} = \frac{\alpha^2}{q^4} \frac{E'}{E} (L^e)^{\mu\nu} W_{\mu\nu}, \quad (8.35)$$

see (8.32) and (6.44). For comparison, recall the cross section for $e\mu^- \rightarrow e\mu^-$ scattering, (6.46),

$$d\sigma = \frac{1}{4ME} \frac{d^3k'}{(2\pi)^3 2E'} \frac{d^3p'}{(2\pi)^3 2p'_0} \left\{ \frac{e^4}{q^4} (L^e)^{\mu\nu} L_{\mu\nu}^{\text{muon}} \right\} (2\pi)^4 \delta^4(p + q - p'), \quad (8.36)$$

which can also be written in the form (8.35) with

$$W_{\mu\nu} = \frac{1}{4\pi M} \left(\frac{1}{2} \sum_s \sum_{s'} \right) \int \frac{d^3p'}{(2\pi)^3 2p'_0} \langle p, s | \tilde{J}_\mu^\dagger | p', s' \rangle \\ \times \langle p', s' | \tilde{J}_\nu | p, s \rangle (2\pi)^4 \delta^4(p + q - p') \quad (8.37)$$

where

$$\langle p', s' | \tilde{J}_\nu | p, s \rangle \equiv \bar{u}^{(s')}(\mathbf{p}') \gamma_\nu u^{(s)}(\mathbf{p}). \quad (8.38)$$

Insertion of (8.37) into (8.35) reproduces (6.49). All we have done is to use a particular regrouping of the factors in our earlier derivation. If we replace (8.38) by $\bar{u}[\]u$ with $[\]$ given by (8.13), then we can recover the result for $ep \rightarrow ep$ scattering. The reason for all this is to note that $W_{\mu\nu}$ for $ep \rightarrow eX$ is nothing but a generalization of (8.37) to the case where a proton breaks up into many particles in the final hadronic state X. It can be formally written as

$$W_{\mu\nu} = \frac{1}{4\pi M} \sum_N \left(\frac{1}{2} \sum_s \right) \int \prod_{n=1}^N \left(\frac{d^3p'_n}{2E'_n (2\pi)^3} \right) \sum_{s_n} \langle p, s | \tilde{J}_\mu^\dagger | X \rangle \\ \times \langle X | \tilde{J}_\nu | p, s \rangle (2\pi)^4 \delta^4\left(p + q - \sum_n p'_n\right), \quad (8.39)$$

where a sum over all possible many-particle (i.e., N particle) states X is included, see Fig. 8.5.

For future reference, it is useful to make a compendium of our results on form factors. We keep to the laboratory kinematics of Fig. 6.10 and neglect the mass of the electron. For all the reactions, the differential cross section in the energy (E') and angle (θ) of the scattered electron can be written in the form

$$\frac{d\sigma}{dE' d\Omega} = \frac{4\alpha^2 E'^2}{q^4} \left\{ \quad \right\}. \quad (8.40)$$

First, for a muon target of mass m (or a quark target of mass m after substitution $\alpha^2 \rightarrow \alpha^2 e_q^2$ where e_q is the quark's fractional charge),

$$\left\{ \right\}_{e\mu \rightarrow e\mu} = \left(\cos^2 \frac{\theta}{2} - \frac{q^2}{2m^2} \sin^2 \frac{\theta}{2} \right) \delta \left(\nu + \frac{q^2}{2m} \right). \quad (8.41)$$

For elastic scattering from a proton target,

$$\left\{ \right\}_{ep \rightarrow ep} = \left(\frac{G_E^2 + \tau G_M^2}{1 + \tau} \cos^2 \frac{\theta}{2} + 2\tau G_M^2 \sin^2 \frac{\theta}{2} \right) \delta \left(\nu + \frac{q^2}{2M} \right), \quad (8.42)$$

where $\tau = -q^2/4M^2$ and M is the mass of the proton. Finally, for the case when the proton target is broken up by the bombarding electron,

$$\left\{ \right\}_{ep \rightarrow eX} = W_2(\nu, q^2) \cos^2 \frac{\theta}{2} + 2W_1(\nu, q^2) \sin^2 \frac{\theta}{2}. \quad (8.43)$$

Making use of the delta function, (8.41) and (8.42) can be integrated over E' with the result [see (6.50)]

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E^2 \sin^4 \frac{\theta}{2}} \frac{E'}{E} \left(\right). \quad (8.44)$$

EXERCISE 8.13 The above results assume (lowest-order) single photon exchange is dominant. If two-photon exchange were significant, convince yourself that the e^-p and e^+p cross sections would not be equal.

8.5 Inelastic Electron Scattering as a (Virtual) Photon–Proton Total Cross Section

It is clear from the above discussion that the important issue is what happens below the dashed line in Fig. 8.5, where a (virtual) photon interacts with a proton. The role of the electron beam is simply that it is responsible for the presence of the virtual photon. It is useful to display these facts in our formalism. We start by writing the total cross section for scattering a *real* photon, with energy $q^0 = \nu \equiv K$ and (transverse) polarization ϵ , off the same unpolarized proton target producing two or more final-state particles. Using the Feynman rules and cross section kinematics which we have developed, we obtain

$$\begin{aligned} \sigma^{\text{tot}}(\gamma p \rightarrow X) &= \frac{1}{(2K)(2M)} \sum_N \left(\frac{1}{2} \sum_s \right) \int \prod_{n=1}^N \left(\frac{d^3 p'_n}{2E'_n (2\pi)^3} \right) \\ &\quad \times \sum_{s_n} (2\pi)^4 \delta^4 \left(p + q - \sum_n p'_n \right) \epsilon^{\mu*} \epsilon^\nu e^2 \langle p, s | \tilde{J}_\mu^\dagger | X \rangle \langle X | \tilde{J}_\nu | p, s \rangle, \end{aligned} \quad (8.45)$$

where, as before, a sum over all final states X is included. If W is the invariant mass of the final state, then

$$W^2 = (p + q)^2 = M^2 + 2MK. \quad (8.46)$$

We immediately note the striking similarity between (8.45) and the formal expression for $W_{\mu\nu}$ given by (8.39). Indeed, (8.45) appears to be simply

$$\sigma^{\text{tot}}(\gamma p \rightarrow X) = \frac{4\pi^2\alpha}{K} \epsilon^{\mu*} \epsilon^\nu W_{\mu\nu}. \quad (8.47)$$

However, there is a crucial proviso: for real photons, we must sum only over the two transverse polarizations of the incident photon. On the other hand, to interpret the $ep \rightarrow eX$ hadronic tensor $W_{\mu\nu}$ as a photon-proton total cross section, it is vital to remember that the photon is *virtual* and not limited to two polarization states. In fact, the cross section for virtual photons is not a well-defined concept. When $q^2 = 0$, the flux factor is the $4MK$ with $K = \nu$; but for virtual photons ($q^2 \neq 0$), the flux is arbitrary. The conventional choice is to require K to continue to satisfy (8.46), that is,

$$K = \frac{W^2 - M^2}{2M} = \nu + \frac{q^2}{2M}, \quad (8.48)$$

in the laboratory frame. This is known as the Hand convention. Another possible choice is $K = |\mathbf{q}|$.

To complete the interpretation of the structure functions, we must specify the polarization vectors ϵ_λ^μ of virtual photons (helicity λ). We take the z axis along $\mathbf{q}$ and use [see (6.92); a further discussion is given in Budnev et al. Phys. Rep C15, 181 (1975)]

$$\lambda = \pm 1: \epsilon_\pm = \mp \sqrt{\frac{1}{2}} (0; 1, \pm i, 0), \quad (8.49)$$

$$\lambda = 0: \epsilon_0 = \frac{1}{\sqrt{-q^2}} (\sqrt{\nu^2 - q^2}; 0, 0, \nu). \quad (8.50)$$

EXERCISE 8.14 Verify that $q \cdot \epsilon = 0$ for each λ , and show that, for a spacelike photon ($q^2 < 0$),

$$\sum (-1)^{\lambda+1} \epsilon^{\mu*} \epsilon^\nu = -g^{\mu\nu} + \frac{q^\mu q^\nu}{q^2}, \quad (8.51)$$

where the sum runs over the three polarization states of (8.49) and (8.50).

For $q^2 > 0$ the factor $(-1)^{\lambda+1}$ is omitted and the sum (8.51) with q^2 replaced by M^2 is identical to that over the spin states of a massive vector particle, see Section 6.12.

We can now evaluate the total cross sections for polarized photons (helicity λ) interacting with unpolarized protons:

$$\sigma_\lambda^{\text{tot}} = \frac{4\pi^2\alpha}{K} \epsilon_\lambda^{\mu*} \epsilon_\lambda^\nu W_{\mu\nu}. \quad (8.52)$$

Using (8.24) for $W_{\mu\nu}$, together with the above polarization vectors, we find that the transverse and longitudinal cross sections are, respectively,

$$\sigma_T \equiv \frac{1}{2}(\sigma_+^{\text{tot}} + \sigma_-^{\text{tot}}) = \frac{4\pi^2\alpha}{K} W_1(\nu, q^2) \quad (8.53)$$

$$\sigma_L \equiv \sigma_0^{\text{tot}} = \frac{4\pi^2\alpha}{K} \left[\left(1 - \frac{\nu^2}{q^2}\right) W_2(\nu, q^2) - W_1(\nu, q^2) \right] \quad (8.54)$$

EXERCISE 8.15 Verify eqs. (8.53) and (8.54). The calculation can be greatly simplified by writing the tensor decomposition for $W_{\mu\nu}$, (8.27), in the laboratory frame, where

$$p = (M; 0, 0, 0),$$

$$q = (\nu; 0, 0, \sqrt{\nu^2 - q^2}).$$

EXERCISE 8.16 Express the $ep \rightarrow eX$ differential cross section (8.34) in terms of $\sigma_{T,L}$. That is, show that

$$\frac{d\sigma}{dE' d\Omega} \Big|_{\text{lab}} = \Gamma(\sigma_T + \epsilon\sigma_L), \quad (8.55)$$

where

$$\Gamma = \frac{\alpha K}{2\pi^2|q^2|} \frac{E'}{E} \frac{1}{1 - \epsilon}, \quad (8.56)$$

$$\epsilon = \left(1 - 2 \frac{\nu^2 - q^2}{q^2} \tan^2 \frac{\theta}{2}\right)^{-1}. \quad (8.57)$$

EXERCISE 8.17 The formalism has been set up in such a way that, when $q^2 \rightarrow 0$,

$$\begin{aligned} \sigma_T &\rightarrow \sigma^{\text{tot}}(\gamma p) \\ \sigma_L &\rightarrow 0, \end{aligned} \quad (8.58)$$

where γ is a real photon and $\sigma^{\text{tot}}(\gamma p)$ is given by (8.45).

Despite its appearance, convince yourself that $W_{\mu\nu}$ must not be singular at $q^2 = 0$. Hence, show that

$$W_2 \rightarrow 0 \quad \text{and} \quad \left(W_1 + \frac{\nu^2}{q^2} W_2 \right) \rightarrow 0 \quad (8.59)$$

as $q^2 \rightarrow 0$, and so establish that σ_L vanishes.

Can we extract additional information about the structure of the proton from these complex events where the proton breaks up? Both the structure of the events and their phenomenological interpretation look quite forbidding. The answer to this question is of great importance and is the subject of the next two chapters.

9

Partons

Equipped with the formalism of Chapter 8, we can now turn to the experimental information and ask the question, “Can small-wavelength photons resolve the quarks inside the proton target?”

9.1 Bjorken Scaling

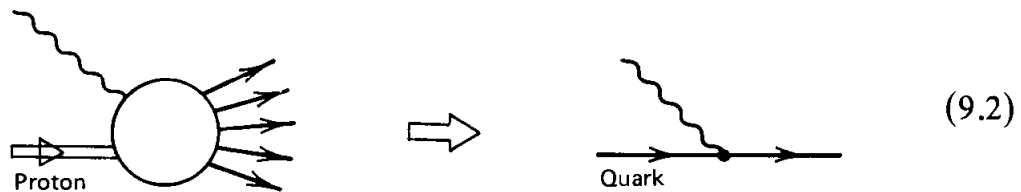
If simple, point-like, spin- $\frac{1}{2}$ quarks reside inside the proton, we should be able to illuminate them with a small-wavelength (large $-q^2$) virtual photon beam (Fig. 9.1). The fact that such photons break up the proton target can be handled by using the inelastic form factors discussed in the previous chapter. The sign that there are structureless particles inside a complex system like a proton is that for small wavelengths, the proton described by (8.43) suddenly starts behaving like a free Dirac particle (a quark) and (8.43) turns into (8.41). The proton structure functions thus become simply

$$\begin{aligned} 2W_1^{\text{point}} &= \frac{Q^2}{2m^2} \delta\left(\nu - \frac{Q^2}{2m}\right), \\ W_2^{\text{point}} &= \delta\left(\nu - \frac{Q^2}{2m}\right). \end{aligned} \quad (9.1)$$

For convenience, we have introduced the positive variable

$$Q^2 \equiv -q^2.$$

Here, m is the quark mass; the “point” notation reminds us the quark is a structureless Dirac particle. Equation (9.1) can be pictured as



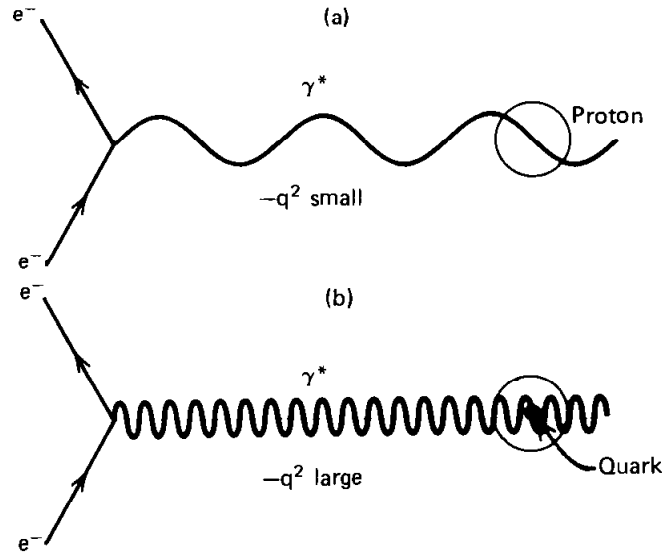


Fig. 9.1 (a) Elastic $ep \rightarrow ep$ scattering in which a large-wavelength “photon beam” measures the size of the proton through the elastic form factor analysis. (b) In deep inelastic scattering a short-wavelength “photon beam” resolves the quarks within the proton provided $\lambda (\approx 1/\sqrt{-q^2}) \ll 1F$.

that is, at large Q^2 , *inelastic* electron–proton scattering is viewed simply as *elastic* scattering of the electron on a “free” quark within the proton. Using the identity $\delta(x/a) = a \delta(x)$, (9.1) may be rearranged to introduce dimensionless structure functions

$$2mW_1^{\text{point}}(\nu, Q^2) = \frac{Q^2}{2m\nu} \delta\left(1 - \frac{Q^2}{2m\nu}\right),$$

$$\nu W_2^{\text{point}}(\nu, Q^2) = \delta\left(1 - \frac{Q^2}{2m\nu}\right). \quad (9.3)$$

These “point” functions now display the intriguing property that they are only functions of the ratio $Q^2/2m\nu$ and *not* of Q^2 and ν independently. This behavior can be contrasted with that for ep elastic scattering. For simplicity, set $\kappa = 0$, so that $G_E = G_M \equiv G$; then, comparing (8.42) and (8.43), we have

$$W_1^{\text{elastic}} = \frac{Q^2}{4M^2} G^2(Q^2) \delta\left(\nu - \frac{Q^2}{2M}\right),$$

$$W_2^{\text{elastic}} = G^2(Q^2) \delta\left(\nu - \frac{Q^2}{2M}\right). \quad (9.4)$$

In contrast to (9.1), the structure functions of (9.4) contain a form factor $G(Q^2)$, and so cannot be rearranged to be functions of a single dimensionless variable. A

mass scale is explicitly present; it is set by the empirical value 0.71 GeV in the dipole formula for $G(Q^2)$ which reflects the inverse size of the proton, see (8.20). As Q^2 increases above $(0.71 \text{ GeV})^2$, the form factor depresses the chance of elastic scattering; the proton is more likely to break up. The point structure functions, on the other hand, depend only on a dimensionless variable $Q^2/2m\nu$, and no scale of mass is present. The mass m merely serves as a scale for the momenta Q^2, ν .

The discussion can be summarized as follows: if large Q^2 virtual photons resolve “point” constituents inside the proton, then

$$\begin{aligned} MW_1(\nu, Q^2) &\xrightarrow{\text{large } Q^2} F_1(\omega), \\ \nu W_2(\nu, Q^2) &\xrightarrow{\text{large } Q^2} F_2(\omega), \end{aligned} \quad (9.5)$$

where

$$\omega = \frac{2q \cdot p}{Q^2} = \frac{2M\nu}{Q^2}. \quad (9.6)$$

Note that in (9.5) we have changed the scale from what it was in (9.3). We have introduced the proton mass instead of the quark mass to define the dimensionless variable ω . The presence of free quarks is signaled by the fact that the inelastic structure functions are independent of Q^2 at a given value of ω [see (9.5)]. This is equivalent to the onset of $\sin^{-4}(\theta/2)$ behavior for large momentum transfers in the Rutherford experiment, which reveals the “point” charge of the nucleus in the atom. A sample of data is shown in Fig. 9.2. νW_2 at $\omega = 4$ is independent of Q^2 ; the photon is indeed interacting with point-like particles. No form factors, leading to additional Q^2 dependence as in (9.4), are present. Are these particles (called partons by Bjorken) the same as the quarks discovered in the spectroscopy of hadrons (Chapter 2)?

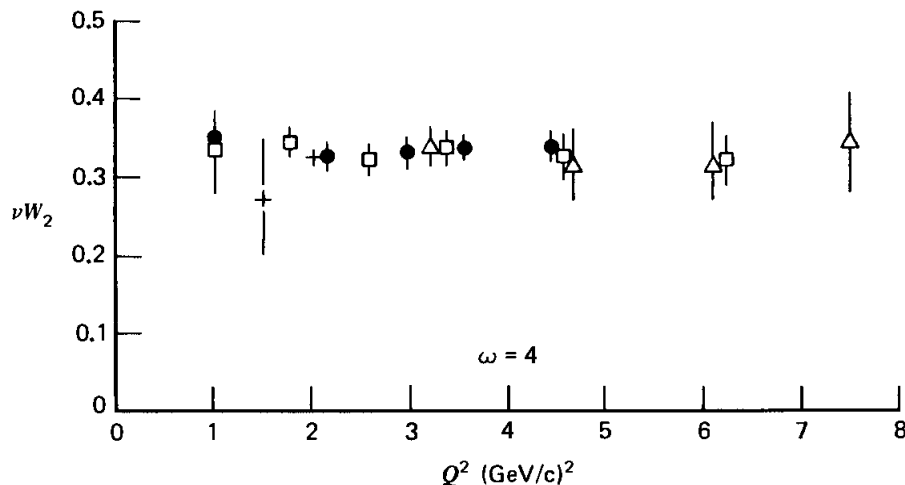
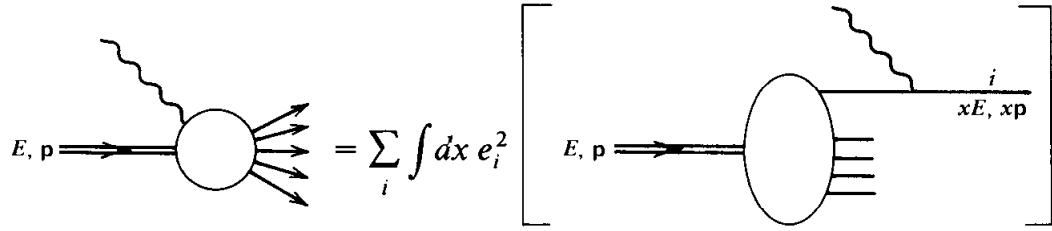


Fig. 9.2 The structure function νW_2 determined by electron-proton scattering as a function of Q^2 for $\omega = 4$. Data are from the Stanford Linear Accelerator.

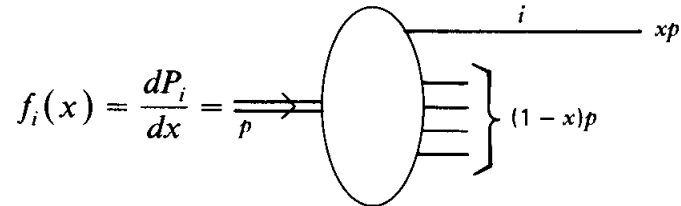
9.2 Partons and Bjorken Scaling

Now that scaling is an approximate experimental fact, we attempt to make the identification of (9.2) explicit:



$$E, p \Rightarrow \text{[Diagram]} = \sum_i \int dx e_i^2 \left[E, p \Rightarrow \text{[Diagram]} \right] \quad (9.7)$$

Equation (9.7) recognizes the fact that various types of “point” partons make up the proton ($i = u, d, \dots$, quarks, with various charges e_i , as well as gluons; the latter do not interact with the photon, of course). They can each carry a different fraction x of the parent proton’s momentum and energy. We introduce the parton momentum distribution



$$f_i(x) = \frac{dP_i}{dx} = \text{[Diagram]} \quad (9.8)$$

which describes the probability that the struck parton i carries a fraction x of the proton’s momentum p . All the fractions x have to add up to 1; therefore,

$$\sum_{i'} \int dx x f_{i'}(x) = 1. \quad (9.9)$$

Here, i' sums over all the partons, not just the charged ones i which interact with the photon. The kinematics can be summarized as follows:

	Proton	Parton	
	↓	↓	
Energy	E	xE	(9.10)
Momentum	p_L	xp_L	
	$p_T = 0$	$p_T = 0$	
Mass	M	$m = (x^2 E^2 - x^2 p_L^2)^{1/2} = xM.$	

Both the proton and its parton progeny move along the z axis (i.e., $p_T = 0$) with longitudinal momenta p_L and $x p_L$. The definition of the reference frame has to be made with more care; we shall return to it later on.

For an electron hitting a parton with momentum fraction x and unit charge, we see from (9.3) and (9.5) that the dimensionless structure functions are

$$F_1(\omega) = \frac{Q^2}{4m\nu x} \delta\left(1 - \frac{Q^2}{2m\nu}\right) = \frac{1}{2x^2\omega} \delta\left(1 - \frac{1}{x\omega}\right),$$

$$F_2(\omega) = \delta\left(1 - \frac{Q^2}{2m\nu}\right) = \delta\left(1 - \frac{1}{x\omega}\right). \quad (9.11)$$

We have used the kinematics of (9.10); ω is the dimensionless variable defined in (9.6). Summing our results for $F_{1,2}$ for one parton, (9.11), over the partons making up a proton, (9.7) and (9.8), we obtain

$$F_2(\omega) = \sum_i \int dx e_i^2 f_i(x) x \delta\left(x - \frac{1}{\omega}\right),$$

$$F_1(\omega) = \frac{\omega}{2} F_2(\omega). \quad (9.12)$$

It is conventional to redefine $F_{1,2}(\omega)$ as $F_{1,2}(x)$ and to express the results in terms of x . Recalling the identification (9.5), we see that (9.12) become, at large Q^2 ,

$$\nu W_2(\nu, Q^2) \rightarrow F_2(x) = \sum_i e_i^2 x f_i(x), \quad (9.13)$$

$$M W_1(\nu, Q^2) \rightarrow F_1(x) = \frac{1}{2x} F_2(x), \quad (9.14)$$

with

$$x = \frac{1}{\omega} = \frac{Q^2}{2M\nu}. \quad (9.15)$$

That is, the momentum fraction is found to be identical to the (dimensionless) kinematic variable x of the virtual photon that we introduced in Chapter 8; see (8.30). In other words, the virtual photon must have just the right value of the variable x to be absorbed by a parton with momentum fraction x . It is the delta function in (9.12) that equates these two distinct physical variables.

The inelastic structure functions $F_{1,2}$ of (9.13) and (9.14) are functions of only one variable, namely, x . They are independent of Q^2 at fixed x . We say they satisfy Bjorken scaling.

EXERCISE 9.1 Prove that $0 \leq x \leq 1$, as it must be if x represents a momentum fraction; recall Exercise 8.11.

Note that the kinematics, (9.10), are a bit funny. Assigning a variable mass xM to the parton is of course out of the question. Clearly, if the parton's momentum

is xp , its energy can only be xE if we put $m = M = 0$. Equivalently, a proton can only emit a parton moving parallel to it ($p_T = 0$ for both) if they both have zero mass. A corollary to this statement is that if a massive particle decays, there is a nonzero angle between its decay products. We justify our previous calculation by working in a Lorentz frame where

$$|\mathbf{p}| \gg m, M, \quad (9.16)$$

so that all masses can be neglected. In this frame, where the proton is moving with infinite momentum, the kinematics of (9.10) and the structure of $F_{1,2}(x)$ given by (9.13) and (9.14) become exact. In this frame, relativistic time dilation slows down the rate at which partons interact with one another; that is, during the short time the virtual photon interacts with the quark [see (9.7)], it is essentially a free particle, not interacting with its friends in the proton. We implicitly used this *incoherence* assumption in the derivation of $F_{1,2}(x)$; (9.7) represents an addition of probabilities (not amplitudes) of scattering from *single* free partons. It is the analogue of the impulse approximation in nuclear physics. However, there is a difference. A struck nucleon can escape from the nucleus as a completely free particle, but the struck colored parton has to recombine with the noninteracting spectator partons to form the colorless hadrons into which the proton breaks up. This has to happen with probability 1, because of color confinement (see Chapter 1); and, due to the size of the proton, it requires a much longer time scale than the quick punch the parton receives from the virtual photon. In summary, we argue that in a hard collision, the parton recoils as if it were free enabling the $ep \rightarrow eX$ cross section to be calculated [see (8.43), (9.13), and (9.14)] and that the subsequent confining final state interactions do not affect the result. This picture is valid when both the Q^2 of the virtual photon and the invariant mass of the final-state hadronic system, W , are large.

EXERCISE 9.2 Convince yourself that the interaction time is much shorter than the time scale over which the partons inside the target interact with one another. Read the explicit derivation in J. D. Bjorken and E. A. Paschos, *Phys. Rev.* 185, 1975 (1969); see also Perl (1974).

EXERCISE 9.3 It is helpful to work through an alternative derivation of the parton model result, (9.13)–(9.15), in terms of the invariant variables of (6.29):

$$s \simeq 2k \cdot p, \quad u \simeq -2k' \cdot p, \quad t \equiv -Q^2 = -2k \cdot k',$$

with particle masses neglected. Set the parton momentum $\hat{p} = xp$, that is, neglect its component transverse to the proton momentum p . We outline the steps below. Using the basic idea of (9.7), we can write

$$\left(\frac{d\sigma}{dt du} \right)_{ep \rightarrow eX} = \sum_i \int dx f_i(x) \left(\frac{d\sigma}{dt du} \right)_{eq_i \rightarrow eq_i}, \quad (9.17)$$

that is, the $ep \rightarrow eX$ rate is simply the incoherent sum over all the contributing partons. Show that the invariant variables for the parton subprocess are given by

$$\hat{s} = xs, \quad \hat{u} = xu, \quad \hat{t} = t.$$

Use these relations, together with the $e\mu$ scattering amplitude of (6.30), to show that

$$\left(\frac{d\sigma}{dt du} \right)_{eq_i \rightarrow eq_i} = x \frac{d\sigma}{d\hat{t} d\hat{u}} = x \frac{2\pi\alpha^2 e_i^2}{t^2} \left(\frac{s^2 + u^2}{s^2} \right) \delta(t + x(s + u)). \quad (9.18)$$

Now, also express the left-hand side of (9.17) in terms of s , t , and u . It is simplest to use (8.31). Verify that

$$\left(\frac{d\sigma}{dt du} \right)_{ep \rightarrow eX} = \frac{4\pi\alpha^2}{t^2 s^2} \frac{1}{s + u} \left[(s + u)^2 x F_1 - us F_2 \right], \quad (9.19)$$

where $F_1 \equiv MW_1$ and $F_2 \equiv \nu W_2$. Insert (9.18) and (9.19) into (9.17). Compare coefficients of us and $s^2 + u^2$ and so obtain the master formula of the parton model:

$$2x F_1(x) = F_2(x) = \sum_i e_i^2 x f_i(x).$$

As before, we see that $F_{1,2}$ are functions only of the scaling variable x , here fixed by the delta function in (9.18):

$$x = \frac{-t}{s + u} = \frac{Q^2}{2M\nu}. \quad (9.20)$$

EXERCISE 9.4 In Chapter 8, we evaluated the $ep \rightarrow eX$ cross section for the electron to be scattered into the $dE' d\Omega$ element in the target proton rest frame (the laboratory frame). Show that

$$dE' d\Omega = \frac{\pi}{EE'} dQ^2 d\nu = \frac{2ME}{E'} \pi y dx dy, \quad (9.21)$$

where x and y are the dimensionless variables

$$x = \frac{Q^2}{2M\nu}, \quad y = \frac{p \cdot q}{p \cdot k_{(\text{lab.})}} = \frac{\nu}{E}, \quad (9.22)$$

see (8.30). The allowed kinematic region ($0 \leq x, y \leq 1$) is shown in Fig. 9.3.

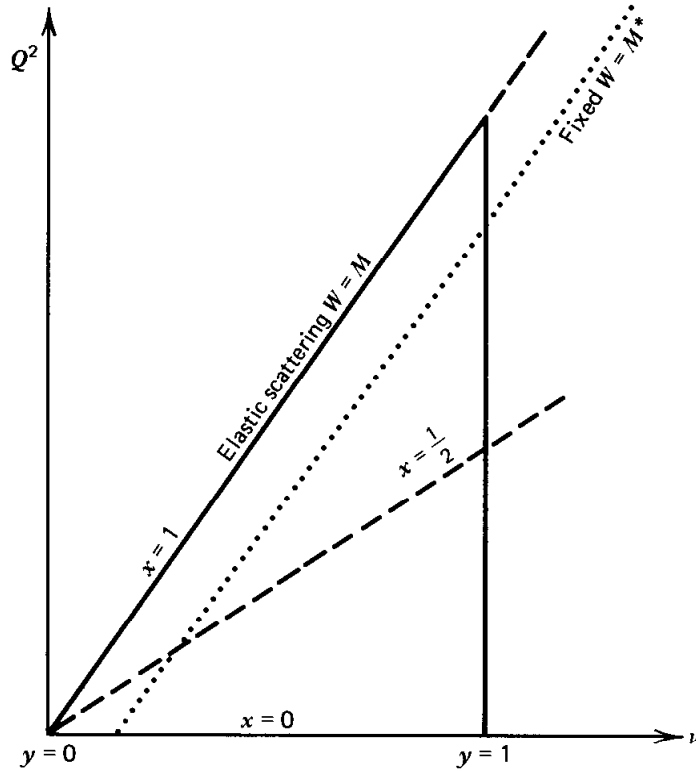


Fig. 9.3 The triangle is the allowed kinematic region for $ep \rightarrow eX$. $\nu_{\max} = E$ in the laboratory frame. W is the invariant mass of the hadronic state X , see (8.29).

Verify that the $ep \rightarrow eX$ cross section may be written in the invariant form

$$M\nu_{\max} \frac{d\sigma}{dx dy} = \frac{2\pi\alpha^2}{x^2 y^2} \left\{ xy^2 F_1 + \left[(1-y) - \frac{Mxy}{2\nu_{\max}} \right] F_2 \right\}, \quad (9.23)$$

where $\nu_{\max} = E$ in the laboratory frame.

EXERCISE 9.5 Use (9.23) to show that the parton model predicts

$$\left(\frac{d\sigma}{dx dy} \right)_{ep \rightarrow eX} = \frac{2\pi\alpha^2}{Q^4} s \left[1 + (1-y)^2 \right] \sum_i e_i^2 x f_i(x) \quad (9.24)$$

if particle masses are neglected.

EXERCISE 9.6 Show that

$$1 - y = \frac{p \cdot k'}{p \cdot k} \simeq \frac{1}{2} (1 + \cos \theta), \quad (9.25)$$

where θ is the scattering angle in the electron-quark center-of-mass frame.

Hence, identify the $(1 - y)^2$ and 1 terms in (9.24) with scattering between an electron and quark with opposite helicities and with the same helicity, respectively.

The parton model result $2xF_1 = F_2$ is known as the Callan–Gross relation. It is a consequence of the quarks having spin $\frac{1}{2}$ and is well borne out by the data.

EXERCISE 9.7 In the limit $\nu, Q^2 \rightarrow \infty$, with x fixed, show that the Callan–Gross relation implies that the virtual photon–quark cross sections of (8.53), (8.54) satisfy

$$\frac{\sigma_L}{\sigma_T} \rightarrow 0. \quad (9.26)$$

EXERCISE 9.8 Starting from (6.51), show that if quarks had spin 0, $F_2(x)$ would still be given by (9.13) but that $F_1(x) = 0$ and hence $\sigma_T = 0$.

Thus, in contrast to (9.26), spin-0 quarks would yield $\sigma_T/\sigma_L = 0$. We can understand this difference by glancing at Fig. 9.4, which shows the head-on collision between the quark and the virtual photon. By conservation of J_z (with z along $\mathbf{p}$), we see that a spin-0 quark cannot absorb a photon of helicity $\lambda = \pm 1$, so $\sigma_T = 0$. Suppose now the quark has spin $\frac{1}{2}$. We recall that its helicity is conserved in a high-energy interaction (see Section 6.6). This can only be achieved by a $\lambda = \pm 1$ photon; hence, $\sigma_L \rightarrow 0$ in this case.

9.3 The Quarks Within the Proton

Does the photon see the proton structure described in Chapter 1, that is, does virtual photon–proton scattering look like Fig. 9.5? Just as measurements of elastic form factors provided us with information on the size of the proton, measurements on the inelastic structure function at large Q^2 reveal the quark structure of the proton. After establishing Bjorken scaling, telling us that the constituents are there, eqs. (9.13) and (9.14) become the tools for extracting further information. The sum in (9.13) runs over the charged partons in the

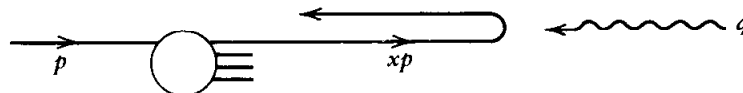


Fig. 9.4 Head-on collision between a constituent quark and the virtual photon.

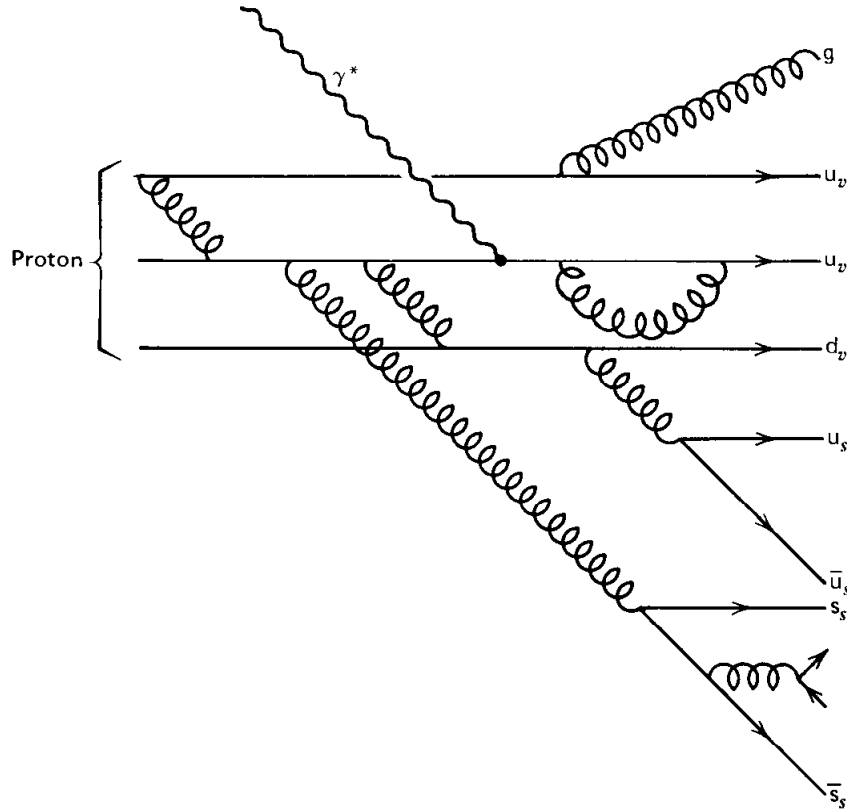


Fig. 9.5 A proton made up of valence quarks, gluons, and slow debris consisting of quark-antiquark pairs.

proton:

$$\begin{aligned} \frac{1}{x} F_2^{ep}(x) = & \left(\frac{2}{3}\right)^2 [u^p(x) + \bar{u}^p(x)] + \left(\frac{1}{3}\right)^2 [d^p(x) + \bar{d}^p(x)] \\ & + \left(\frac{1}{3}\right)^2 [s^p(x) + \bar{s}^p(x)], \end{aligned} \quad (9.27)$$

where $u^p(x)$ and $\bar{u}^p(x)$ are the probability distributions of u quarks and antiquarks within the proton. We have neglected the possibility of a sizable presence of charm and heavier quarks inside the proton.

We have six unknown quark structure functions, $f_i(x)$. However, the inelastic structure functions for neutrons are experimentally accessible by scattering electrons from a deuterium target. The counterpart to (9.27) is

$$\frac{1}{x} F_2^{en} = \left(\frac{2}{3}\right)^2 [u^n + \bar{u}^n] + \left(\frac{1}{3}\right)^2 [d^n + \bar{d}^n] + \left(\frac{1}{3}\right)^2 [s^n + \bar{s}^n]; \quad (9.28)$$

and as the proton and neutron are members of an isospin doublet, their quark content is related. There are as many u quarks in a proton as d quarks in a

neutron, and so on. So

$$\begin{aligned} u^p(x) &= d^n(x) \equiv u(x), \\ d^p(x) &= u^n(x) \equiv d(x), \\ s^p(x) &= s^n(x) \equiv s(x). \end{aligned} \quad (9.29)$$

EXERCISE 9.9 Show that the above expressions lead to the bounds

$$\frac{1}{4} \leq \frac{F_2^{en}(x)}{F_2^{ep}(x)} \leq 4$$

whatever the value of x . The lower (upper) limit would be realized if only u (d) quarks were present in the proton.

Further constraints on the quark structure functions $f_i(x)$ result from the fact that the quantum numbers of the proton must be exactly those of the uud combination of “valence” quarks of Chapter 2. As shown in Fig. 9.5, we describe the proton as three-constituent or three-valence quarks $u_v u_v d_v$ accompanied by many quark–antiquark pairs $u_s \bar{u}_s$, $d_s \bar{d}_s$, $s_s \bar{s}_s$, and so on. These are known as “sea” quarks. If we picture them as being radiated by the valence quarks, as in Fig. 9.5, then as a first approximation we may assume that the three lightest flavor quarks (u, d, s) occur in the “sea” with roughly the same frequency and momentum distribution, and neglect the heavier flavor quark pairs $c_s \bar{c}_s$, and so on. This picture of the proton can then be summarized as follows:

$$u_s(x) = \bar{u}_s(x) = d_s(x) = \bar{d}_s(x) = s_s(x) = \bar{s}_s(x) = S(x), \quad (9.30a)$$

$$u(x) = u_v(x) + u_s(x), \quad (9.30b)$$

$$d(x) = d_v(x) + d_s(x), \quad (9.30c)$$

where $S(x)$ is the sea quark distribution common to all quark flavors. Clearly, the heavier strange quarks are penalized in the radiation process by some threshold suppression so that (9.30a) is only approximately true.

By summing over all contributing partons, we must recover the quantum numbers of the proton: charge 1, baryon number 1, strangeness 0. It follows that

$$\begin{aligned} \int_0^1 [u(x) - \bar{u}(x)] dx &= 2, \\ \int_0^1 [d(x) - \bar{d}(x)] dx &= 1, \\ \int_0^1 [s(x) - \bar{s}(x)] dx &= 0. \end{aligned} \quad (9.31)$$

These sum rules express the requirement that the net number of each kind of valence quark corresponds to the uud combination of constituents discussed in

Chapter 2. Equation (9.31) clearly follows from (9.30) with

$$\begin{aligned} u - \bar{u} &= u - \bar{u}_s = u - u_s = u_v, \\ d - \bar{d} &= d - \bar{d}_s = d - d_s = d_v, \\ s - \bar{s} &= s_s - \bar{s}_s = 0. \end{aligned}$$

Note however that the sum rules are true in any picture where the sea is taken to be made of quark–antiquark pairs, and so does not affect the quantum numbers of the proton, which are exclusively determined by the valence quarks, as expressed by (9.31).

Combining (9.30) with (9.27) and (9.28), we obtain

$$\begin{aligned} \frac{1}{x} F_2^{ep} &= \frac{1}{9} [4u_v + d_v] + \frac{4}{3} S, \\ \frac{1}{x} F_2^{en} &= \frac{1}{9} [u_v + 4d_v] + \frac{4}{3} S, \end{aligned} \quad (9.32)$$

where $\frac{4}{3}$ is the sum of e_i^2 over the six sea quark distributions. Since gluons create the $q\bar{q}$ pairs in the sea, we expect $S(x)$ to have a bremsstrahlung-like spectrum at small x , so that the number of sea quarks grows logarithmically as $x \rightarrow 0$ (see Exercise 9.10).

EXERCISE 9.10 Assume that the virtual photon–proton total cross section of Section 8.5 behaves like a constant as $x \rightarrow 0$, $\nu \rightarrow \infty$ for fixed Q^2 , and hence show that

$$f_i(x) \xrightarrow{x \rightarrow 0} \frac{1}{x}. \quad (9.33)$$

Thus, we have a logarithmic growth of partons at small x .

When probing the small-momentum ($x \approx 0$) debris of the proton, we therefore anticipate that the presence of the three valence quarks will be overshadowed by these multiple, low-momentum $q\bar{q}$ pairs that make up the sea $S(x)$. According to (9.32), this means that

$$\frac{F_2^{en}(x)}{F_2^{ep}(x)} \xrightarrow{x \rightarrow 0} 1. \quad (9.34)$$

This is indeed the case experimentally, as can be checked from the data in Fig. 9.6. On the other hand, when probing the large-momentum part of the proton structure ($x \approx 1$), the fast-valence quarks u_v, d_v leave little momentum unoccupied for sea pairs. In this limit, the valence quarks dominate in (9.32), and therefore

$$\frac{F_2^{en}(x)}{F_2^{ep}(x)} \xrightarrow{x \rightarrow 1} \frac{u_v + 4d_v}{4u_v + d_v}. \quad (9.35)$$

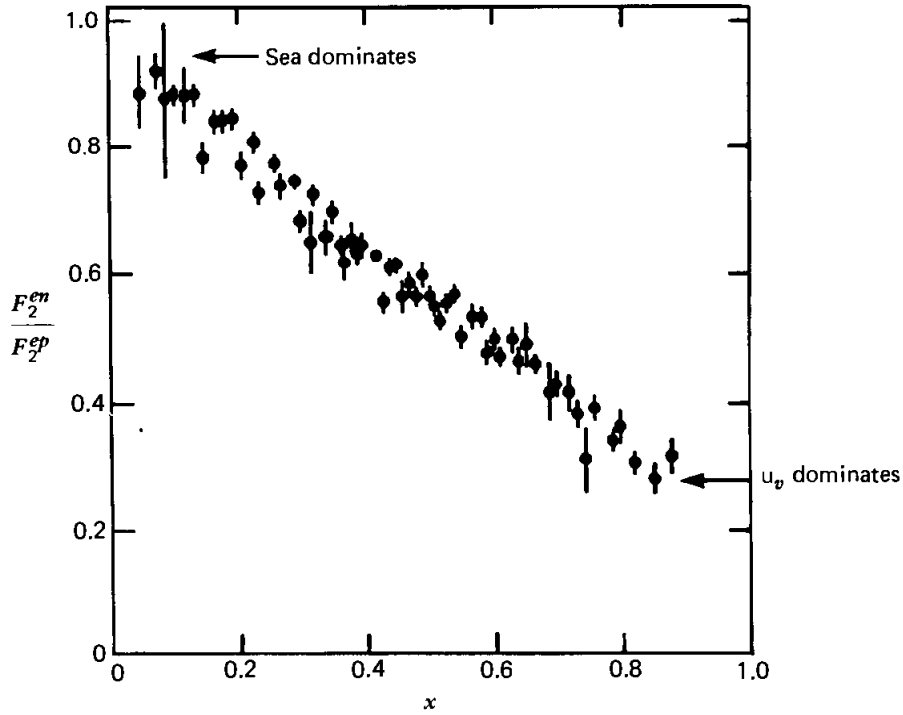


Fig. 9.6 The ratio F_2^{en}/F_2^{ep} as a function of x , measured in deep inelastic scattering. Data are from the Stanford Linear Accelerator.

For the proton, there is evidence that $u_v \gg d_v$ at large x , and the ratio (9.35) tends to $\frac{1}{4}$, as hinted at in Fig. 9.6.

EXERCISE 9.11 Discuss, on physical grounds, the behavior of $f_i(x)$ in the limit as $x \rightarrow 1$ when parton i carries all the momentum of the proton. Counting rules have been proposed which argue that

$$f_i(x) \xrightarrow{x \rightarrow 1} (1-x)^{2n_s-1},$$

where n_s is the number of spectator valence quarks which share between them the residual, vanishingly small momentum of the proton. Contrast the $x \rightarrow 1$ behavior of $u^p(x)$ with that of $u^\pi(x)$, the u -quark structure function of a π^+ -meson.

What should the complete $F_2(x)$ look like according to our picture of the proton (Fig. 9.5)? Its shape can be guessed by successive approximation; see Fig. 9.7. Figure 9.7 is self-explanatory. The transition from scenario 2 to 3 is of course due to the fact that once the quarks interact, they can redistribute the momenta among themselves, and the sharply defined momentum $x = \frac{1}{3}$ is washed out, becoming a distribution of momenta peaked around $x = \frac{1}{3}$. Data on $F_2^{ep}(x)$ at large Q^2 do indeed have the general shape of scenario 4 corresponding to Fig. 9.5.

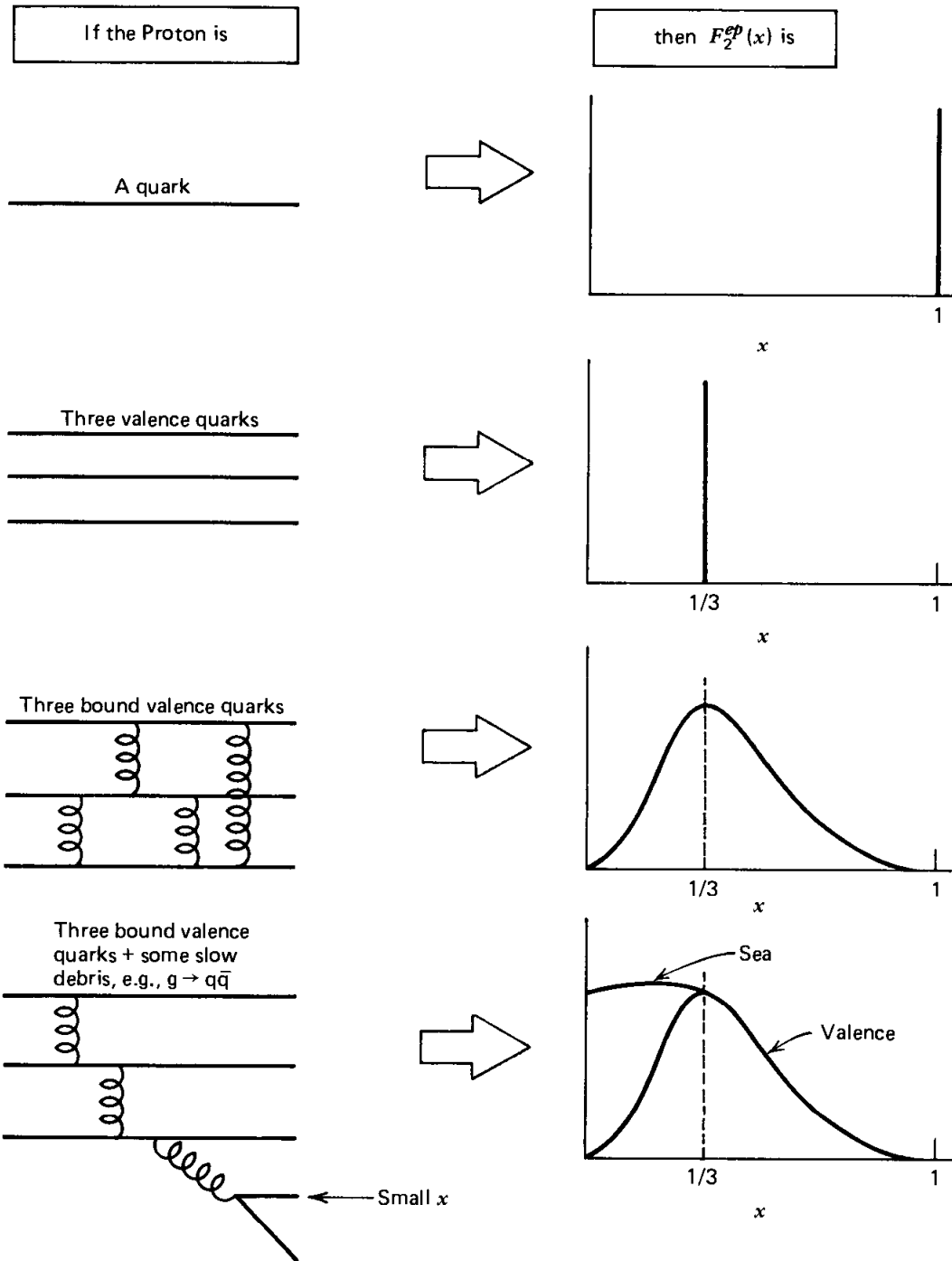


Fig. 9.7 The structure function pictured corresponding to different compositions assumed for the proton.

However, by subtracting eqs. (9.32),

$$\frac{1}{x} [F_2^{ep}(x) - F_2^{en}(x)] = \frac{1}{3} [u_v(x) - d_v(x)], \quad (9.36)$$

we can observe the valence quarks without their sea quark partners. The result should look like scenario 3 of Fig. 9.7 and should peak around $\frac{1}{3}$. It does, as can be seen in Fig. 9.8.

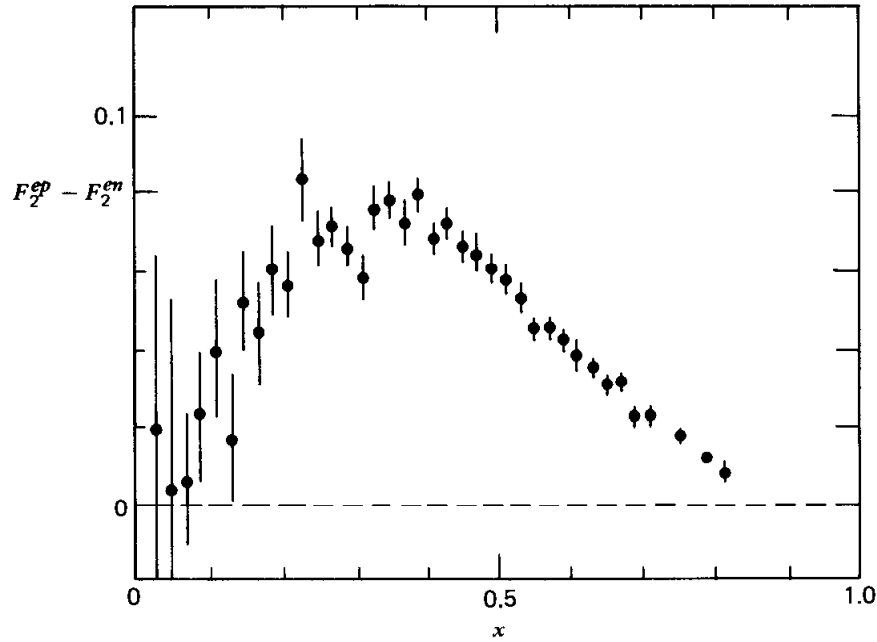


Fig. 9.8 The difference $F_2^{ep} - F_2^{en}$ as a function of x , as measured in deep inelastic scattering. Data are from the Stanford Linear Accelerator.

An alternative phenomenological approach is to parametrize all large Q^2 data on $F_2^{ep, en}(x)$ in terms of the valence and sea distributions and extract the quark structure functions at each x subject to the sum rules (9.31). The result of such an analysis is shown in Fig. 9.9. Fig. 9.9a shows the $\bar{q}$ distribution, which has been subjected also to assumption (9.30). We see how $u(x) = u_v(x) + u_s(x)$ approaches $\bar{u}(x)$ at small x as $xu_v(x) \rightarrow 0$. Figure 9.9b displays the general shape of the total valence and sea quark components, corresponding to scenarios 3 and 4 in Fig. 9.7. Note how slow the sea quarks are as compared to their valence partners.

9.4 Where Are the Gluons?

If we sum over the momenta of all the partons, we must reconstruct the total momentum p of the proton [see (9.8)],

$$\int_0^1 dx (xp) [u + \bar{u} + d + \bar{d} + s + \bar{s}] = p - p_g,$$

or, dividing by p ,

(9.37)

$$\int_0^1 dx x (u + \bar{u} + d + \bar{d} + s + \bar{s}) = 1 - \epsilon_g.$$

The momentum fraction $\epsilon_g \equiv p_g/p$ carried by the gluons is not directly exposed

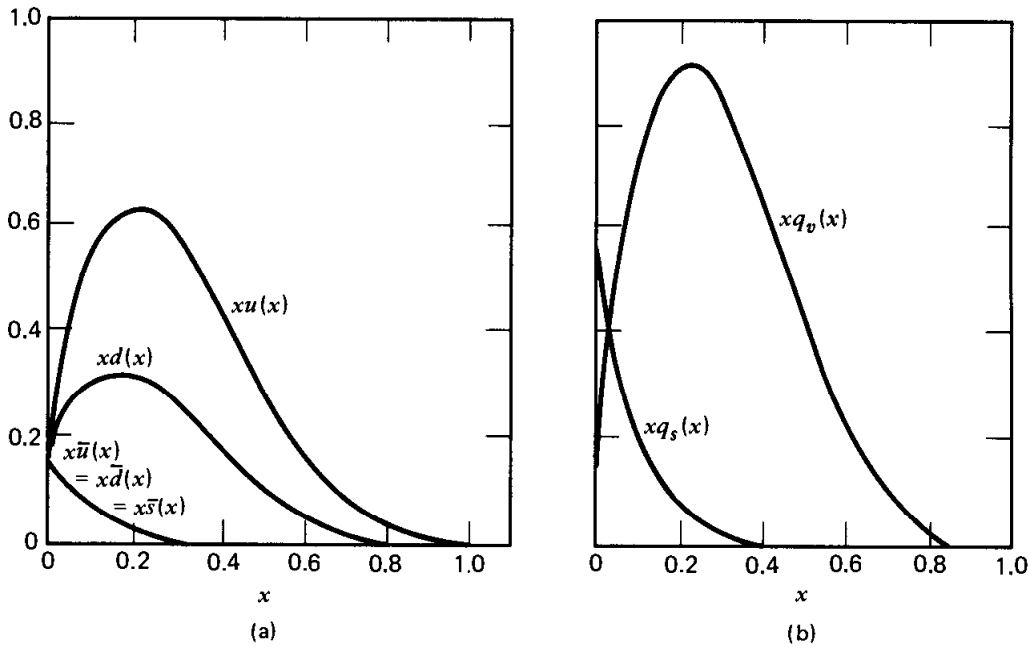


Fig. 9.9 The quark structure functions extracted from an analysis of deep inelastic scattering data. Figure (b) shows the total valence and sea quark contributions to the structure of the proton.

by the photon probe (since gluons carry no electric charge) and is therefore subtracted from the right-hand side. Integrating over the experimental data on $F_2^{ep, en}(x)$ gives us the following information:

$$\begin{aligned} \int dx F_2^{ep}(x) &= \frac{4}{9}\epsilon_u + \frac{1}{9}\epsilon_d = 0.18, \\ \int dx F_2^{en}(x) &= \frac{1}{9}\epsilon_u + \frac{4}{9}\epsilon_d = 0.12, \end{aligned} \quad (9.38)$$

where

$$\epsilon_u \equiv \int_0^1 dx x(u + \bar{u})$$

is the momentum carried by u quarks and antiquarks, and similarly for ϵ_d . Equation (9.38) follows from (9.27) and (9.28) after neglecting the strange quarks which carry a small fraction of the nucleon's momentum. From (9.37), we have

$$\epsilon_g = 1 - \epsilon_u - \epsilon_d,$$

and on solving (9.38), we obtain

$$\epsilon_u = 0.36, \quad \epsilon_d = 0.18, \quad \epsilon_g = 0.46. \quad (9.39)$$

Hence, the gluons carry about 50% of the momentum, which was unaccounted for by the charged quarks.

In summary, an analysis of data on deep inelastic scattering of leptons by nucleons reveals the presence of point-like Dirac particles inside hadrons through Bjorken scaling. A study of the quantum numbers of these partons allows us to identify them with the quarks introduced in the study of the hadron spectrum in Chapter 2. The momentum distribution of the quarks forces us to the conclusion that a substantial fraction of the proton's momentum is carried by neutral partons, not by quarks. These are the gluons of QCD.

10

Quantum Chromodynamics

10.1 The Dual Role of Gluons

We have seen in the previous chapter that deep inelastic scattering measurements actually require the existence of electrically neutral as well as charged constituents of the proton. We concluded that the charged partons could be identified with the colored quarks postulated in Chapter 2 to explain the observed systematics of the hadron spectrum. It is tempting to identify the neutral partons with gluons. Is this identification justified? That is, does experiment actually require the existence of gluons independent of the formal arguments which were the original motivation for postulating their existence?

Recall that the color charge of quarks was originally introduced to remedy a statistics problem in constructing the Δ^{++} wavefunction (see Section 2.11). It is interesting to reflect on the fact that although it is economical to associate the same color charge with the charge of the strong interaction and although it is helpful that such a force, mediated by the exchange of gluons, is asymptotically free (see Sections 1.3 and 7.9) so that we can apply perturbation theory, none of these arguments constitutes direct evidence that quantum chromodynamics (QCD) is the correct physical theory.

It is clearly crucial to check that the gluons, introduced in Chapters 1 and 2, can indeed be identified with the neutral partons discovered as “missing momentum” in deep inelastic scattering. Subsequently, we can ask more probing questions and verify that their dynamical properties correspond to those of the carriers of the color force.

In order to proceed, it is not necessary at this stage to formally develop QCD as a color gauge theory. It is sufficient to recall the essential properties of the theory. We have already discussed them in Chapters 1 and 2.

- Quarks carry color as well as electric charge; there are three colors, R, G, and B.
- Color is exchanged by eight bicolored gluons (see Fig. 1.4).
- Color interactions are assumed to be “a copy of electromagnetic interactions.” To be precise, quark–gluon interactions are computed by the rules of QED with the substitution $\sqrt{\alpha} \rightarrow \sqrt{\alpha_s}$ at each vertex (see Fig. 1.9) and

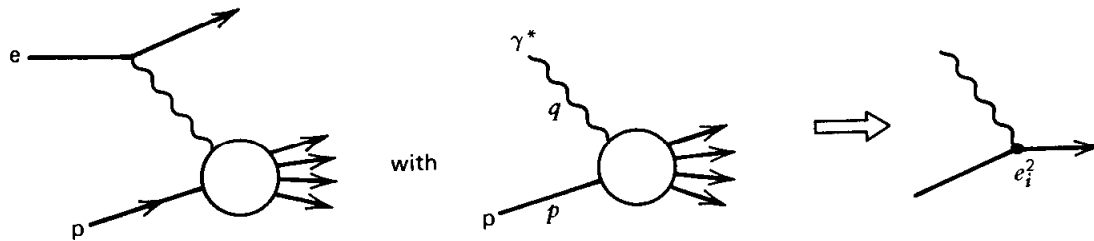


Fig. 10.1 Pictorial representation of the parton model for $ep \rightarrow eX$.

the introduction of a color factor, which may be computed by the methods of Section 2.15. That is, the $q\bar{q}g$ vertex has the same structure as the $e\bar{e}\gamma$ vertex. The (eight) gluons are massless and have spin 1.

- Gluons themselves carry color charge, and so they can interact with other gluons. That is, there is a ggg as well as a qqg vertex in the theory (see Fig. 1.4d).
- At short distances, α_s is sufficiently small so that we can compute color interactions using the perturbative techniques familiar from QED.

The crucial statement that “color interactions are a copy of electromagnetic interactions” will be given a formal meaning in Chapter 14, where both QED and QCD are shown to be a consequence of local gauge symmetries.

How does the color dynamics of the partons (quarks *and* gluons) affect our discussion of deep inelastic scattering in the two previous chapters? The parton model of Chapter 9, symbolically represented by Fig. 10.1, completely ignores the dynamical role of gluons as the carriers of the strong force associated with colored quarks. We have, for instance, neglected the fact that quarks can radiate gluons. We must therefore allow for the possibility that the quark in Fig. 10.1 may radiate a gluon before or after being struck by the virtual photon, γ^* . These possibilities are shown in Fig. 10.2. Moreover, a gluon constituent in the target can contribute to deep inelastic scattering via $\gamma^*g \rightarrow q\bar{q}$ pair production as shown in Fig. 10.3. In a computational sense, the processes in Figs. 10.2 and 10.3 are $O(\alpha_s)$ contributions to the cross section, whereas the leading contribution in Fig. 10.1 is $O(\alpha)$.

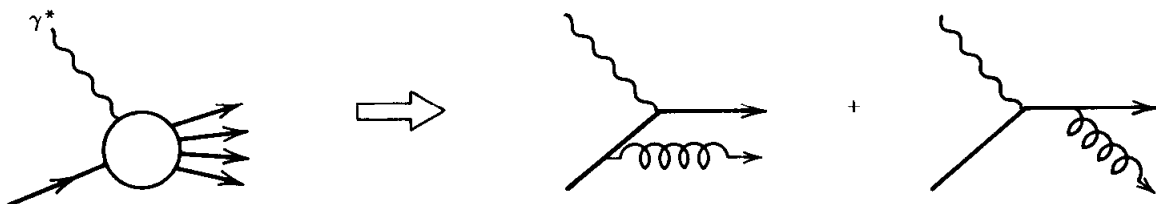


Fig. 10.2 $O(\alpha_s)$ contributions ($\gamma^*q \rightarrow qg$) to $ep \rightarrow eX$.

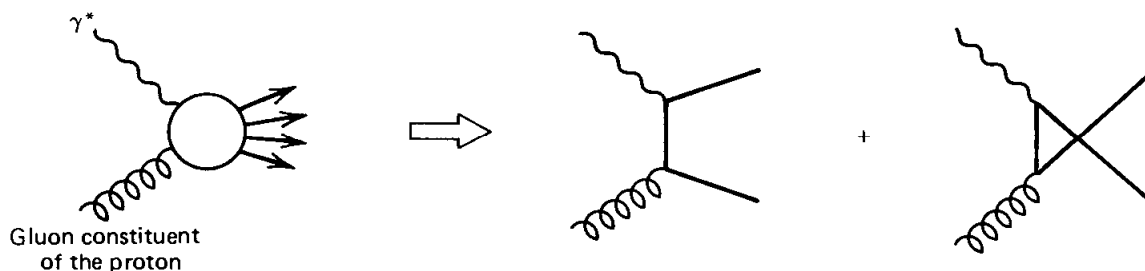


Fig. 10.3 $O(\alpha_s)$ gluon-initiated “hard” scattering contributions ($\gamma^* g \rightarrow q\bar{q}$) to $ep \rightarrow eX$.

The inclusion of QCD diagrams of the type shown in Figs. 10.2 and 10.3 has two experimentally observable consequences: (1) the scaling property of the structure functions will no longer be true, and (2) the outgoing quark (and therefore the direction of its hadron jet) will no longer be collinear with the virtual photon. Point (2) is visualized in Fig. 10.4. In the parton model of Chapter 9, final-state hadrons in the jet of the struck quark are produced in the direction of the virtual photon (which is experimentally measured) as sketched in Fig. 10.4a. But if gluons are emitted (Fig. 10.4b), the quark can recoil against a radiated gluon and two jets are produced (indicated by arrows) each of which has a transverse momentum p_T relative to the virtual photon.

Quark–gluon color theory (QCD) allows us to compute the contribution from the diagrams of Figs. 10.2 and 10.3. Using QCD, we can therefore predict the scaling violations as well as the jet angular distributions relative to the virtual photon. They can then be compared with experiment and so the quark–gluon dynamics can be confronted with data in a quantitative and completely dynamical sense.

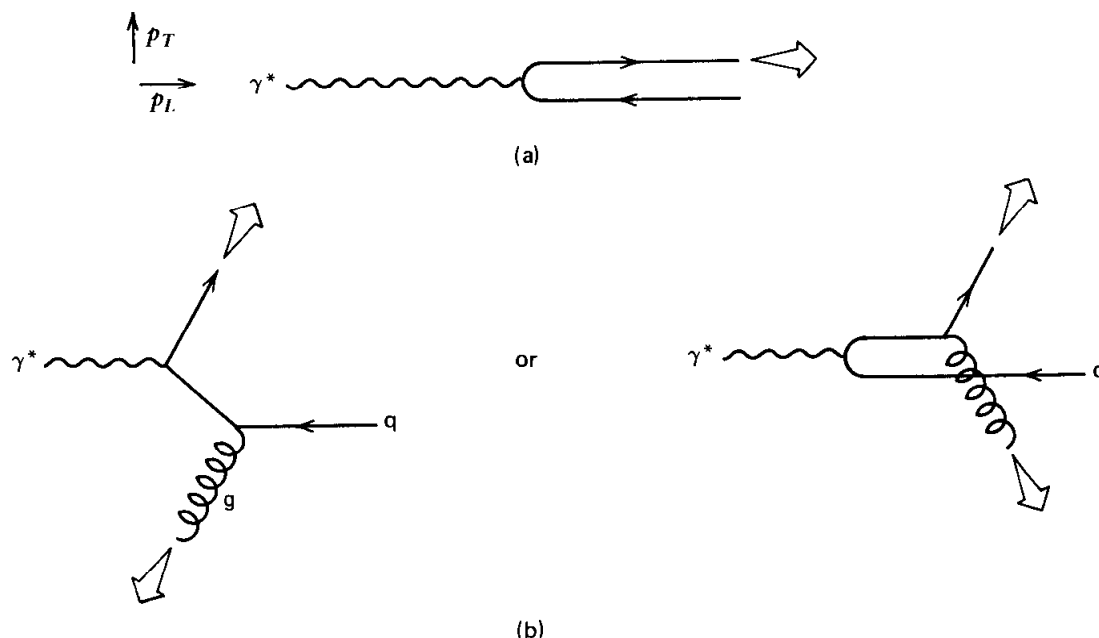


Fig. 10.4 (a) Parton model diagram for $\gamma^* q \rightarrow q$, producing a jet with $p_T = 0$. (b) Gluon emission diagrams which produce jets with $p_T \neq 0$.

10.2 Embedding γ^* -Parton Processes in Deep Inelastic Scattering

How do we find the contribution of these γ^* -parton scattering diagrams to the cross section for deep inelastic scattering? A useful starting point is to recall that the $ep \rightarrow eX$ cross section is given in terms of the structure functions $W_{1,2}$ or, equivalently,

$$\begin{aligned} F_1 &= MW_1(\nu, Q^2), \\ F_2 &= \nu W_2(\nu, Q^2), \end{aligned} \quad (10.1)$$

see (8.34). Since we are going beyond the parton model, we shall find $F_{1,2}$ no longer scale; that is, they are functions of both

$$\nu \equiv \frac{P \cdot q}{M} \quad \text{and} \quad Q^2 = -q^2, \quad (10.2)$$

rather than simply the ratio $x = Q^2/2M\nu$. In Chapter 8, we interpreted the structure functions in terms of the virtual photon–proton cross sections; see (8.53) and (8.54). In the deep inelastic limit, these relations simplify to

$$2F_1 = \frac{\sigma_T}{\sigma_0} \quad (10.3)$$

$$\frac{F_2}{x} = \frac{\sigma_T + \sigma_L}{\sigma_0} \quad (10.4)$$

where σ_T and σ_L are the γ^*p total cross sections for transverse and longitudinal virtual photons, respectively, and

$$\sigma_0 \equiv \frac{4\pi^2\alpha}{2MK} \simeq \frac{4\pi^2\alpha}{s}. \quad (10.5)$$

EXERCISE 10.1 Derive (10.3)–(10.5). It is easiest to work in the laboratory frame; the variables are given in Section 6.8. In the deep inelastic limit, the electron beam energy $E \gg E'$. Then, as $\nu^2 \sim E^2$ and $Q^2 \simeq 4EE'\sin^2(\theta/2)$, we have $\nu^2 \gg Q^2$. We also have introduced the γ^* -proton center-of-mass energy squared:

$$s = (q + p)^2 = M^2 + 2M\nu - Q^2 \simeq 2MK, \quad (10.6)$$

where K is associated with the γ^* -flux factor of (8.48).

In calculating the γ^* -parton contributions to σ_T and σ_L , we immediately face a problem. Equations (10.3) and (10.4) refer to γ^* -proton and not γ^* -parton cross sections. For example, consider single gluon emission ($\gamma^*q \rightarrow qg$), which is shown

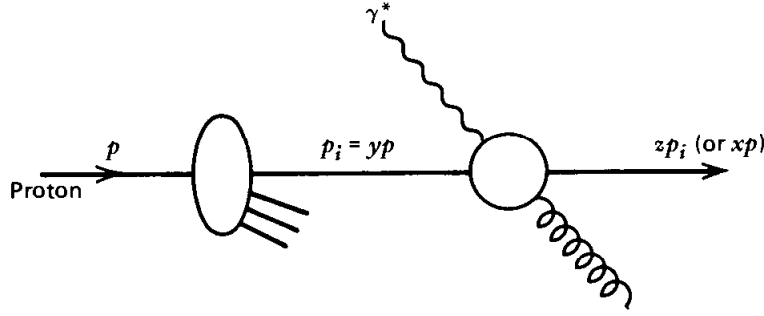


Fig. 10.5 Embedding $\gamma^*q \rightarrow qg$ in $\gamma^*p \rightarrow X$.

in Fig. 10.5. The relation between the two frames is given by

γ^* -Proton Frame	$\longrightarrow$	γ^* -Parton Frame
p	$\longrightarrow$	$p_i = yp$
$x = \frac{Q^2}{2p \cdot q}$	$\longrightarrow$	$z = \frac{Q^2}{2p_i \cdot q} = \frac{x}{y}$

Here, we have relied extensively on the fact that both collinear frames move with infinite momentum.

We are now ready to relate cross section ratios in the two systems. For instance, for the ratio (10.3), we have

$$\left(\frac{\sigma_T(x, Q^2)}{\sigma_0} \right)_{\gamma^*p} = \sum_i \int_0^1 dz \int_0^1 dy f_i(y) \delta(x - zy) \left(\frac{\hat{\sigma}_T(z, Q^2)}{\hat{\sigma}_0} \right)_{\gamma^*i} \quad (10.7)$$

where $f_i(y)$ are the parton structure functions (which give the probability that there is a parton i carrying a fraction y of the proton's momentum p) and $\hat{\sigma}_T$ is the cross section for the absorption of a transverse photon of momentum q by a parton of momentum p_i ; x is fixed, and we have to integrate over all z, y subject to the constraint $x = zy$. Using the delta function to perform the z integration, we obtain

$$\boxed{\left(\frac{\sigma_T(x, Q^2)}{\sigma_0} \right) = \sum_i \int_x^1 \frac{dy}{y} f_i(y) \left(\frac{\hat{\sigma}_T(x/y, Q^2)}{\hat{\sigma}_0} \right)}. \quad (10.8)$$

Throughout this chapter, we use $\hat{\sigma}$, $\hat{s}$, $\hat{t}$, and so on, to distinguish γ^* -parton quantities from those of the parent process, σ , s , t , and so on.

10.3 The Parton Model Revisited

If all gluon effects were absent, we should be able to recover the parton model result from (10.8). In this case, $\hat{\sigma}_T$ and $\hat{\sigma}_L$ are given by the diagram of Fig. 10.1,

$\gamma^*q \rightarrow q$. If we neglect the mass of the outgoing quark, $(q + p_i)^2 = 0$. Hence,

$$z = \frac{Q^2}{2p_i \cdot q} = 1, \quad (10.9)$$

and $\hat{\sigma}(z, Q^2)$ is proportional to $\delta(1 - z)$.

EXERCISE 10.2 Show that the parton model diagram, Fig. 10.1, gives

$$\frac{\hat{\sigma}_T(z, Q^2)}{\hat{\sigma}_0} = e_i^2 \delta(1 - z) \quad (10.10)$$

$$\hat{\sigma}_L(z, Q^2) = 0. \quad (10.11)$$

We outline the various stages of the calculation. First, show that for $\gamma^*(q)q(p) \rightarrow q(p')$,

$$|\overline{\mathcal{M}}|^2 = 2e_i^2 e^2 p \cdot q, \quad (10.12)$$

where we have averaged over transverse polarization states of the incoming γ^* . From Section 4.3, we have

$$F d\hat{\sigma}_T = |\overline{\mathcal{M}}|^2 (2\pi)^4 \delta^{(4)}(p' - p - q) \frac{d^3 p'}{2p'_0 (2\pi)^3}. \quad (10.13)$$

where F is the γ^*q flux factor. Calculate $F\hat{\sigma}_T$ by making use of (6.47). Use $F\hat{\sigma}_0 = 8\pi^2\alpha$, see (10.5).

To determine the parton model prediction for F_2/x of (10.4), we input in (10.8) the following cross section ratio for $\gamma^*q \rightarrow q$:

$$\frac{1}{\hat{\sigma}_0} (\hat{\sigma}_T + \hat{\sigma}_L) = e_i^2 \delta(1 - z), \quad (10.14)$$

see (10.10) and (10.11). After substitution, we obtain

$$\frac{F_2(x, Q^2)}{x} = \sum_i e_i^2 \int_x^1 \frac{dy}{y} f_i(y) \delta\left(1 - \frac{x}{y}\right) = \sum_i e_i^2 f_i(x). \quad (10.15)$$

An identical expression is found for $2F_1$. The parton model results of (9.13) and (9.14) are indeed reproduced.

10.4 The Gluon Emission Cross Section

We are now ready to include the gluon emission diagrams of Fig. 10.2. To calculate $\gamma^*q \rightarrow qg$, we take over our results for Compton scattering. In Section

6.14, we evaluated the closely related QED process $\gamma^*e \rightarrow \gamma e$ and found

$$|\overline{\mathcal{M}}|^2 = 32\pi^2\alpha^2 \left(-\frac{u}{s} - \frac{s}{u} + \frac{2tQ^2}{su} \right), \quad (10.16)$$

using for the γ^* -polarization sum $\sum \epsilon_\mu^* \epsilon_\nu = -g_{\mu\nu}$, see (6.98). The result for $\gamma^*q \rightarrow qg$ follows directly on substitution of $\alpha^2 \rightarrow e_i^2\alpha\alpha_s$ and insertion of the color factor $\frac{4}{3}$, and $u \leftrightarrow t$ on account of the different ordering of the outgoing particles; so,

$$|\overline{\mathcal{M}}|^2 = 32\pi^2(e_i^2\alpha\alpha_s)\frac{4}{3} \left(-\frac{\hat{t}}{\hat{s}} - \frac{\hat{s}}{\hat{t}} + \frac{2\hat{u}Q^2}{\hat{s}\hat{t}} \right). \quad (10.17)$$

The invariant variables are denoted $\hat{s}, \hat{t}, \hat{u}$ to indicate that we are considering a parton subprocess.

The factor $\frac{4}{3}$ takes into account the summation/averaging over final/initial colors. It can readily be deduced from counting the color lines shown in Fig. 10.6. Each color line can take one of three possible colors, so there are $(3 \times 3 - 1)$ different configurations, if we subtract the color singlet configuration as explained in Section 2.15. We now average over the three initial quark colors and obtain $\frac{8}{3}$. However, this result has to be divided by a factor 2 because of the unfortunate historical definition of α_s (see Section 2.15).

Most simple QCD diagrams are exactly analogous to QED diagrams, and the QCD cross section is obtained by the replacement

$$\alpha^n \rightarrow C_F \alpha_s^n, \quad (10.18)$$

where n is the number of quark–gluon (or gluon–gluon) vertices in the diagram. The color factor C_F is obtained by simply summing and averaging in much the same way as we do for spin. However, without changing the cross section, the color factor can be altered by a redefinition of α_s . For example,

$$\alpha_s^n C_F \rightarrow (2\alpha_s)^n (C_F/2^n) = (\alpha'_s)^n C'_F. \quad (10.19)$$

The conventional definition of α_s corresponds to the second convention in (10.19). The final rule to obtain the conventional color factor is simple: “count color on your fingers” and divide the result by 2^n . This is why we divided the color factor $\frac{8}{3}$ of Fig. 10.6 by 2^1 .

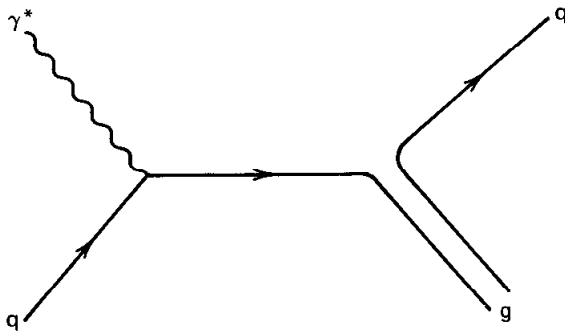


Fig. 10.6 Color lines for $\gamma^*q \rightarrow qg$.

Now back to the main problem. In preparation for the calculation of the $\gamma^* q \rightarrow qg$ cross section from (10.17), we gather together the relevant kinematics.

EXERCISE 10.3 In the center-of-mass frame of the parton process $\gamma^* q_1 \rightarrow q_2 g$ of Fig. 10.7, show that

$$\hat{s} = 2k^2 + 2kq_0 - Q^2 = 4k'^2, \quad (10.20)$$

$$\hat{t} = -Q^2 - 2k'q_0 + 2kk' \cos \theta = -2kk'(1 - \cos \theta), \quad (10.21)$$

$$\hat{u} = -2kk'(1 + \cos \theta), \quad (10.22)$$

where k, k' are the magnitudes of the center-of-mass momenta $\mathbf{k}, \mathbf{k}'$. Note that for the virtual photon, $q_0^2 = k^2 - Q^2$. A useful result is

$$4kk' = -\hat{t} - \hat{u} = \hat{s} + Q^2. \quad (10.23)$$

The interesting quantity is the transverse momentum of the outgoing quark, $p_T = k' \sin \theta$. Show that

$$p_T^2 = \frac{\hat{s}\hat{t}\hat{u}}{(\hat{s} + Q^2)^2} \quad (10.24)$$

or, in the limit of small-angle scattering, $-\hat{t} \ll \hat{s}$, that

$$p_T^2 = \frac{\hat{s}(-\hat{t})}{\hat{s} + Q^2}. \quad (10.25)$$

Further, show that for small scattering angles ($\cos \theta \simeq 1$),

$$d\Omega = \frac{4\pi}{\hat{s}} dp_T^2. \quad (10.26)$$

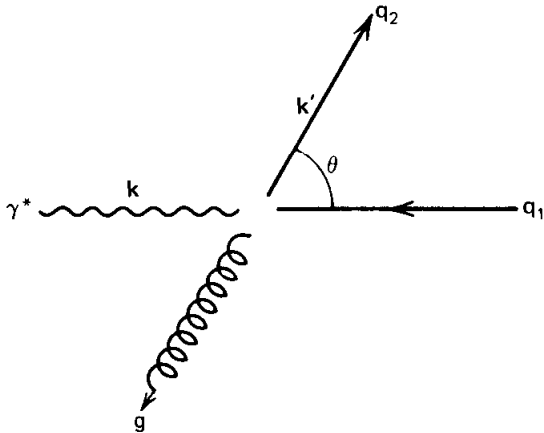


Fig. 10.7 Center-of-mass frame for $\gamma^* q_1 \rightarrow q_2 g$.

It is clear from (10.17) that at high energy ($\hat{s}$ large), the $\gamma^*q \rightarrow qg$ cross section peaks as $-\hat{t} \rightarrow 0$. Referring back to Fig. 4.8 and the related discussion, we see that this is due to quark exchange in the t channel. We can therefore approximate the cross section by its forward peak. For forward scattering, we find, from (10.26) and (4.35), that

$$\frac{d\hat{\sigma}}{dp_T^2} \simeq \frac{1}{16\pi\hat{s}^2} |\overline{\mathcal{M}}|^2. \quad (10.27)$$

EXERCISE 10.4 Derive (10.27). Make use of (10.26) and (10.20), together with (4.34). Show that the γ^*q flux factor is given by $2\hat{s}$ using the convention of (8.48).

On substituting (10.17) into (10.27), the $\gamma^*q \rightarrow qg$ cross section becomes

$$\frac{d\hat{\sigma}}{dp_T^2} = \frac{8\pi e_i^2 \alpha \alpha_s}{3\hat{s}^2} \left(\frac{1}{-\hat{t}} \right) \left[\hat{s} + \frac{2(\hat{s} + Q^2)Q^2}{\hat{s}} \right], \quad (10.28)$$

where we have used $-\hat{t} \ll \hat{s}$. Using (10.25) and

$$z \equiv \frac{Q^2}{2p_i \cdot q} = \frac{Q^2}{(p_i + q)^2 - q^2} = \frac{Q^2}{\hat{s} + Q^2}, \quad (10.29)$$

equation (10.28) can finally be rewritten as

$$\boxed{\frac{d\hat{\sigma}}{dp_T^2} \simeq e_i^2 \hat{\sigma}_0 \frac{1}{p_T^2} \frac{\alpha_s}{2\pi} P_{qq}(z)}, \quad (10.30)$$

where $\hat{\sigma}_0 = 4\pi^2\alpha/\hat{s}$, see (10.5), and where

$$P_{qq}(z) = \frac{4}{3} \left(\frac{1+z^2}{1-z} \right) \quad (10.31)$$

represents the probability of a quark emitting a gluon and so becoming a quark with momentum reduced by a fraction z . The $z \rightarrow 1$ singularity is associated with the emission of a “soft” massless gluon. It is an example of an infrared divergence (mentioned in Chapter 7). We explain how it is canceled by virtual gluon diagrams in Section 10.8.

The cross section (10.30) is also singular as $p_T^2 \rightarrow 0$. Now the diagrams of Figs. 10.1 and 10.3 either do not contribute (remember that for the parton diagram the p_T of the outgoing parton relative to the virtual photon always vanishes) or are negligible in comparison to $\gamma^*q \rightarrow qg$ in the limit $-\hat{t} \ll \hat{s}$. This is the case for the pair production diagrams of Fig. 10.3. Therefore, in the region $-\hat{t} \ll \hat{s}$, (10.30) represents the full p_T^2 distribution of the final-state parton jets.

What is the experimental signature of this result? We refer to Fig. 10.4. The presence of gluon emission is signaled by a quark jet and gluon jet in the final

state, neither of which is moving along the direction of the virtual photon. The transverse momentum p_T of the jet or of the bremsstrahlung hadrons contained in this jet (see Chapter 1) is nonzero; in fact, we expect the p_T distribution to be given by (10.30). It has to be embedded into the electron–proton system, using (10.8), exactly as was done for the parton cross section in Section 10.3. Figure 10.8 shows the result of comparing such a calculation with data.

Several remarks about this comparison are in order. The calculation shown in Fig. 10.8 actually includes all diagrams in Figs. 10.1, 10.2, and 10.3, and the limit $(-\hat{t}) \ll \hat{s}$ was not made. Moreover, one has to model the way a quark or gluon fragments into hadrons or, alternatively, one can infer this information from a different experiment. How this is done is explained in the next chapter. What is important is the qualitative observation that hadrons emerge with $p_T \neq 0$ signaling the presence of gluon emission. In a parton model without gluons, all final-state jets would be collinear with the virtual photon. Their hadron fragments will therefore be nearly collinear with the photon, too, that is, with a spread of p_T of about 300 MeV as required by the uncertainty principle for confined quarks. This prediction is represented by the dashed line in Fig. 10.8. The data clearly establish an excess of large p_T hadrons which are the fragments of the quark and gluon jets recoiling against one another.

This “Rutherford experiment of QCD” can be repeated in many disguises such as $e^+e^- \rightarrow$ hadrons or $pp \rightarrow$ large- p_T hadrons (see Chapter 1). The ideas and computational techniques are very similar to the example discussed here. Finally, the large Q^2 of the photon guarantees that we are dealing with a short-distance

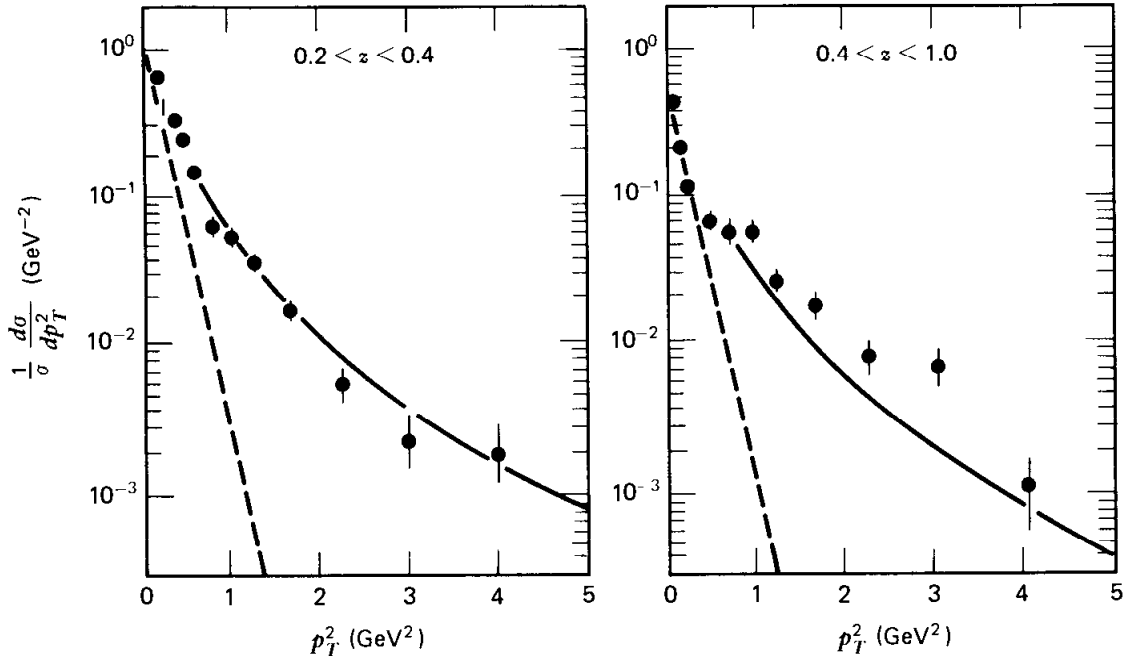


Fig. 10.8 The p_T^2 distribution of hadrons produced in μN interactions relative to the direction of the virtual photon. The dashed line is the expectation in the absence of gluon emission. Data are from the EMC collaboration at CERN. (μN and $e N$ interactions give the same curves.)

interaction where α_s is small (see Chapters 1 and 7). The number of large- p_T hadrons in Fig. 10.8 measures the normalization of the cross section of (10.30), that is, it measures $\alpha_s(Q^2)$. At these values of Q^2 the data imply that $\alpha_s \simeq 0.2$.

10.5 Scaling Violations. The Altarelli–Parisi Equation

How does the gluon bremsstrahlung diagram of (10.30) contribute to the structure functions? Recall that the structure functions are related to parton cross sections via (10.8) [and (10.3), (10.4)]. This involves the integrated γ^* -parton cross section. We therefore have to compute

$$\begin{aligned}\hat{\sigma}(\gamma^*q \rightarrow qg) &= \int_{\mu^2}^{\hat{s}/4} dp_T^2 \frac{d\hat{\sigma}}{dp_T^2} \\ &\simeq e_i^2 \hat{\sigma}_0 \int_{\mu^2}^{\hat{s}/4} \frac{dp_T^2}{p_T^2} \frac{\alpha_s}{2\pi} P_{qq}(z) \\ &\simeq e_i^2 \hat{\sigma}_0 \left(\frac{\alpha_s}{2\pi} P_{qq}(z) \log \frac{Q^2}{\mu^2} \right).\end{aligned}\quad (10.32)$$

EXERCISE 10.5 Show that the maximal transverse momentum for the two-body interaction $\gamma^*q \rightarrow qg$ is given by

$$(p_T^2)_{\max} = \frac{\hat{s}}{4} = Q^2 \frac{1-z}{4z}.\quad (10.33)$$

In deriving (10.32), we integrated up to the maximum p_T of the gluon and then used (10.33) to write $\log(\hat{s}/4) \simeq \log Q^2$ in the large Q^2 limit. The lower limit μ on the transverse momentum is introduced as a cutoff to regularize the divergence when $p_T^2 \rightarrow 0$.

Adding $\hat{\sigma}(\gamma^*q \rightarrow qg)$ to the parton model cross section, (10.14), we find QCD modifies (10.15) to

$$\begin{aligned}\frac{F_2(x, Q^2)}{x} &= \left| \text{diagram 1} \right|^2 + \left| \text{diagram 2} + \text{diagram 3} \right|^2 \\ &= \sum_q e_q^2 \int_x^1 \frac{dy}{y} q(y) \left(\delta\left(1 - \frac{x}{y}\right) + \frac{\alpha_s}{2\pi} P_{qq}\left(\frac{x}{y}\right) \log \frac{Q^2}{\mu^2} \right),\end{aligned}\quad (10.34)$$

where we have introduced the notation that the quark structure function $q(y) \equiv$

$f_q(y)$. The presence of the $\log Q^2$ factor means that the parton model scaling prediction for the structure functions should be violated. That is, in QCD, F_2 is a function of Q^2 as well as of x , but the variation with Q^2 is only logarithmic. The violation of Bjorken scaling is a signature of gluon emission.

EXERCISE 10.6 Study the origin of the $\log Q^2$ term. Recall that the $\gamma^*q \rightarrow qg$ cross section, $d\hat{\sigma}/dp_T^2$, is dominated by the forward peak. The t -channel quark propagator leads to a factor $1/p_T^4$. Show that helicity conservation at the gluon vertex weakens this singularity by introducing a factor p_T^2 in the numerator.

Equation (10.34) may be regarded as the first two terms in a power series in α_s ; α_s is a useful expansion parameter at large Q^2 since $\alpha_s \sim (\log Q^2)^{-1}$. But comparing the leading and next-order terms in (10.34), we find that the expansion parameter α_s is multiplied by $\log Q^2$. From (7.65), we know that $\alpha_s(Q^2) \log(Q^2/\mu^2)$ does not vanish at large Q^2 , and so (10.34) does not seem very useful as it stands. How should we proceed? Can we absorb the $\log Q^2$ term into a modified quark probability distribution? To this end, we rewrite (10.34) in the “parton-like” form

$$\begin{aligned} \frac{F_2(x, Q^2)}{x} &\equiv \sum_q e_q^2 \int_x^1 \frac{dy}{y} (q(y) + \Delta q(y, Q^2)) \delta\left(1 - \frac{x}{y}\right) \\ &= \sum_q e_q^2 (q(x) + \Delta q(x, Q^2)) \end{aligned} \quad (10.35)$$

where

$$\Delta q(x, Q^2) \equiv \frac{\alpha_s}{2\pi} \log\left(\frac{Q^2}{\mu^2}\right) \int_x^1 \frac{dy}{y} q(y) P_{qq}\left(\frac{x}{y}\right). \quad (10.36)$$

The quark densities $q(x, Q^2)$ now depend on Q^2 . We interpret this as arising from a photon with larger Q^2 probing a wider range of p_T^2 within the proton.

We can picture this as follows. As Q^2 is increased to $Q^2 \sim Q_0^2$, say, the photon starts to “see” evidence for the point-like valence quarks within the proton; see Fig. 10.9a. If the quarks were noninteracting, no further structure would be resolved as Q^2 was increased and exact scaling [described by $q(x)$] would set in, and the parton model would be satisfactory. However, QCD predicts that on increasing the resolution ($Q^2 \gg Q_0^2$), we should “see” that each quark is itself surrounded by a cloud of partons. We have calculated one particular diagram, shown in Fig. 10.9b, but there are of course other diagrams with a greater number of partons. The number of resolved partons which share the proton’s momentum increases with Q^2 . There is an increased probability of finding a quark at small x , and a decreased chance of finding one at high x , because high-momentum quarks lose momentum by radiating gluons.

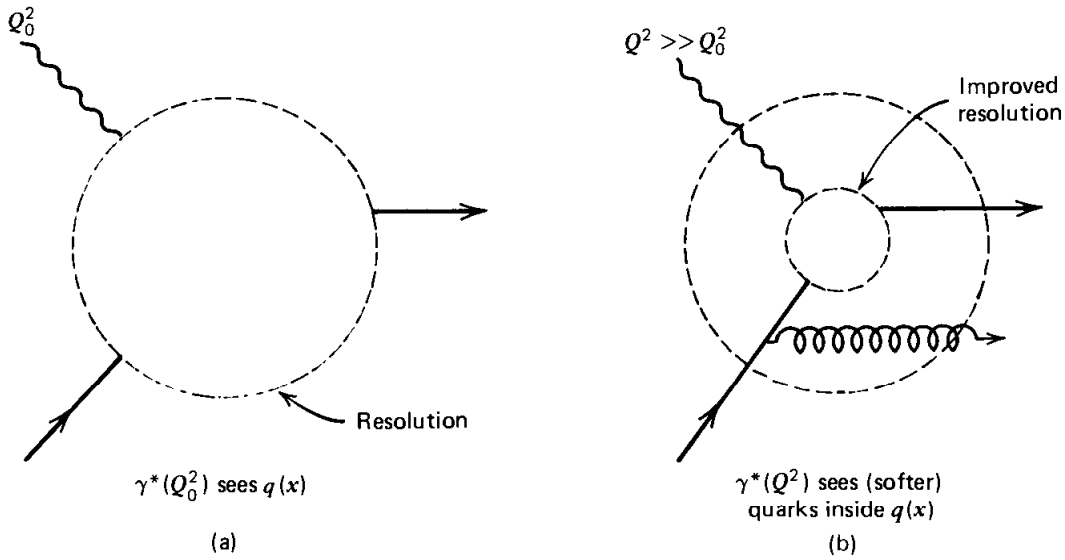


Fig. 10.9 The quark structure of the proton as seen by a virtual photon as Q^2 increases.

The Q^2 evolution of the quark densities is determined by QCD through (10.36). By considering the change in the quark density, $\Delta q(x, Q^2)$, when one probes a further interval of $\Delta \log Q^2$, (10.36) can be rewritten as an integro-differential equation for $q(x, Q^2)$:

$$\frac{d}{d \log Q^2} q(x, Q^2) = \frac{\alpha_s}{2\pi} \int_x^1 \frac{dy}{y} q(y, Q^2) P_{qq}\left(\frac{x}{y}\right). \quad (10.37)$$

This is an “Altarelli–Parisi evolution equation.” The equation mathematically expresses the fact that a quark with momentum fraction x [$q(x, Q^2)$ on the left-hand side] could have come from a parent quark with a larger momentum fraction y [$q(y, Q^2)$ on the right-hand side] which has radiated a gluon. The probability that this happens is proportional to $\alpha_s P_{qq}(x/y)$. The integral in (10.37) is the sum over all possible momentum fractions $y(> x)$ of the parent.

To summarize: QCD predicts the breakdown of scaling and allows us to compute explicitly the dependence of the structure function on Q^2 . Given the quark structure function at some reference point $q(x, Q_0^2)$, we can compute it for any value of Q^2 using the Altarelli–Parisi equation (10.37). The experimental results for $q(x, Q^2)$, or, to be precise, $F_2(x, Q^2)$, are displayed in Fig. 10.10. Moment analysis is often used to show that the Q^2 variation of the structure function is described by the differential equation (10.37). This procedure is purely technical and of no interest to us (see, however, Exercise 10.16). The systematics of the Q^2 dependence should be noted, however. Around $x = 0.25$ ($\omega = 4$), the structure function is found to scale, and Fig. 9.2 displays the absence of Q^2 dependence at this particular x value. But for $x \leq 0.25$, the structure function increases with Q^2 , while for $x \geq 0.25$, it decreases. Another way to state this result is that we resolve increasing numbers of “soft” quarks with increasing Q^2 .

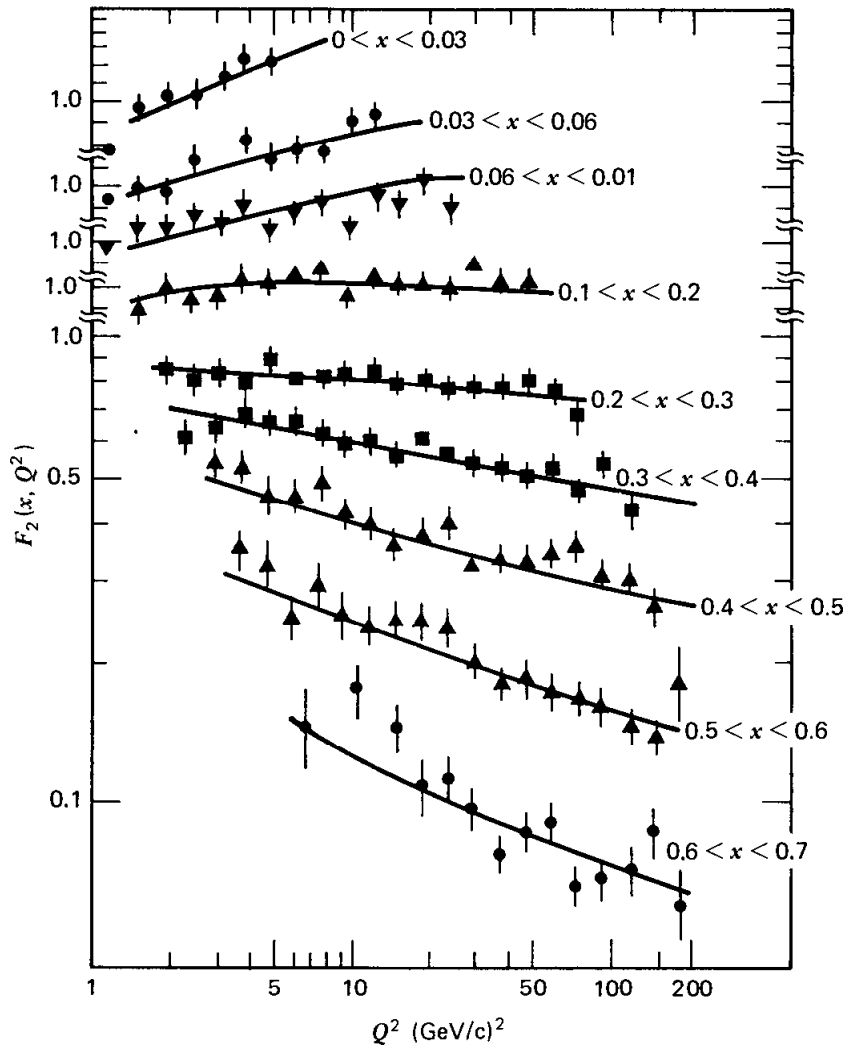


Fig. 10.10 Deviations from scaling. With increasing Q^2 , the structure function $F_2(x, Q^2)$ increases at small x and decreases at large x . The data are from the CDHS counter experiment at CERN.

The large-momentum quark component ($x \approx 1$) is depleted and shifted toward low momentum ($x \approx 0$). This agrees with our earlier discussion that large- Q^2 , high-resolution photons have more chance of seeing “softer” quarks whose momentum has been degraded by gluon emission.

EXERCISE 10.7 The origin of the scaling violation of $q(x, Q^2)$ given by (10.37) can be traced back to (10.32). There, we assumed that α_s is a constant. Show that (10.37) is also obtained for a running coupling constant. Assume that α_s in (10.32) is $\alpha_s(p_T^2)$ as given by (7.65).

Comment The reason for choosing p_T^2 as the argument of α_s cannot be exhibited without a discussion of the higher-order diagrams. However, note that we are working in a kinematic regime with two large quantities, p_T^2 and

Q^2 , and the dominant region is $p_T^2 \ll Q^2$. In this limit, a discussion of higher orders introduces p_T^2 as the argument of α_s [see, for example, Reya (1981) or Dokshitzer et al. (1980)].

10.6 Including Gluon Pair Production

So far, we have only incorporated contributions to deep inelastic scattering, $ep \rightarrow eX$, from the quark-initiated processes $\gamma^*q \rightarrow q$ and $\gamma^*q \rightarrow qg$. To order α_s , we should also include contributions where a gluon in the initial proton produces a quark–antiquark pair to which the virtual photon then couples, that is, the process $\gamma^*g \rightarrow q\bar{q}$ of Fig. 10.3. This is similar to the Compton diagrams of Section 10.4.

EXERCISE 10.8 Show that the color factor for $\gamma^*g \rightarrow q\bar{q}$ is $\frac{1}{2}$.

EXERCISE 10.9 Verify that for $\gamma^*g \rightarrow q\bar{q}$,

$$|\overline{\mathcal{M}}|^2 = 32\pi^2 (e_q^2 \alpha_s) \frac{1}{2} \left(\frac{\hat{u}}{\hat{t}} + \frac{\hat{t}}{\hat{u}} - \frac{2\hat{s}Q^2}{\hat{t}\hat{u}} \right), \quad (10.38)$$

using $\sum \epsilon_\mu^* \epsilon_\nu = -g_{\mu\nu}$ for the γ^* -polarization sum. Hence, show that (10.34) for the proton structure function contains the additional contribution

$$\left. \frac{F_2(x, Q^2)}{x} \right|_{\gamma^*g \rightarrow q\bar{q}} = \left| \begin{array}{c} \text{Diagram 1} \\ \text{Diagram 2} \end{array} \right|^2 \quad (10.39)$$

$$= \sum_q e_q^2 \int_x^1 \frac{dy}{y} g(y) \frac{\alpha_s}{2\pi} P_{qg}\left(\frac{x}{y}\right) \log \frac{Q^2}{\mu^2}, \quad (10.40)$$

where $g(y)$ is the gluon density in the proton and where

$$P_{qg}(z) = \frac{1}{2} (z^2 + (1-z)^2) \quad (10.41)$$

represents the probability that a gluon annihilates into a $q\bar{q}$ pair such that the quark has a fraction z of its momentum. Detailed measurements of the scaling violations of $F_2(x, Q^2)$ probe the gluon distribution inside the proton through (10.40).

10.7 Complete Evolution Equations for the Parton Densities

If we include the pair production contribution to the evolution of the quark density, (10.37) becomes

$$\frac{dq_i(x, Q^2)}{d \log Q^2} = \frac{\alpha_s}{2\pi} \int_x^1 \frac{dy}{y} \left(q_i(y, Q^2) P_{qq}\left(\frac{x}{y}\right) + g(y, Q^2) P_{qg}\left(\frac{x}{y}\right) \right) \quad (10.42)$$

for each quark flavor i . The second term, previously omitted, considers the possibility that a quark with momentum fraction x is the result of $q\bar{q}$ pair creation by a parent gluon with momentum fraction $y (> x)$. The probability is $\alpha_s P_{qg}(x/y)$. The framework is obviously still incomplete since we require an equation for the evolution of the gluon density in the proton. Repeating our previous arguments, we can give a symbolic representation of the gluon evolution equation

$$\frac{d}{d \log Q^2} (g(x, Q^2)) = \sum_i q_i(y, Q^2) P_{gq}\left(\frac{x}{y}\right) + g(y, Q^2) P_{gg}\left(\frac{x}{y}\right) \quad (10.43)$$

which tells us that

$$\frac{dg(x, Q^2)}{d \log Q^2} = \frac{\alpha_s}{2\pi} \int_x^1 \frac{dy}{y} \left(\sum_i q_i(y, Q^2) P_{gq}\left(\frac{x}{y}\right) + g(y, Q^2) P_{gg}\left(\frac{x}{y}\right) \right), \quad (10.44)$$

where the sum $i = 1, \dots, 2n_f$ runs over quarks and antiquarks of all flavors. P_{gq} does not depend on the index i if the quark masses can be neglected.

EXERCISE 10.10 How would you set about verifying that

$$P_{gq}(z) = \frac{4}{3} \frac{1 + (1 - z)^2}{z}, \quad (10.45)$$

$$P_{gg}(z) = 6 \left(\frac{1 - z}{z} + \frac{z}{1 - z} + z(1 - z) \right)? \quad (10.46)$$

EXERCISE 10.11 Express (10.42) in symbolic form.

EXERCISE 10.12 Obtain the evolution equations for the combinations

$$q_{NS} \equiv q_i - q_j \quad (10.47)$$

$$q_S \equiv \sum_i q_i. \quad (10.48)$$

The subscripts are conventional and, in fact, are used to indicate that the combinations refer to nonsinglet and singlet combinations of the quark flavor group.

10.8 Physical Interpretation of the P Functions

Let us take stock of where QCD has taken us. At large Q^2 , the structure functions for $ep \rightarrow eX$ are given by a “parton model-like” formula with parton densities which do not scale but evolve with Q^2 in a way which is calculable from QCD. Although we have arrived at these Q^2 -dependent parton densities by studying deep inelastic scattering, these densities should simply characterize the target proton and should not depend on the nature of the probe. In other words, the parton densities are universal in the sense that they can relate the structure functions found in different processes.

The evolution is governed by the P_{qq} , P_{gq} , P_{qg} , and P_{gg} functions. The physical meaning of the P functions becomes transparent if we go back to (10.35) and rewrite it in the form of (10.7):

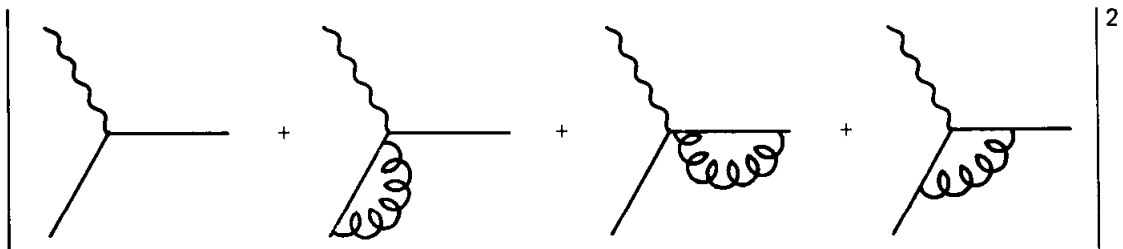
$$q(x, Q^2) + \Delta q(x, Q^2) = \int_0^1 dy \int_0^1 dz q(y, Q^2) \mathcal{P}_{qq}(z, Q^2) \delta(x - zy) \quad (10.49)$$

where

$$\mathcal{P}_{qq}(z, Q^2) \equiv \delta(1 - z) + \frac{\alpha_s}{2\pi} P_{qq}(z) \log \frac{Q^2}{\mu^2}. \quad (10.50)$$

It is natural to interpret $\mathcal{P}_{qq}(z)$ as the *probability density* of finding a quark inside a quark with fraction z of the parent quark momentum to first order in α_s . The $\delta(1 - z)$ term corresponds to there being no change in $q(x, Q^2)$. Clearly, this probability for a quark to remain unchanged will be reduced when the $O(\alpha_s)$ contributions are included.

At this point, it is crucial to include some virtual gluon diagrams which we have neglected until now. When these diagrams are included, the first term on the right-hand side of (10.34) is enlarged to



There is thus an $O(\alpha_s)$ contribution from the interference of the parton diagram with the (three) diagrams containing virtual gluons. These additional interference contributions are also singular at $z = 1$. It turns out that their singularity will exactly cancel the $z = 1$ singularity present in the incomplete $O(\alpha_s)$ calculation (10.50).

Rather than calculate these contributions explicitly, we can easily see what they must give. The additional contribution is entirely concentrated at $z = 1$ and is of the form $\delta(1 - z)$. It must be such that the total probability $P_{qq}(z)$ satisfies the constraint

$$\int_0^1 P_{qq}(z) dz = 0. \quad (10.51)$$

This constraint expresses the fact that the total number of quarks minus anti-quarks is conserved; the probability, $\mathcal{P}_{qq}$, of finding a quark in a quark integrated over all z must add up to 1. From (10.50), we see that if the integral of $\mathcal{P}_{qq}(z)$ is to be unity, we need the condition (10.51).

The virtual diagrams regularize the $1/(1 - z)$ singularity in $P_{qq}(z)$ of (10.31) so that (10.51) holds. This modification to $P_{qq}(z)$ can be conveniently expressed in terms of the so-called “+ prescription” for regularization in which $1/(1 - z)$ is replaced by $1/(1 - z)_+$ defined so that

$$\int_0^1 dz \frac{f(z)}{(1 - z)_+} = \int_0^1 dz \frac{f(z) - f(1)}{1 - z} \quad (10.52)$$

where $(1 - z)_+ = (1 - z)$ for $z < 1$ but is infinite at $z = 1$.

EXERCISE 10.13 Use (10.51) and (10.52) to show that

$$P_{qq}(z) = \frac{4}{3} \frac{1 + z^2}{(1 - z)_+} + 2\delta(1 - z). \quad (10.53)$$

EXERCISE 10.14 Use momentum conservation to justify

$$\int_0^1 dz z (q_S(z, Q^2) + g(z, Q^2)) = 1. \quad (10.54)$$

Hence, determine the $\delta(1 - z)$ term in P_{gg} and verify

$$P_{gg}(z) = 6 \left(\frac{1 - z}{z} + \frac{z}{(1 - z)_+} + z(1 - z) \right) + \left(\frac{11}{2} - \frac{n_f}{3} \right) \delta(1 - z), \quad (10.55)$$

where n_f is the number of quark flavors.

EXERCISE 10.15 Show that momentum conservation at the QCD vertex requires (for $z < 1$)

$$\begin{aligned} P_{qq}(z) &= P_{gq}(1 - z), \\ P_{qg}(z) &= P_{qg}(1 - z), \\ P_{gg}(z) &= P_{gg}(1 - z). \end{aligned} \tag{10.56}$$

Check that the explicit formulas for the “splitting” functions satisfy these relations.

EXERCISE 10.16 If $\alpha_s(Q^2) = c/\log Q^2$, show that (10.37) leads to

$$\frac{\int x^{n-1} q(x, Q^2) dx}{\int x^{n-1} q(x, Q_0^2) dx} = \left(\frac{\log Q^2}{\log Q_0^2} \right)^{A_n},$$

where

$$\begin{aligned} A_n &= \frac{c}{2\pi} \int_0^1 x^{n-1} P_{qq}(x) dx \\ &= \frac{c}{2\pi} \frac{4}{3} \left(-\frac{1}{2} + \frac{1}{n(n+1)} - 2 \sum_{j=2}^n \frac{1}{j} \right). \end{aligned}$$

That is, in QCD, the moments ($n \geq 1$) of the quark structure functions decrease as calculable powers of $\log Q^2$. c is given by (7.65).

The observant reader may have noticed what appears to be a contradiction in our interpretation of the P functions. For example, we regard the P_{qq} term as the correction factor to the quark density that arises from allowing for gluon emission. However, there are two diagrams: one with the gluon emitted from the initial quark line and the other with the gluon radiated from the final quark line; see Fig. 10.2. Our picture is only valid if the first diagram dominates. Then, the emitted gluon can be considered as part of the proton structure. It is a “parton-like” diagram. It turns out that both diagrams are required to ensure gauge invariance of the amplitude, but that the second only plays the role of canceling the contributions from the unphysical polarization states of the gluon. Adopting a physical gauge, in which we sum only over transverse gluons, only the first diagram remains.

10.9 The Altarelli–Parisi Techniques Also Apply to Leptons and Photons: The Weizsäcker–Williams Formula

We conclude this chapter by emphasizing the astonishing simplicity of the Altarelli–Parisi formalism. Let us return to (10.30), which computes the probability for producing a gluon with momentum fraction $1 - z$ and transverse momentum p_T in the process $\gamma^* q \rightarrow qg$. We should really have written it as a double-differential cross section

$$\frac{d\hat{\sigma}}{dz dp_T^2} = (e_i^2 \hat{\sigma}_0) \gamma_{qq}(z, p_T^2), \quad (10.57)$$

with

$$\gamma_{qq}(z, p_T^2) = \frac{\alpha_s}{2\pi} \frac{1}{p_T^2} P_{qq}(z). \quad (10.58)$$

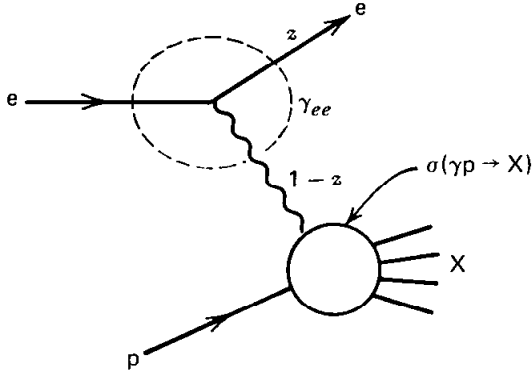
This cross section for $\gamma^* q \rightarrow qg$ can be pictured symbolically as

$$\frac{d\hat{\sigma}}{dz dp_T^2} = \begin{array}{c} \text{Diagram: A wavy line (photon) enters from the top left, hits a vertex. From this vertex, a solid line (quark) goes to the right, and another solid line (quark) goes down. The downward quark line emits a wavy line (gluon) to the right. The gluon line is labeled '1 - z'. The quark line after emission is labeled 'z'. Two dashed circles are shown: one around the top vertex labeled 'e_i^2 \hat{\sigma}_0' and one around the gluon emission vertex labeled '\gamma_{qq}(z, p_T^2)'. The entire diagram is equated to the left-hand side of equation (10.57'). \end{array} \quad (10.57')$$

(That z is indeed the fractional momentum of the quark after gluon radiation can be easily seen by adding its momentum zp_i to that of the photon and noting the mass-shell condition of the resulting outgoing quark, $(q + zp_i)^2 = 0$. This shows that z is given by (10.29), our previous definition.)

In (10.57') we see that the $O(\alpha_s)$ cross section factors into the $O(\alpha)$ parton model cross section $(e_i^2 \hat{\sigma}_0)$ and the probability γ_{qq} that the quark radiates a gluon with fraction $1 - z$ of its momentum and with transverse momentum p_T . How is this possible? Cross sections are calculated from probability *amplitudes*, where the different amplitudes for the process are added and then the square modulus of the sum is taken. What we have found is that, in the limit that the gluons have a p_T that is not too large, the calculation of (10.57') can be viewed as two sequential events, that is, provided we retain only the singular part, $1/p_T^2$, of the full p_T distribution as we did in deriving (10.30). The probabilities for the interaction $e_i^2 \hat{\sigma}_0$ and the gluon emission γ_{qq} can then be calculated separately and multiplied. That is, the probabilistic parton picture thus applies to gluon emission.

This technique is not special to quarks and gluons (QCD). In fact, it was known, since the work of Weizsäcker and Williams in 1934, that it applies equally well to leptons and photons (QED). Consider, for example, the process $ep \rightarrow eX$. The cross section may be written



$$\frac{d\sigma}{dz dp_T^2} =$$

$$= \gamma_{ee}(z, p_T^2) [\sigma(\gamma p \rightarrow X)]_{E_\gamma = (1-z)E} \quad (10.59)$$

with

$$\gamma_{ee}(z, p_T^2) = \frac{\alpha}{\pi} \frac{1}{p_T^2} P_{ee}(z), \quad (10.60)$$

where z and p_T are, respectively, the momentum fraction and transverse momentum of the outgoing electron, and

$$P_{ee}(z) = \frac{1+z^2}{1-z}; \quad (10.61)$$

see, for example, Chen and Zerwas (1975) *Phys. Rev. D* 12, 187. Equation (10.60) follows from (10.58) after taking account of the factor 2 mismatch in the definitions of α and α_s , namely, $\alpha_s \rightarrow 2\alpha$, see (10.19). $P_{ee}(z)$ is just $P_{qq}(z)$ as given by (10.31) but without the color factor. Equation (10.60) is known in QED as the equivalent photon distribution. Similarly, one can define the QED equivalent of P_{qg} given by (10.41):

$$P_{e\gamma}(z) = z^2 + (1-z)^2. \quad (10.62)$$

In the next chapter, we further illustrate the power of this technique by calculating the cross section for the process $e^+e^- \rightarrow q\bar{q}g$, which we view as $e^+e^- \rightarrow q\bar{q}$ followed by the emission of a gluon from the quark (or antiquark). $\gamma_{qq}(z, p_T^2)$ has been computed once and for all and can just be substituted into other diagrams.

11

e^+e^- Annihilation and QCD

In the previous chapter, we studied QCD in the framework of a truly historic type of experiment, namely, the deep inelastic scattering of leptons by hadrons. The large- Q^2 , short-wavelength, virtual photon, prepared by the inelastic scattering of the lepton (see Fig. 11.1a), probes the proton, revealing its constituents (Chapter 9) and their color interactions (Chapter 10). The resulting picture is easy to interpret, because the short-distance (small α_s) nature of the quark–gluon interactions allows us to confront the experimental results with quantitative perturbative calculations.

High-resolution photons can also be prepared by colliding high-energy electron and positron beams head-on (see Fig. 11.1b). The exceptional power of this experimental technique is illustrated by the gallery of diagrams in Fig. 11.2: e^+e^- colliders can be used to study QED, weak interactions, quarks, and gluons and also to study or search for heavy quarks and leptons. Moreover, e^+e^- annihilation is a “clean” process in the sense that leptons (rather than hadrons, which are complex structures made of partons) appear in the initial state. For these reasons, we choose e^+e^- processes as our main working example to illustrate how the ideas and techniques of Chapters 9 and 10 carry over to other experimental situations.

11.1 e^-e^+ Annihilation into Hadrons: $e^-e^+ \rightarrow q\bar{q}$

The bulk of hadrons produced in e^-e^+ annihilations are fragments of a quark and antiquark produced by the process $e^-e^+ \rightarrow q\bar{q}$. (We shall justify this statement by computing the higher-order process $e^-e^+ \rightarrow q\bar{q}g$ in the next section.) The cross section for the (QED) process $e^-e^+ \rightarrow q\bar{q}$ of Fig. 11.2c is readily obtained from that for the process of Fig. 11.2a,

$$\sigma(e^-e^+ \rightarrow \mu^-\mu^+) = \frac{4\pi\alpha^2}{3Q^2}, \quad (11.1)$$

a result we obtained in (6.33). Here, the center-of-mass energy squared is

$$s = Q^2 = 4E_b^2, \quad (11.2)$$

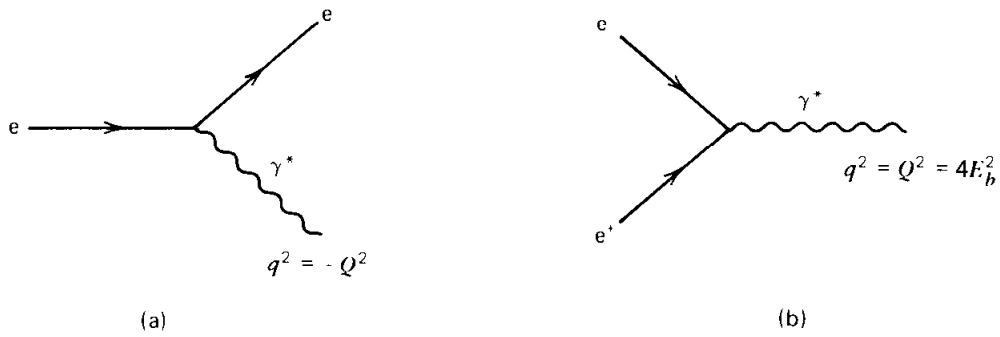


Fig. 11.1 Virtual photon probes prepared in (a) deep inelastic scattering, and (b) the head-on collision of electron and positron beams each of energy E_b .

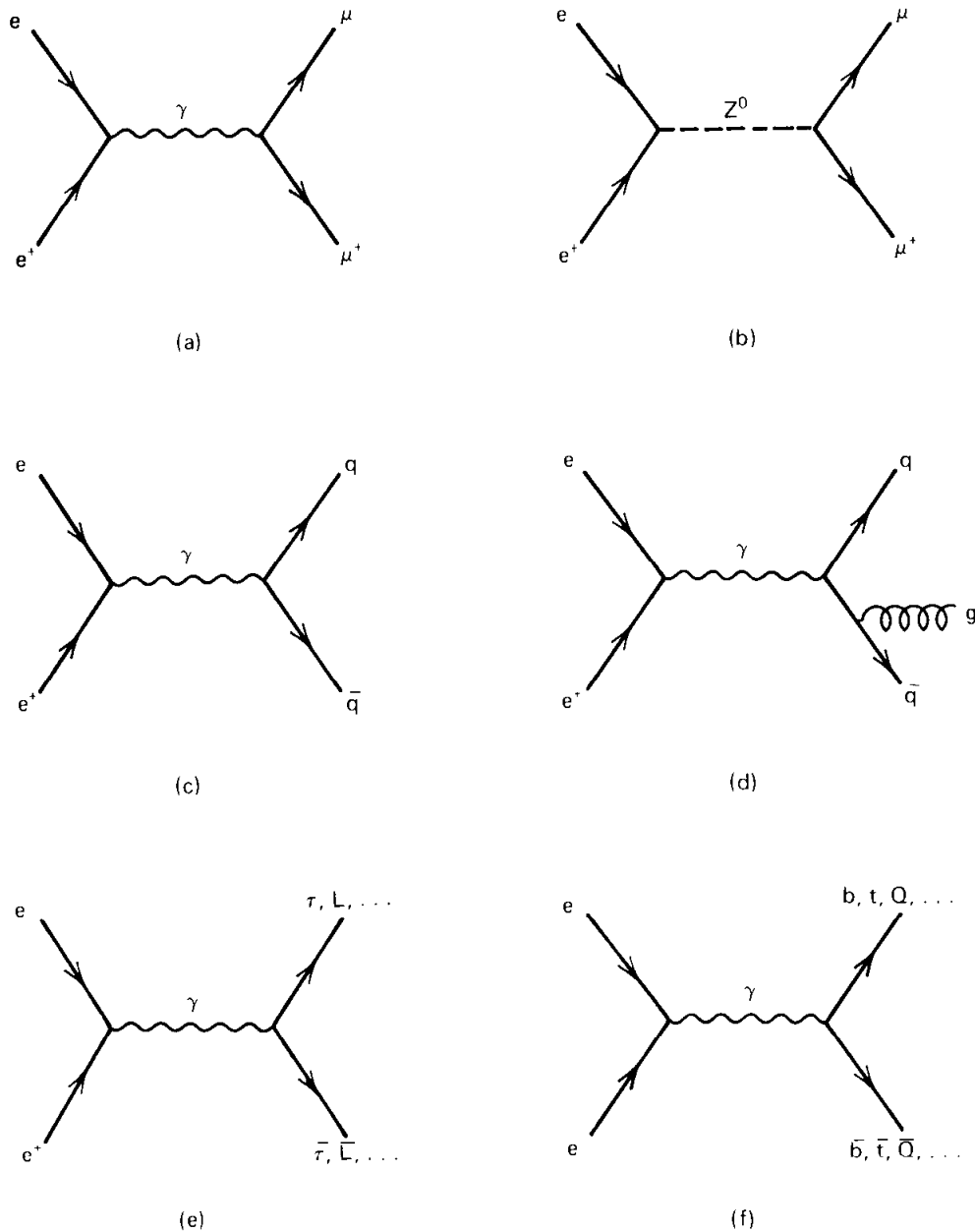


Fig. 11.2 Some experimental possibilities resulting from e^+e^- annihilation.

see Fig. 11.1b. The required cross section is

$$\sigma(e^-e^+ \rightarrow q\bar{q}) = 3e_q^2\sigma(e^-e^+ \rightarrow \mu^-\mu^+), \quad (11.3)$$

where we have taken account of the fractional charge of the quark, e_q . The extra factor of 3 arises because we have a diagram for each quark color and the cross sections have to be added. To obtain the cross section for producing all types of hadrons, we must sum over all quark flavors $q = u, d, s, \dots$, and therefore

$$\sigma(e^-e^+ \rightarrow \text{hadrons}) = \sum_q \sigma(e^-e^+ \rightarrow q\bar{q}) \quad (11.4)$$

$$= 3 \sum_q e_q^2 \sigma(e^-e^+ \rightarrow \mu^-\mu^+). \quad (11.5)$$

This simple calculation thus leads to the dramatic prediction

$$R \equiv \frac{\sigma(e^-e^+ \rightarrow \text{hadrons})}{\sigma(e^-e^+ \rightarrow \mu^-\mu^+)} = 3 \sum_q e_q^2. \quad (11.6)$$

As $\sigma(e^-e^+ \rightarrow \mu^-\mu^+)$ is well known (see Fig. 6.6), a measurement of the total e^-e^+ annihilation cross section into hadrons therefore directly counts the number of quarks, their flavors, as well as their colors. We have

$$\begin{aligned} R &= 3 \left[\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 \right] = 2 && \text{for } u, d, s, \\ &= 2 + 3\left(\frac{2}{3}\right)^2 = \frac{10}{3} && \text{for } u, d, s, c, \\ &= \frac{10}{3} + 3\left(\frac{1}{3}\right)^2 = \frac{11}{3} && \text{for } u, d, s, c, b. \end{aligned} \quad (11.7)$$

In Fig. 11.3, these predictions are compared to the measurements of R . The value $R \simeq 2$ is apparent below the threshold for producing charmed particles at $Q = 2(m_c + m_u) \simeq 3.7$ GeV. Above the threshold for all five quark flavors ($Q > 2m_b \simeq 10$ GeV), $R \simeq \frac{11}{3}$ as predicted. These measurements confirm that there are three colors of quark, since $R = \frac{11}{3}$ would be reduced by a factor 3 if there was only one color, see (11.3).

These results for R will be modified when interpreted in the context of QCD. Equation (11.4) is based on the (leading order) process $e^-e^+ \rightarrow q\bar{q}$. However, we should also include contributions from diagrams where the quark and/or anti-quark radiate gluons. To $O(\alpha_s)$, the result (11.6) is then modified to

$$R = 3 \sum_q e_q^2 \left(1 + \frac{\alpha_s(Q^2)}{\pi} \right).$$

That is, the scaling result, (11.6), that R is independent of Q^2 is violated logarithmically through the $\log Q^2$ behavior of α_s , see (7.65). At present, experiments are unable to detect this additional contribution to R , and our previous comparison with the data remains a good approximation. We postpone the

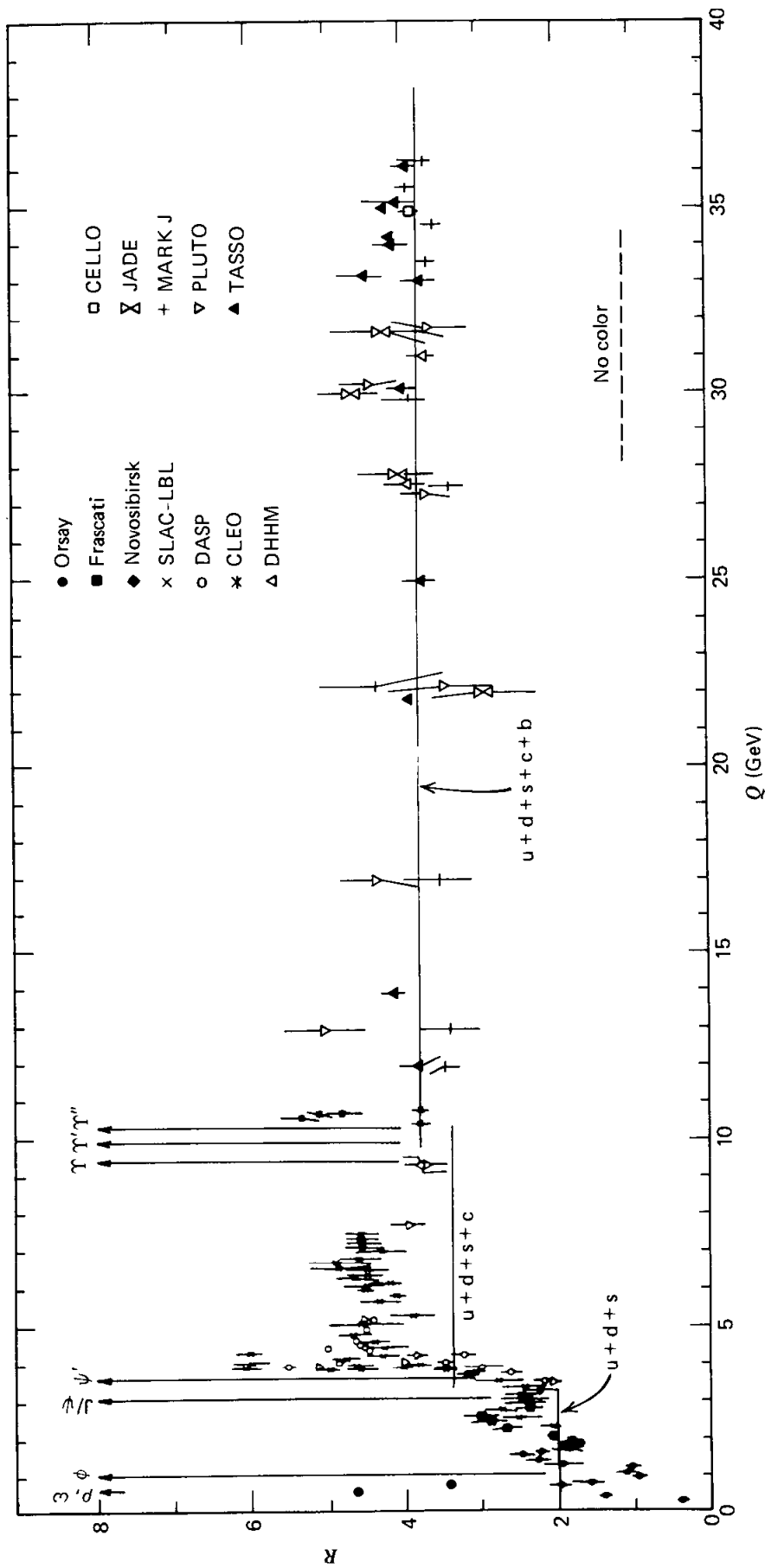


Fig. 11.3 Ratio R of (11.6) as a function of the total e^-e^+ center-of-mass energy. (The sharp peaks correspond to the production of narrow 1^- resonances just below or near the flavor thresholds.)

derivation of the α_s/π correction until Section 11.7, where it will emerge as a by-product of the study of the three-jet events $e^-e^+ \rightarrow q\bar{q}g$.

11.2 Fragmentation Functions and Their Scaling Properties

So far, we have not faced the problem of how the quarks turn into the hadrons that hit the detector. It was sufficient to state that the quarks must fragment into hadrons with unit probability. This gives (11.4). For more detailed calculations, this problem cannot be sidestepped.

In the center-of-mass frame, the produced quark and antiquark separate with equal and opposite momentum and materialize into two back-to-back jets of hadrons which have momenta roughly collinear with the original q and $\bar{q}$ directions. The hadrons may be misaligned by a momentum transverse to the q or $\bar{q}$ direction by an amount not exceeding about 300 MeV.

In Chapter 1, we visualized jet formation as hadron bremsstrahlung once the q and $\bar{q}$ separate by a distance of around 1 fm. Then, α_s becomes large, and strong color forces pull on the separating q and $\bar{q}$. The potential energy becomes so large that one or more $q\bar{q}$ pairs are created (see Fig. 1.14). Eventually, all the energy is degraded into two jets of hadrons moving more or less in the direction of the q and $\bar{q}$.

To describe the fragmentation of quarks into hadrons, we use an analogous formalism to that introduced in Chapter 9 to describe the quarks inside hadrons. Figure 11.4 shows the observation of a hadron h , whose energy is measured to be E_h . The corresponding differential cross section can be written as

$$\frac{d\sigma}{dz}(e^-e^+ \rightarrow hX) = \sum_q \sigma(e^-e^+ \rightarrow q\bar{q}) [D_q^h(z) + D_{\bar{q}}^h(z)]. \quad (11.8)$$

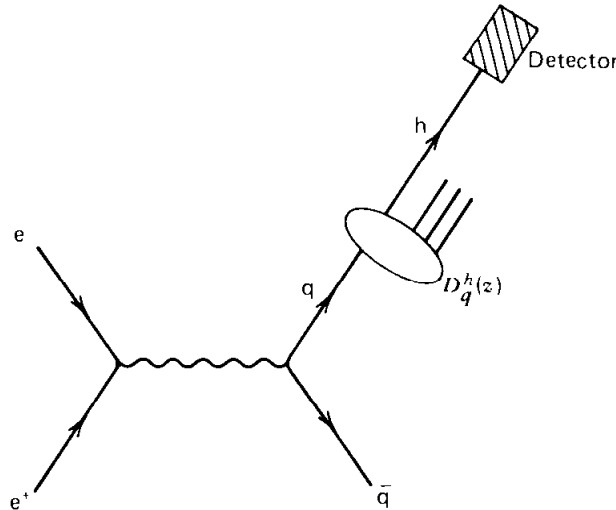


Fig. 11.4 A hadron h observed with a fraction z of the quark's energy; $z = 2E_h/Q$.

It describes Fig. 11.4 as two sequential events: the production of a $q\bar{q}$ pair, followed by the fragmentation of either the q or the $\bar{q}$ producing the detected hadron h . The D functions therefore represent the probability that the hadron h is found in the debris of a quark (or antiquark) carrying a fraction z of its energy. That is,

$$z \equiv \frac{E_h}{E_q} = \frac{E_h}{E_b} = \frac{2E_h}{Q}. \quad (11.9)$$

The summation in (11.8) is over all quark flavors and recognizes the fact that the detector is unaware of the quantum numbers of the parent of the observed hadron.

The fragmentation function, $D(z)$, describes the transition (parton $\rightarrow$ hadron) in the same way that the structure function, $f(x)$, of Chapter 9 describes the embedding (hadron $\rightarrow$ parton). Like the f functions, the D functions are subject to constraints imposed by momentum and probability conservation:

$$\sum_h \int_0^1 z \dot{D}_q^h(z) dz = 1 \quad (11.10)$$

$$\sum_q \int_{z_{\min}}^1 [D_q^h(z) + D_{\bar{q}}^h(z)] dz = n_h, \quad (11.11)$$

where $z_{\min}$ is the threshold energy $2m_h/Q$ for producing a hadron of mass m_h , and n_h is the average multiplicity of hadrons of type h . Equation (11.10) simply states that the sum of the energies of all hadrons is the energy of the parent quark. Clearly, the same relation holds for $D_{\bar{q}}^h(z)$. Equation (11.11) says that the number n_h of hadrons of type h is given by the sum of probabilities of obtaining h from all possible parents, namely, from q or $\bar{q}$ of any flavor.

EXERCISE 11.1 Fragmentation functions are often parametrized by the form

$$D_q^h(z) = N \frac{(1-z)^n}{z}$$

where n and N are constants. Show that

$$N = (n+1)\langle z \rangle,$$

where $\langle z \rangle$ is the average fraction of the quark energy carried by hadrons of type h after fragmentation. Further, show that

$$n_h \sim \log \left(\frac{Q}{2m_h} \right)$$

for the two-jet process of Fig. 11.4. That is, the multiplicity of hadrons h grows logarithmically with the annihilation energy.

Taking the ratio of (11.8) and (11.4), and using (11.3), we find

$$\frac{1}{\sigma} \frac{d\sigma}{dz}(e^-e^+ \rightarrow hX) = \frac{\sum_q e_q^2 [D_q^h(z) + D_{\bar{q}}^h(z)]}{\sum_q e_q^2} \quad (11.12)$$

$$= \mathcal{F}(z) \quad (11.13)$$

That is, the inclusive cross section $d\sigma/dz$ divided by the total annihilation cross section into hadrons, σ , is predicted to scale. The cross sections σ and $d\sigma/dz$ depend on the annihilation energy Q , but (11.12) predicts that the ratio is

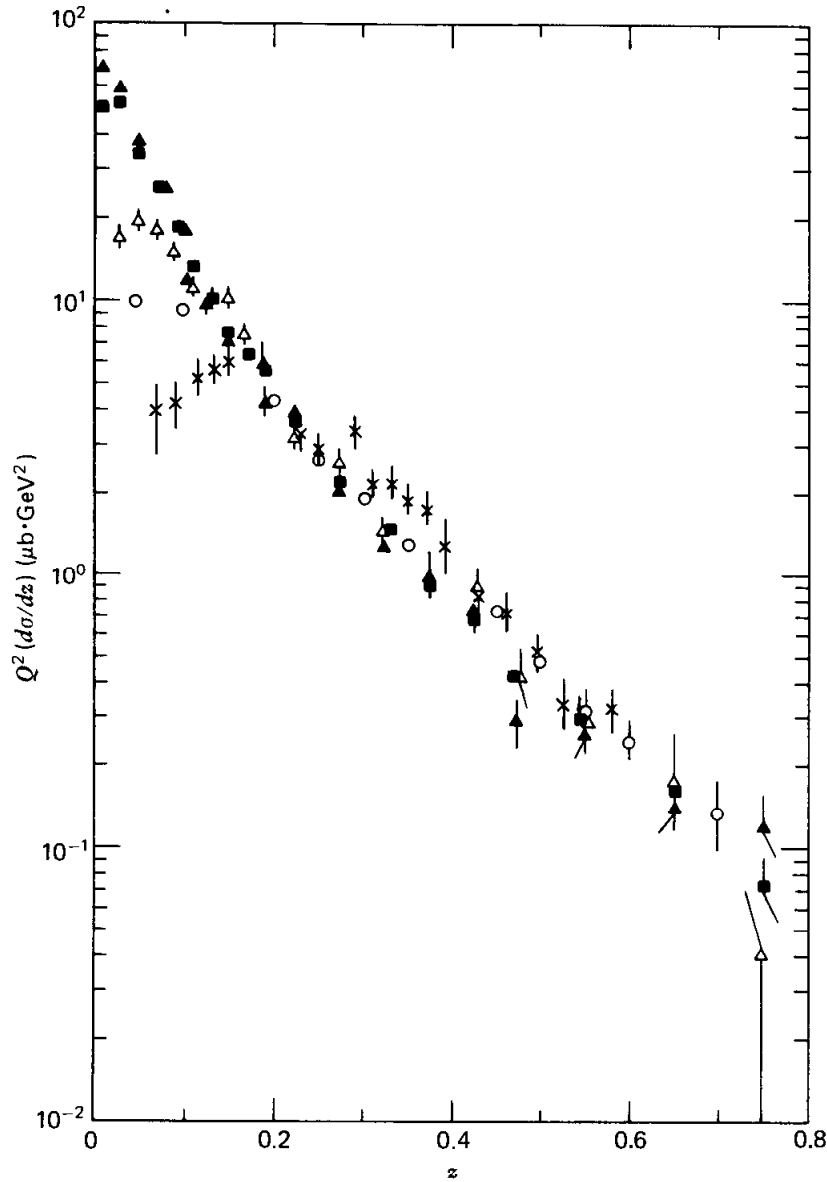


Fig. 11.5 $Q^2(d\sigma/dz) \sim 1/\sigma(d\sigma/dz)$ for $e^-e^+ \rightarrow hX$ as measured as a function z at different center-of-mass energies Q : ($\times$) 5 GeV; ($\circ$) 7.4 GeV; (Δ) 12 GeV; ($\blacksquare$) 27.4–31.6 GeV; ($\blacktriangle$) 35.0–36.6 GeV. Data are from the Stanford Linear Accelerator and PETRA.

independent of Q . Such a scaling result is not a complete surprise, because we have relied on the scaling parton model to derive (11.12), see Fig. 11.4.

Figure 11.5 shows $(1/\sigma)(d\sigma/dz)$ as a function of z for different values of Q^2 . The scaling is not perfect. Gluon emission from the q or $\bar{q}$ will introduce $\log Q^2$ scaling violations in (11.13). Their qualitative trend is the same as in electroproduction, that is, $\mathcal{F}(z, Q^2)$ will increase at small z with increasing values of Q^2 but decrease for z near 1. The large violations of scaling for $z \lesssim 0.2$, seen in Fig. 11.5, are not exclusively due to gluon emission, however, and are the subject of the next section.

EXERCISE 11.2 The fragmentation functions $D(z)$ describe properties of partons and are therefore the same, no matter how the partons are produced. Consider the inclusive leptonproduction cross section $\sigma(ep \rightarrow hX)$ and show that

$$\frac{1}{\sigma} \frac{d\sigma}{dz}(ep \rightarrow hX) = \frac{\sum_q e_q^2 f_q(x) D_q^h(z)}{\sum_q e_q^2 f_q(x)},$$

where $f_q(x)$ are the proton structure functions of Chapter 9, see Fig. 11.6. The sum runs over the quarks and antiquarks that can be a parent of h .

EXERCISE 11.3 Using charge conjugation and isospin invariance, show that

$$D_u^{\pi^+} = D_{\bar{u}}^{\pi^-} = D_d^{\pi^-} = D_{\bar{d}}^{\pi^+},$$

$$D_u^{\pi^-} = D_{\bar{u}}^{\pi^+} = D_d^{\pi^+} = D_{\bar{d}}^{\pi^-},$$

$$D_s^{\pi^+} = D_s^{\pi^-}.$$

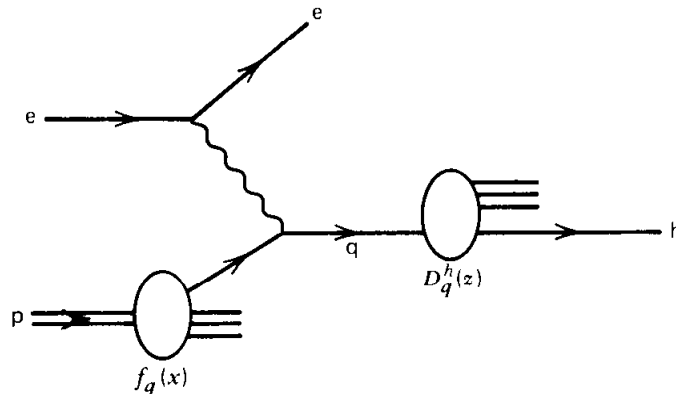


Fig. 11.6 Deep inelastic leptonproduction of hadron h ; $ep \rightarrow hX$.

EXERCISE 11.4 Using the notation

$$N_p^\pi(z) \equiv \frac{1}{\sigma} \frac{d\sigma}{dz} (ep \rightarrow \pi X),$$

show that, in the valence quark approximation for p, n,

$$\frac{\int dz [N_n^{\pi^+} - N_n^{\pi^-}]}{\int dz [N_p^{\pi^+} - N_p^{\pi^-}]} = \frac{2}{7}.$$

11.3 A Comment on Heavy Quark Production

Although the similarities between e^-e^+ annihilation and leptonproduction are becoming more and more evident, we must not forget one major difference. In leptonproduction, u, d, and s quarks play a dominant role because they are plentiful inside the nucleon target. Charm quarks occur in roughly one in ten events, and were indeed ignored in the phenomenological discussion of Chapter 9. In e^-e^+ annihilation, the situation is quite different. Beyond threshold ($Q^2 > 4m_c^2$), the cross section rises steeply and readily attains a sizable fraction of its asymptotic value, see Fig. 11.3. It is clear from (11.3) that c and u quarks are produced with the same cross section, since both have $e_q = +\frac{2}{3}$. However, close to threshold, the final-state hadron structure of these relatively frequent charm-quark events is very different from the two-jet structure of a typical event involving light quarks. Then, the c and $\bar{c}$ are produced almost at rest, and subsequently decay weakly into a rather large number of soft hadrons with low z. Therefore, an increase of low-z events is associated with the crossing of the charm threshold. This leads to a violation of scaling which will confuse, indeed simulate, the violations resulting from gluon emission.

The drastic violations of scaling, seen for $z \lesssim 0.2$ in Fig. 11.5, are associated with the production of c and b quarks, resulting in events with a large multiplicity and low z values when the c and b thresholds are crossed. This mechanism has a positive aspect. The characteristic features of heavy quark events can serve as an experimental signature in the search for yet heavier quarks, such as the t quark in Table 1.5, as the energy of e^-e^+ colliders is increased. A “step” in R also signals a new quark, for example, charm in Fig. 11.3. But for $e_q = -\frac{1}{3}$, this step is four times smaller and not easy to observe experimentally. Looking for deviations from the two-jet structure, characteristic of light quarks, is often a more sensitive test.

11.4 Three-Jet Events: $e^-e^+ \rightarrow q\bar{q}g$

From the viewpoint of perturbative QCD, we have only considered the leading $O(\alpha^2)$ contribution to $\sigma(e^-e^+ \rightarrow \text{hadrons})$. In order $\alpha^2\alpha_s$, the q or $\bar{q}$ can emit a

gluon, see Fig. 11.2d; and these $e^-e^+ \rightarrow q\bar{q}g$ events have three fragmenting jets in the final state. The additional jet has a gluon as its parent parton. For larger values of the annihilation energy Q , such that $\alpha_s(Q^2) \simeq 0.1-0.2$, we expect that three-jet events will account for roughly 10% of the final states. Our first task is to introduce kinematical variables to describe such events.

The momentum vectors of the q , $\bar{q}$, and g , which are produced by a virtual photon (γ^*) at rest, are displayed in Fig. 11.7. As in (11.9), we work with the energies, and with the longitudinal and transverse momenta of the partons, scaled to the e^- (and e^+) beam energy. That is, we introduce

$$x_q \equiv \frac{2E_q}{Q}, \quad x_{\bar{q}} \equiv \frac{2E_{\bar{q}}}{Q}, \quad x_g \equiv \frac{2E_g}{Q} \quad (11.14)$$

and

$$x_T \equiv \frac{2p_T}{Q}. \quad (11.15)$$

Like z in (11.9), all these ratios are bounded by 0 and 1. The four-momentum fractions in Fig. 11.7 are

$$\begin{aligned} (x_q; 0, 0, -x_q) & \quad \text{for } q, \\ (x_{\bar{q}}; x_T, 0, x_L) & \quad \text{for } \bar{q}, \\ (x_g; -x_T, 0, x_q - x_L) & \quad \text{for } g. \end{aligned} \quad (11.16)$$

The variables are defined relative to the most energetic jet, for example, the q jet in Fig. 11.7. Its direction is referred to as the “thrust axis.” The q , $\bar{q}$, and g are coplanar in the plane $y = 0$. Longitudinal and transverse momentum conservation are embodied in (11.16), but energy conservation introduces the additional requirement that

$$x_q + x_{\bar{q}} + x_g = 2. \quad (11.17)$$

The zero mass of the $\bar{q}$ and g leads to the further constraints [see (11.16)]

$$\begin{aligned} x_{\bar{q}}^2 - x_T^2 - x_L^2 &= 0, \\ x_g^2 - x_T^2 - (x_L - x_q)^2 &= 0. \end{aligned} \quad (11.18)$$

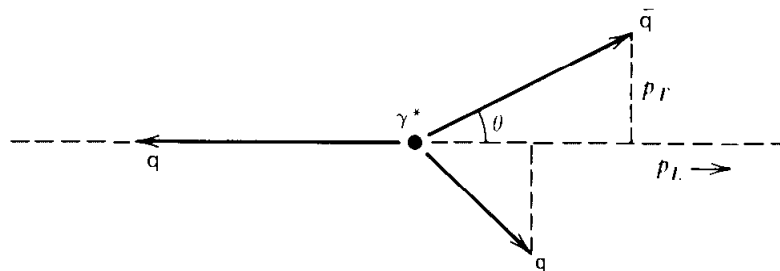


Fig. 11.7 The process $e^-e^+ \rightarrow \gamma^* \rightarrow q\bar{q}g$ in the center-of-mass frame.

From (11.18) and (11.17), it follows that

$$x_T^2 = \frac{4}{x_q^2} (1 - x_q)(1 - x_{\bar{q}})(1 - x_g). \quad (11.19)$$

EXERCISE 11.5 Derive (11.19). Also show that the angle θ between the q and $\bar{q}$ directions in Fig. 11.7 is determined by the relation

$$x_{\bar{q}} = \frac{2(1 - x_q)}{2 - x_q - x_q \cos \theta}. \quad (11.20)$$

Let us now compute the cross section corresponding to Fig. 11.7. For this particular graph, the $\bar{q}$ emits a softer gluon, so that

$$x_q \geq x_{\bar{q}} \geq x_g. \quad (11.21)$$

The most obvious experimental signature of gluon emission is that the q and $\bar{q}$ are no longer produced back to back. The $\bar{q}$ is produced with a transverse momentum fraction x_T relative to the direction of the quark. The relevant observable quantity is therefore $d\sigma/dx_T^2$. This cross section can readily be obtained using the Altarelli-Parisi-Weizsäcker-Williams technique of Section 10.9. Referring to Fig. 11.8, we obtain

$$\frac{d\sigma}{dx_{\bar{q}} dp_T^2} = \sigma(e^-e^+ \rightarrow q\bar{q}) \gamma_{\bar{q}\bar{q}}(x_{\bar{q}}, p_T^2), \quad (11.22)$$

see (10.57), where σ gives the probability for producing a $q\bar{q}$ pair and $\gamma_{\bar{q}\bar{q}}$ is the probability that the $\bar{q}$ subsequently emits a gluon with a fraction $(1 - x_{\bar{q}})$ of its momentum and a transverse momentum $|p_T|$. From (11.3) and (10.58), we have

$$\begin{aligned} \sigma(e^-e^+ \rightarrow q\bar{q}) &= \frac{4\pi\alpha^2}{Q^2} e_q^2, \\ \gamma_{\bar{q}\bar{q}}(x_{\bar{q}}, p_T^2) &= \gamma_{qq}(x_{\bar{q}}, p_T^2) = \frac{\alpha_s}{2\pi} \frac{1}{p_T^2} P_{qq}(x_{\bar{q}}). \end{aligned} \quad (11.23)$$

On substitution of (11.23) into (11.22), we find

$$\frac{1}{\sigma} \frac{d\sigma}{dx_{\bar{q}} dx_T^2} = \frac{\alpha_s}{2\pi} \frac{1}{x_T^2} P_{qq}(x_{\bar{q}}). \quad (11.24)$$

To calculate $d\sigma/dx_T^2$, it remains to integrate over all possible $\bar{q}$ energy fractions $x_{\bar{q}}$. Using (10.31) for P_{qq} , we obtain

$$\frac{1}{\sigma} \frac{d\sigma}{dx_T^2} = 2 \frac{\alpha_s}{2\pi} \frac{1}{x_T^2} \int_{(x_{\bar{q}})_{\min}}^{(x_{\bar{q}})_{\max}} dx \frac{4}{3} \left(\frac{1+x^2}{1-x} \right). \quad (11.25)$$

An extra factor of 2 is included to allow for the equally probable diagram with $q \leftrightarrow \bar{q}$.

The integrand in (11.25) diverges when $x_{\bar{q}} \rightarrow 1$. The kinematic situation where $x_{\bar{q}}$ reaches its maximum value is therefore of special interest. From (11.21), we see

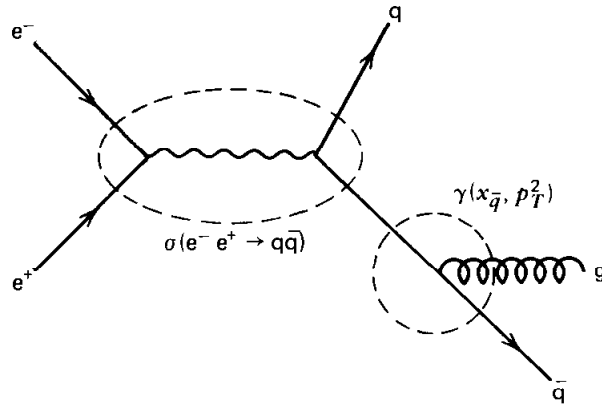


Fig. 11.8 The calculation of $e^-e^+ \rightarrow q\bar{q}g$ using the probabilistic techniques of §10.9.

that the largest value allowed for $x_{\bar{q}}$ is

$$x_{\bar{q}} = x_q. \quad (11.26)$$

This value can be approached if we make the emitted gluon as soft as possible, which, remembering that x_T is fixed, occurs when

$$x_g = x_T. \quad (11.27)$$

This kinematic configuration is shown in Fig. 11.9. Thus, from (11.17), we have

$$(x_q)_{\min} = (x_{\bar{q}})_{\max} = 1 - \frac{x_T}{2}. \quad (11.28)$$

Momentum is conserved for θ or x_T not too large, but x_g is only exactly equal to x_T when $\theta \rightarrow 0$. Such approximations are implicit in the Altarelli–Parisi calculation of (11.25), see Chapter 10. Using (11.28), (11.25) becomes

$$\frac{1}{\sigma} \frac{d\sigma}{dx_T^2} \simeq \frac{8\alpha_s}{3\pi} \frac{1}{x_T^2} \int_{(x_{\bar{q}})_{\min}}^{1 - \frac{1}{2}x_T} \frac{dx}{1-x}, \quad (11.29)$$

where we have approximated $1 + x^2$ by 2. Finally, omitting all but the leading logarithmic term, we obtain

$$\boxed{\frac{1}{\sigma} \frac{d\sigma}{dx_T^2} \simeq \frac{4\alpha_s}{3\pi} \frac{1}{x_T^2} \log\left(\frac{1}{x_T^2}\right)}. \quad (11.30)$$

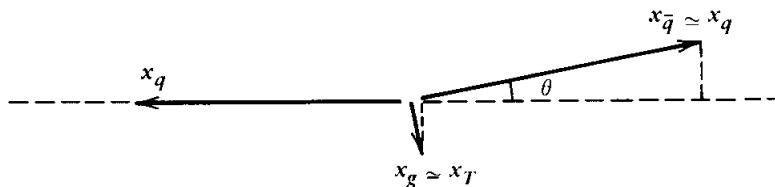


Fig. 11.9 The kinematic configuration giving the maximum value of $x_{\bar{q}}$ for a given x_T .

11.5 An Alternative Derivation of the $e^-e^+ \rightarrow q\bar{q}g$ Cross Section

The cross section (11.30) for $e^-e^+ \rightarrow \gamma^* \rightarrow q\bar{q}g$ can also be obtained using the Feynman rules of Chapter 6. We have calculated similar processes before. For instance, we found that $\gamma^*q \rightarrow qg$ is given by (10.17) and that $\gamma^*g \rightarrow q\bar{q}$ is given by (10.38). Proceeding in the same way, we find that the square of the amplitude for the sum of the $\gamma^* \rightarrow q\bar{q}g$ diagrams of Fig. 11.10 is

$$|\mathfrak{M}|^2 = N \left(\frac{t}{s} + \frac{s}{t} + \frac{2uQ^2}{st} \right) \quad (11.31)$$

where N represents the normalization factors and coupling constants and where

$$\begin{aligned} s &= (p_\gamma - p_q)^2, \\ t &= (p_\gamma - p_{\bar{q}})^2, \\ u &= (p_\gamma - p_g)^2, \end{aligned} \quad (11.32)$$

where $Q^2 \equiv p_\gamma^2$. Strictly speaking, for a parton process we should have denoted the Mandelstam variables by $\hat{s}, \hat{t}, \hat{u}$.

In order to rewrite (11.31) in terms of the energy fraction variables x_i of (11.14), we use

$$\begin{aligned} s &= Q^2(1 - x_q), \\ t &= Q^2(1 - x_{\bar{q}}), \\ u &= Q^2(1 - x_g). \end{aligned} \quad (11.33)$$

These relations follow from four-momentum conservation, which gives

$$p_\gamma^2 = (p_q + p_{\bar{q}} + p_g)^2 = 2p_q \cdot p_{\bar{q}} + 2p_q \cdot p_g + 2p_{\bar{q}} \cdot p_g. \quad (11.34)$$

For instance,

$$\begin{aligned} s &= (p_{\bar{q}} + p_g)^2 = 2p_{\bar{q}} \cdot p_g = p_\gamma^2 - 2p_q \cdot (p_{\bar{q}} + p_g) \\ &= p_\gamma^2 - 2p_q \cdot p_\gamma = Q^2 \left(1 - \frac{2E_q}{Q} \right) = Q^2(1 - x_q). \end{aligned}$$

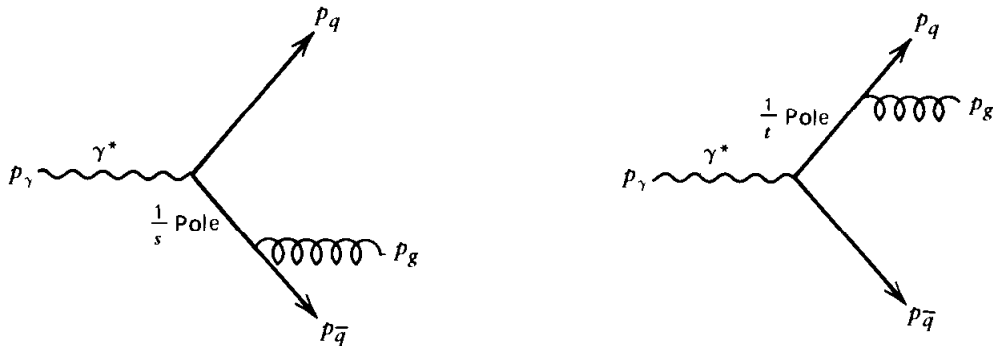


Fig. 11.10 The diagrams for $\gamma^* \rightarrow q\bar{q}g$ showing the particle four-momenta.

Substituting (11.33) into (11.31), we obtain for $\gamma^* \rightarrow q\bar{q}g$

$$|\mathcal{M}|^2 = N \frac{x_q^2 + x_{\bar{q}}^2}{(1 - x_q)(1 - x_{\bar{q}})}, \quad (11.35)$$

where we have used (11.17) to eliminate x_g . If we “attach” the e^-e^+ pair, then, up to a factor (which changes $N \rightarrow N'$), (11.35) gives the cross section for $e^-e^+ \rightarrow q\bar{q}g$

$$\boxed{\frac{d\sigma}{dx_q dx_{\bar{q}}} = N' \frac{x_q^2 + x_{\bar{q}}^2}{(1 - x_q)(1 - x_{\bar{q}})}}. \quad (11.36)$$

To verify that (11.36) is equivalent to our previous result (11.24), we must change the variable x_q to x_T^2 . We have

$$\frac{d\sigma}{dx_{\bar{q}} dx_T^2} = \frac{d\sigma}{dx_{\bar{q}} dx_q} \frac{dx_q}{dx_T^2}. \quad (11.37)$$

It is sufficient to use the small p_T approximation inherent in the Altarelli–Parisi result (11.24). Using (11.19), we find

$$\left| \frac{dx_T^2}{dx_q} \right| \simeq 4x_{\bar{q}}(1 - x_{\bar{q}}) \quad \text{for } x_q \simeq 1. \quad (11.38)$$

Thus, (11.36) may be written

$$\frac{d\sigma}{dx_{\bar{q}} dx_T^2} \simeq N' \left(\frac{1 + x_{\bar{q}}^2}{1 - x_{\bar{q}}} \right) \left[\frac{1}{4(1 - x_q)(1 - x_{\bar{q}})x_{\bar{q}}} \right], \quad (11.39)$$

where here again we have assumed $x_q \simeq 1$. In this limit, $x_{\bar{q}} \simeq (1 - x_g)$, and so the factor in square brackets is just x_T^{-2} , as can be seen from (11.19). Thus, (11.39) becomes

$$\frac{d\sigma}{dx_{\bar{q}} dx_T^2} \simeq N' \frac{3}{4} P_{qq}(x_{\bar{q}}) \frac{1}{x_T^2}, \quad (11.40)$$

where P_{qq} is given by (10.31). This is the same as the Altarelli–Parisi result of (11.24), and, indeed, we can identify the normalization coefficient to be

$$N' = \frac{2\alpha_s}{3\pi} \sigma.$$

The exact $O(\alpha_s)$ result is therefore

$$\boxed{\frac{1}{\sigma} \frac{d\sigma}{dx_q dx_{\bar{q}}} = \frac{2\alpha_s}{3\pi} \frac{x_q^2 + x_{\bar{q}}^2}{(1 - x_q)(1 - x_{\bar{q}})}}. \quad (11.41)$$

whereas (11.24) is the leading logarithmic approximation.

11.6 A Discussion of Three-Jet Events

The $e^-e^+ \rightarrow q\bar{q}g$ events led to three hadronic jets in the final state. The distribution (11.30) can be written as

$$\frac{1}{\sigma} \frac{d\sigma}{dp_T^2} \sim \alpha_s \frac{1}{p_T^2} \log \left(\frac{Q^2}{4p_T^2} \right), \quad (11.42)$$

where p_T is the transverse momentum between the q and $\bar{q}$ as a result of the emission of the gluon, recall Fig. 11.7. Only when the $\bar{q}$ (or q) recoils against g can its p_T relative to the q (or $\bar{q}$) be nonzero. For two-jet events, $e^-e^+ \rightarrow q\bar{q}$, we have $p_T = 0$. Now (11.42) shows that, for a fixed p_T , the cross section ratio increases with increasing Q^2 . That is, the number of $\bar{q}$ jets with a transverse momentum p_T relative to the q jet increases with Q^2 . This is a result of the increased probability of emitting a gluon with a given p_T value when the annihilation energy increases. The physics is identical to that in electroproduction. There, also, the cross section

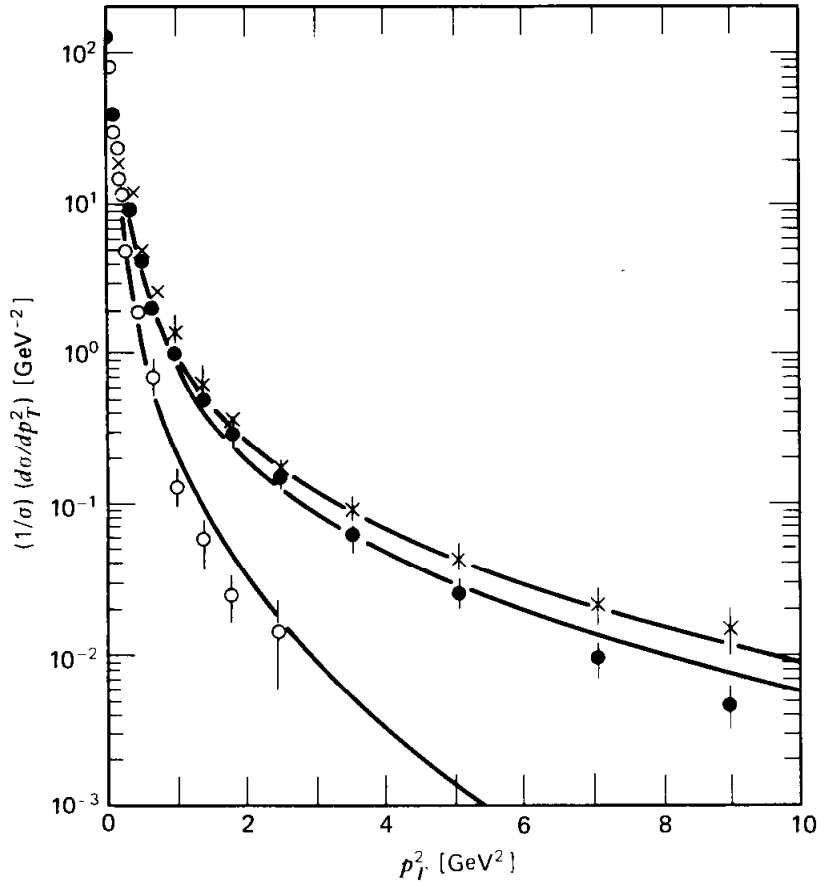


Fig. 11.11 The transverse momentum distribution $d\sigma/dp_T^2$ of hadrons relative to the thrust axis for different e^+e^- center-of-mass energies Q : (○) $Q = 12$ GeV; (●) $27.4 \leq Q \leq 31.6$ GeV; (×) $35.0 \leq Q \leq 36.6$ GeV. The curves are a QCD calculation. Data are from PETRA.

for producing jets with transverse momentum p_T relative to the γ^* -direction increased as $\log Q^2/p_T^2$ [compare (10.30) and (11.24)].

The hadron fragments of the $\bar{q}$ jet will also have large p_T relative to the quark direction since the p_T distribution of these hadrons should follow the trend of the p_T distribution of jets. The two distributions can be explicitly related using the D functions introduced in Section 11.2. The resulting p_T and Q^2 dependence of hadrons relative to the thrust axis (whichever parton it refers to) is shown in Fig. 11.11. In some events, all three jets will be well separated despite the $k_T \approx 300$ MeV of the daughter hadrons relative to their parent jet. One such event is shown in Fig. 11.12.

The assumption that α_s is constant in (11.42) requires explanation. If we had taken $\alpha_s(Q^2) \sim 1/\log Q^2$, then the above discussion would be meaningless. The

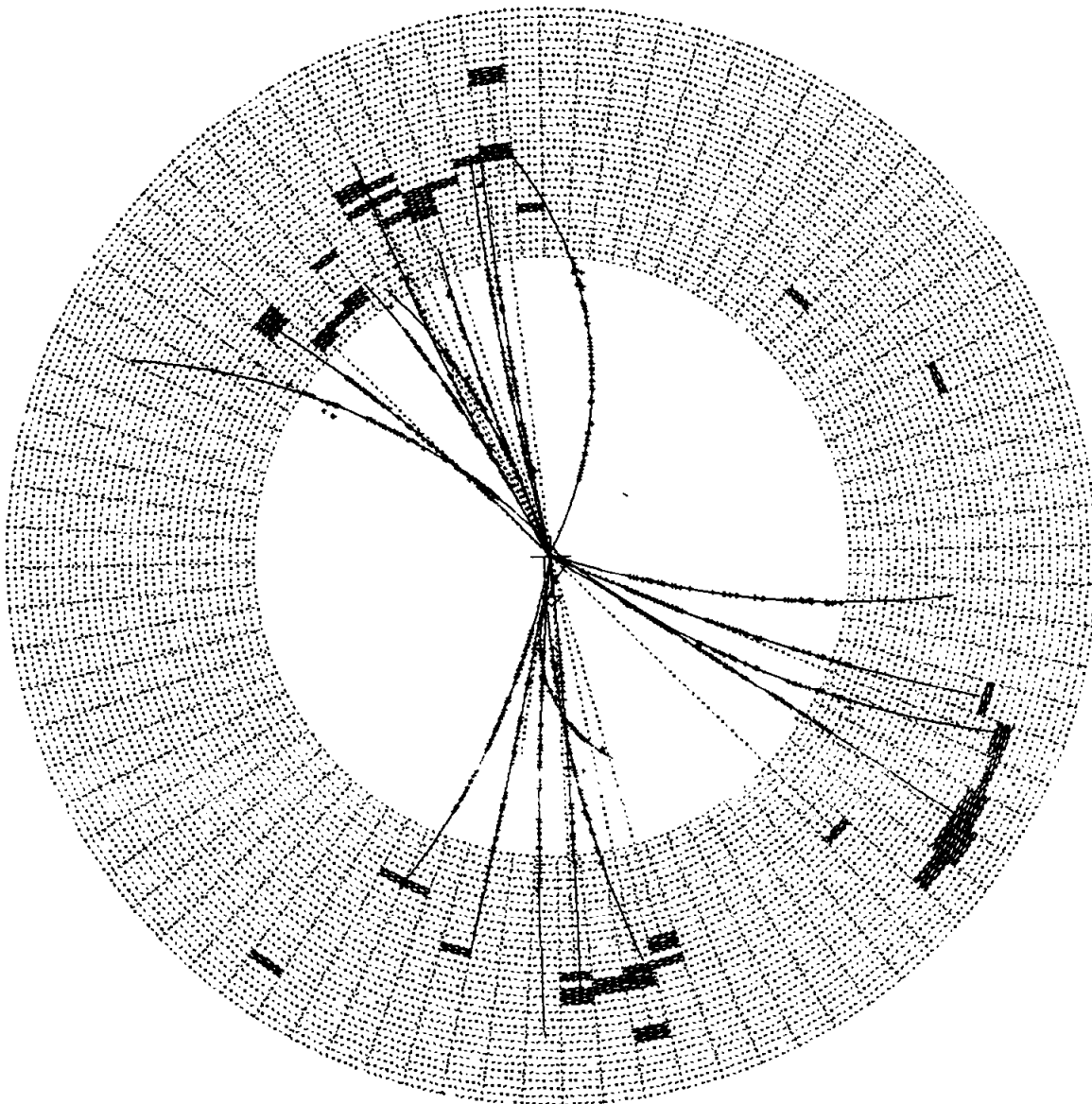


Fig. 11.12 A three-jet event observed by the JADE detector at PETRA.

crucial point is that we are discussing a process with two momentum scales, p_T^2 and Q^2 with $p_T^2 \ll Q^2$, a situation already encountered in Exercise 10.7. We argued there that α_s is in fact $\alpha_s(p_T^2)$, but noted that keeping α_s constant gave the correct result to leading order.

EXERCISE 11.6 The x_T distribution (11.30) can be translated into an “acollinearity” distribution $d\sigma/d\theta$, where θ is the angle between the q and $\bar{q}$ jet directions defined in (11.20) and Fig. 11.7. Show that for θ not too large,

$$\frac{1}{\sigma} \frac{d\sigma}{d\theta} \approx \frac{8\alpha_s}{3\pi} \frac{1}{\theta} \log\left(\frac{1}{\theta^2}\right).$$

An exact result can be obtained using (11.41) instead of (11.30).

We have repeatedly drawn attention to the fact that hadrons fragmenting from a quark, or any other parton, form a cone around the direction $\mathbf{p}_q$. It is an experimental fact that the $\langle k_T \rangle$ for the hadron fragments is about 300 MeV, where T refers to the direction transverse to $\mathbf{p}_q$. We might anticipate that at higher energies the fragmentation cone would narrow,

$$\langle \theta \rangle \approx \frac{\langle k_T \rangle}{p_q} \approx \frac{0.3 \text{ GeV}}{Q/2}, \quad (11.43)$$

and the jets become narrow bundles of energetic particles when Q increases. This is not the case. Gluon radiation results in an increase of the $\langle k_T \rangle$ of the hadrons which we associate with the original quark or antiquark jet. It is increasingly probable that in a very high-energy two-jet event, one of the observed jets is actually the fragmentation product of a qg or $\bar{q}g$ state as a result of gluon emission by the q or $\bar{q}$. The experimentalist will recognize such a jet as being “fatter” or having increased $\langle k_T \rangle$. This dynamical broadening of $\langle \theta \rangle$ almost compensates for the kinematic narrowing given by (11.43). As a consequence, it turns out that the narrowing of jets with increasing Q^2 is a logarithmic and not a linear effect, with

$$\langle \theta \rangle \sim \frac{1}{\log Q^2}. \quad (11.44)$$

This result has to be derived with care. We shall only briefly sketch how it comes about. By definition,

$$\langle k_T \rangle \sim \int_0^Q k_T \frac{d\sigma}{dk_T}. \quad (11.45)$$

Let us assume that the k_T of the hadrons in the broadened jet just reflects the relative k_T of the q or $\bar{q}$ and the emitted gluon. Then, $d\sigma/dk_T$ in (11.45) is nothing but the familiar transverse momentum distribution of gluon emission

$$\frac{d\sigma}{dk_T} \sim \frac{\alpha_s}{k_T}.$$

We here neglect all logarithms and therefore also take α_s to be constant. In this approximation, $d\sigma/dk_T$ is the same as (11.42). Substituting into (11.45) gives

$$\langle k_T \rangle \sim \alpha_s \int_0^Q dk_T \sim \alpha_s Q,$$

and from (11.43) we see that $\langle \theta \rangle \sim \alpha_s$. The logarithmic narrowing of the jets (11.44) comes about when we incorporate the running of $\alpha_s \sim 1/\log Q^2$. A proper justification of this result is complicated (e.g., Dokshitzer et al., 1980). However, the phenomenological message is clear: jet identification which at low energies is only possible through clever statistical analysis of $e^+e^- \rightarrow \text{hadron}$ events, should be simpler for high-energy experiments due to the increased collimation predicted by (11.44).

11.7 QCD Corrections to $e^-e^+ \rightarrow \text{Hadrons}$

The parton model result (11.6),

$$R \equiv \frac{\sigma(e^-e^+ \rightarrow \text{hadrons})}{\sigma(e^-e^+ \rightarrow \mu^-\mu^+)} = 3 \sum_q e_q^2, \quad (11.46)$$

which we derived from $\sigma(e^-e^+ \rightarrow q\bar{q})$, will be modified by the possibility of gluon emission from the q or $\bar{q}$. The order α_s diagrams are shown in Fig. 11.10, and the cross section $d\sigma/dx_q dx_{\bar{q}}$ is given in (11.41). To calculate the order α_s correction to R , we must integrate over both x_q and $x_{\bar{q}}$ from 0 to 1. In doing this, we encounter a problem which is common in perturbative QCD calculations. The integrand, (11.41), diverges as x_q or $x_{\bar{q}}$ goes to 1. To trace the origin of the problem, consider, for instance, the factor $1 - x_q$ in the denominator. Using (11.33),

$$\begin{aligned} 1 - x_q &= \frac{s}{Q^2} = \frac{2p_{\bar{q}} \cdot p_g}{Q^2} \\ &= \frac{2}{Q^2} E_{\bar{q}} E_g (1 - \cos \theta_{\bar{q}g}) \end{aligned} \quad (11.47)$$

and so $1 - x_q$ vanishes when the gluon becomes soft ($E_g \rightarrow 0$) or when $\bar{q}$ and g become collinear ($\cos \theta_{\bar{q}g} \rightarrow 1$). The first type of divergence is called an *infrared divergence* (see Chapter 7), and the second is called a *collinear divergence* (or mass singularity since, if the quark or the gluon had nonzero mass, $\cos \theta_{\bar{q}g} = 1$ would be kinematically impossible). In QED, where leptons do have a mass, such mass singularities do not occur.

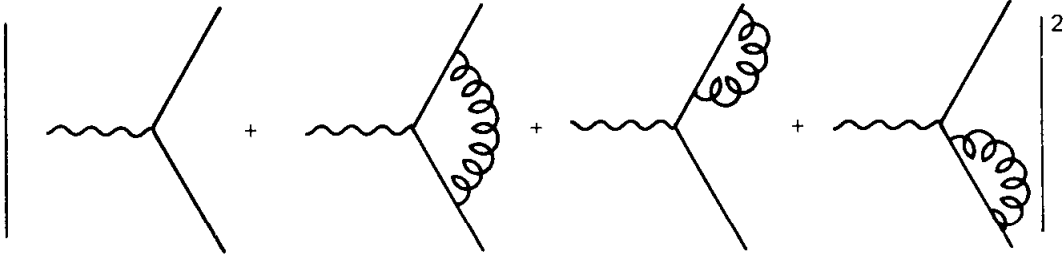
To proceed, we must regularize these infrared and mass singularities. One way to accomplish this is to give the gluon a fictitious mass m_g and to repeat the calculation of the Feynman diagrams of Fig. 11.10 which led to (11.41). A

straightforward but lengthy calculation yields

$$\begin{aligned}
 \sigma_{\text{real}} &= \left| \text{diagram 1} + \text{diagram 2} \right|^2 \\
 &= \int dx_q dx_{\bar{q}} \frac{d\sigma}{dx_q dx_{\bar{q}}} \\
 &= \sigma_q \frac{\alpha_s}{2\pi} \frac{4}{3} \left\{ \log^2 \left(\frac{m_g}{Q} \right) + 3 \log \left(\frac{m_g}{Q} \right) - \frac{\pi^2}{3} + 5 \right\}, \quad (11.48)
 \end{aligned}$$

where σ_q denotes $\sigma(e^-e^+ \rightarrow q\bar{q})$. As expected, (11.48) is divergent when $m_g \rightarrow 0$.

Obviously, (11.48) cannot be the final answer since it depends on the fictitious mass m_g , whereas the correct result should not. However, there is another $O(\alpha_s)$ contribution. This additional $O(\alpha_s)$ term occurs in



and corresponds to the interference of the $\gamma^* \rightarrow q\bar{q}$ diagram, with the sum of the three diagrams containing virtual gluon loops (see Section 10.8). This interference term gives a contribution

$$\sigma_{\text{virtual}} = \sigma_q \frac{\alpha_s}{2\pi} \frac{4}{3} \left\{ -\log^2 \left(\frac{m_g}{Q} \right) - 3 \log \left(\frac{m_g}{Q} \right) + \frac{\pi^2}{3} - \frac{7}{2} \right\}. \quad (11.49)$$

The total order α_s contribution is

$$\sigma = \sigma_{\text{real}} + \sigma_{\text{virtual}} = \sigma_q \frac{\alpha_s}{\pi}, \quad (11.50)$$

which is finite and independent of m_g , as indeed it must be.

The cancellation of the singularities between the contributions with the emission of real and virtual gluons is not just a property of this particular process. It occurs over and over again. For example, in Section 10.8, we already discussed the analogous cancellation in deep inelastic scattering. These particular cancellations are examples of a general theorem due to Kinoshita, Lee, and Nauenberg. The reason that we did not meet this mechanism earlier in the chapter is because we choose to compute the cross section for the production of gluons with a fixed,

nonzero p_T . Then, x_q and $x_{\bar{q}}$ are unable to reach 1, and the denominator in (11.41) cannot vanish. This choice also excludes the diagrams with the virtual gluon loops for which $p_T = 0$.

Including the order α_s correction to R of (11.46), we obtain

$$R = 3 \sum_q e_q^2 \left[1 + \frac{\alpha_s(Q^2)}{\pi} \right]. \quad (11.51)$$

For a typical Q^2 where $\alpha_s \sim 0.2$, the correction is small and cannot easily be distinguished within the errors of the present measurements. Note, however, that R is no longer exactly constant since α_s is a function of Q^2 , see (7.65). This is another example of a scaling parton model result being modified by $\log Q^2$ corrections arising from the emission of gluons.

11.8 Perturbative QCD

We have seen that hadrons with unexpectedly large transverse momenta in both deep inelastic scattering ($ep \rightarrow hX$) and e^-e^+ annihilation ($e^-e^+ \rightarrow h$) have a common origin. This is an example of parton diagrams for one process being used, after crossing, in another process. In this way, the same underlying QCD process can be tested in totally different experimental situations. Some examples are shown in Fig. 11.13. The ingredients are q , $\bar{q}$, g , γ , and γ^* ; a lepton pair (i.e., e^-e^+ , $\mu^-\mu^+$); the structure functions $f(x)$, giving the probability of finding partons in the parent hadron; and, finally, the fragmentation functions $D(z)$, denoting the probability of final-state hadrons emerging from the partons. Figure 11.13 shows just some of the possible ways of combining these ingredients.

To be a useful test of QCD, it is important that the process is a short-distance interaction, so that $\alpha_s(Q^2)$ is small enough for perturbation theory to be valid. The processes of Fig. 11.13 satisfy this requirement. We must also recall from Chapter 9 the criteria for the validity of the parton model (which is the lowest-order approximation to QCD). A large energy and a large momentum transfer are required to justify the impulse approximation used in the parton model calculation. Through the uncertainty principle, the large energy requirement guarantees that the time scale of the parton interaction is short, so that we can ignore the interactions with “spectator” partons (that is, with constituents which are not directly involved in the QCD subprocess). A large momentum (e.g., Q^2 , p_T^2 , a heavy quark mass) guarantees that the process occurs at a short range so that α_s is small.

As an example, consider processes (d) and (e) of Fig. 11.13. Even when the initial protons collide with very high energy, the momentum transferred in the process also has to be large for perturbative QCD to be valid. This can be ensured by requiring that the photon be produced with a large transverse momentum,

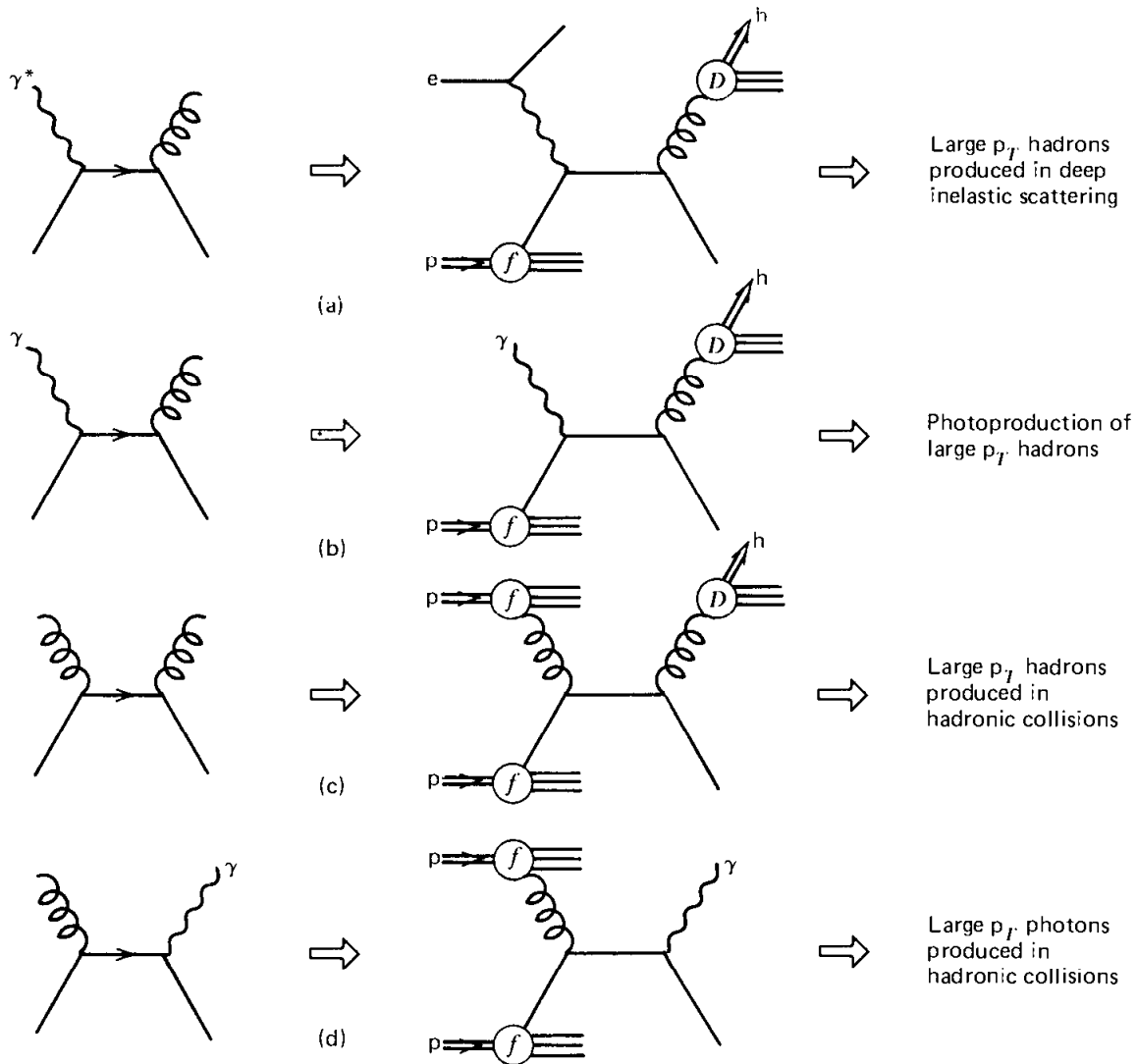


Fig. 11.13 Various observable processes which contain γ (or gluon)-quark Compton scattering, or the crossed reaction, as a parton subprocess; f and D denote structure and fragmentation functions, respectively.

process (d), or by giving it a large Q^2 by detecting a lepton pair in the final state with a large invariant mass, process (e).

Figure 11.13 shows only one parton diagram for each process. To the same order of α_s , other diagrams have to be computed to obtain a result which can be compared with experiment. For instance, in process (c), another possibility is quark-quark scattering via gluon exchange, whereby quarks in the beam scatter at large angles off target quarks producing two quark jets in the final state with large transverse momentum (see also Exercise 11.7).

There are many applications of perturbative QCD. Reviews are given, for example, by Field (1979), Ellis and Sachrajda (1979), Reya (1981), Altarelli (1982), Collins and Martin (1982), and Pennington (1983). The phenomenological evidence is that the theory can successfully confront experiment in all such

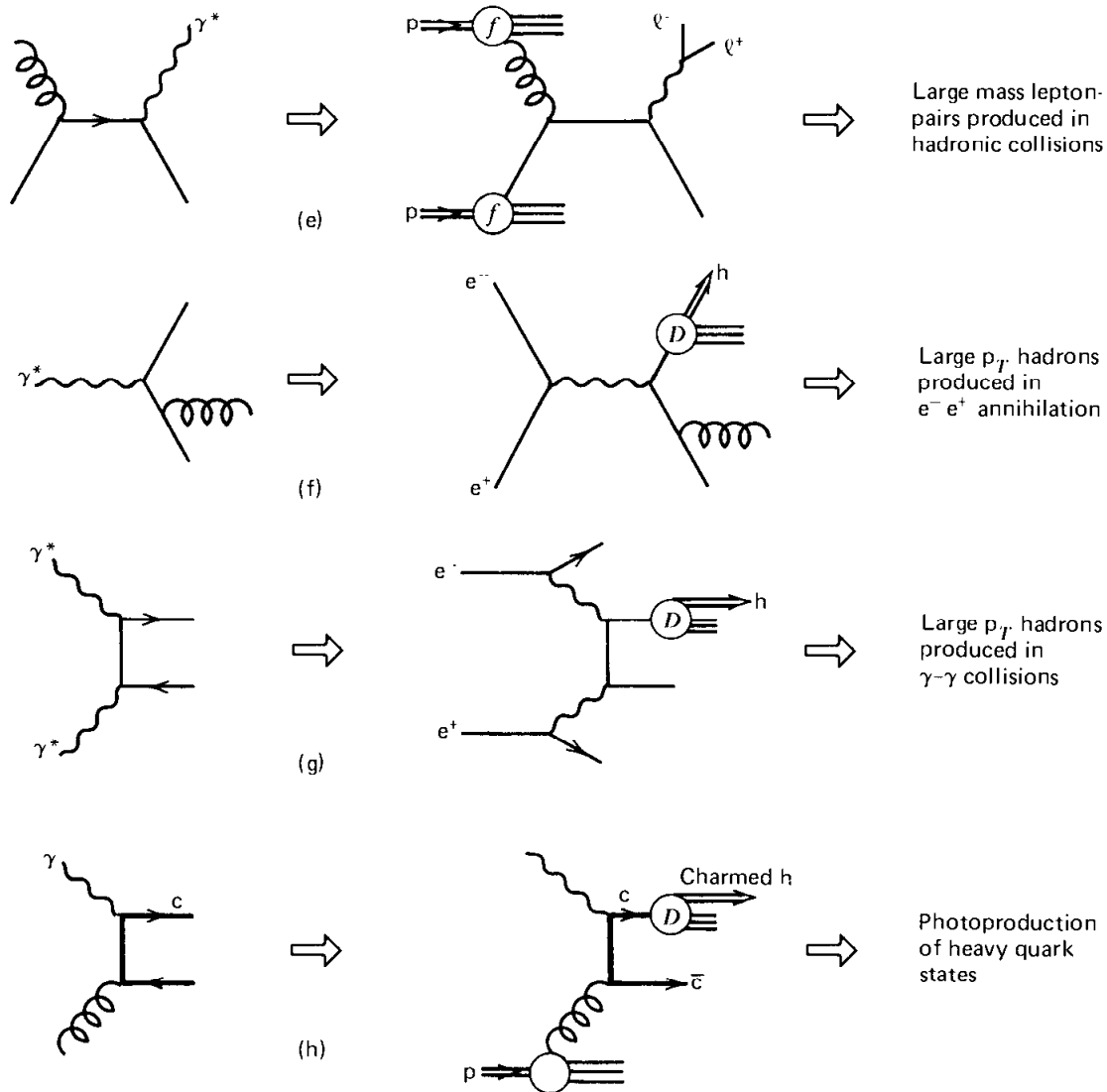


Fig. 11.13 (Continued)

situations. Unfortunately, however, due primarily to color confinement, no QCD tests exist (or can even be envisaged) that are comparable to the accuracy of the lepton magnetic moment calculation in QED.

EXERCISE 11.7 List all parton processes, in addition to those shown in Fig. 11.13, that can contribute to reaction (c) to the same order of α_s . Comment on the relative strengths of the subprocesses, comparing, in particular, pp - and $p\bar{p}$ -initiated reactions.

11.9 A Final Example: The Drell–Yan Process

Diagram (e) of Fig. 11.13 gives an $O(\alpha^2\alpha_s)$ contribution to the cross section for the hadronic production of lepton pairs. However, this is a correction to the

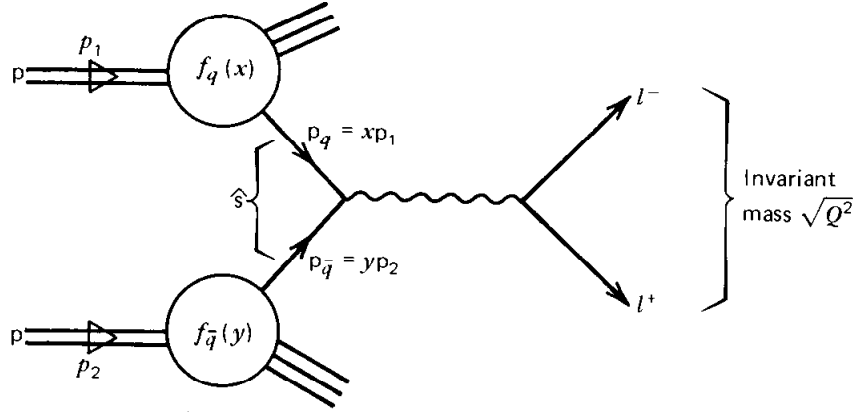


Fig. 11.14 The Drell-Yan process, $pp \rightarrow l^- l^+ X$.

lower-order process $q\bar{q} \rightarrow \gamma^* \rightarrow l^- l^+$ shown in Fig. 11.14, which is known as the Drell-Yan process. Together with e^-e^+ annihilation and deep inelastic leptonproduction, it has played an important role in determining the structure functions and in testing the parton model and its QCD corrections.

To calculate the cross section corresponding to Fig. 11.14, we begin with that for the parton subprocess,

$$\hat{\sigma}(q\bar{q} \rightarrow l^- l^+) = \frac{4\pi\alpha^2}{3Q^2} e_q^2, \quad (11.52)$$

see (11.3). In order to embed (11.52) in the hadronic process, we first rewrite it as a differential cross section, $d\sigma/dQ^2$, for the production of lepton pairs having invariant mass $\sqrt{Q^2}$, where

$$Q^2 = \hat{s} = (p_q + p_{\bar{q}})^2. \quad (11.53)$$

That is,

$$\frac{d\hat{\sigma}}{dQ^2} = \frac{4\pi\alpha^2}{3Q^2} e_q^2 \delta(Q^2 - \hat{s}). \quad (11.54)$$

The hadronic cross section can now be obtained using the structure functions $f_i(x)$, familiar from deep inelastic scattering (Chapter 9). We obtain

$$\frac{d\sigma}{dQ^2} (pp \rightarrow l\bar{l}X) = \left(\frac{1}{3}\right)\left(\frac{1}{3}\right) 3 \sum_q \int dx \int dy f_q(x) f_{\bar{q}}(y) \frac{d\hat{\sigma}}{dQ^2}, \quad (11.55)$$

where the sum is over quark flavors. The factors of $\frac{1}{3}$ average over the initial q and $\bar{q}$ colors and the factor 3 sums over the $q\bar{q}$ color combinations which can annihilate to form a colorless virtual photon. The q and $\bar{q}$ carry fractions x and y of the momenta of the protons, respectively, and so (11.53) becomes

$$\hat{s} = (xp_1 + yp_2)^2 \simeq xys, \quad (11.56)$$

where $s \approx 2 p_1 \cdot p_2$ is the center-of-mass energy squared of the colliding protons. From (11.54)–(11.56), we obtain

$$\boxed{\frac{d\sigma}{dQ^2}(\text{pp} \rightarrow l\bar{l}X) = \frac{4\pi\alpha^2}{9Q^4} \sum_q e_q^2 \int dx \int dy f_q(x) f_{\bar{q}}(y) \delta\left(1 - xy \frac{s}{Q^2}\right).}$$

(11.57)

In lowest order, with no gluon emission, we expect a scaling result. This is hidden in (11.57). Although the cross section is a function of both the collision energy $\sqrt{s}$ and the lepton pair mass $\sqrt{Q^2}$, the quantity

$$Q^4 \frac{d\sigma}{dQ^2} = \mathcal{F}\left(\frac{s}{Q^2}\right) \quad (11.58)$$

is a function only of the ratio s/Q^2 . Figure 11.15 shows that the data satisfy this scaling law rather well.

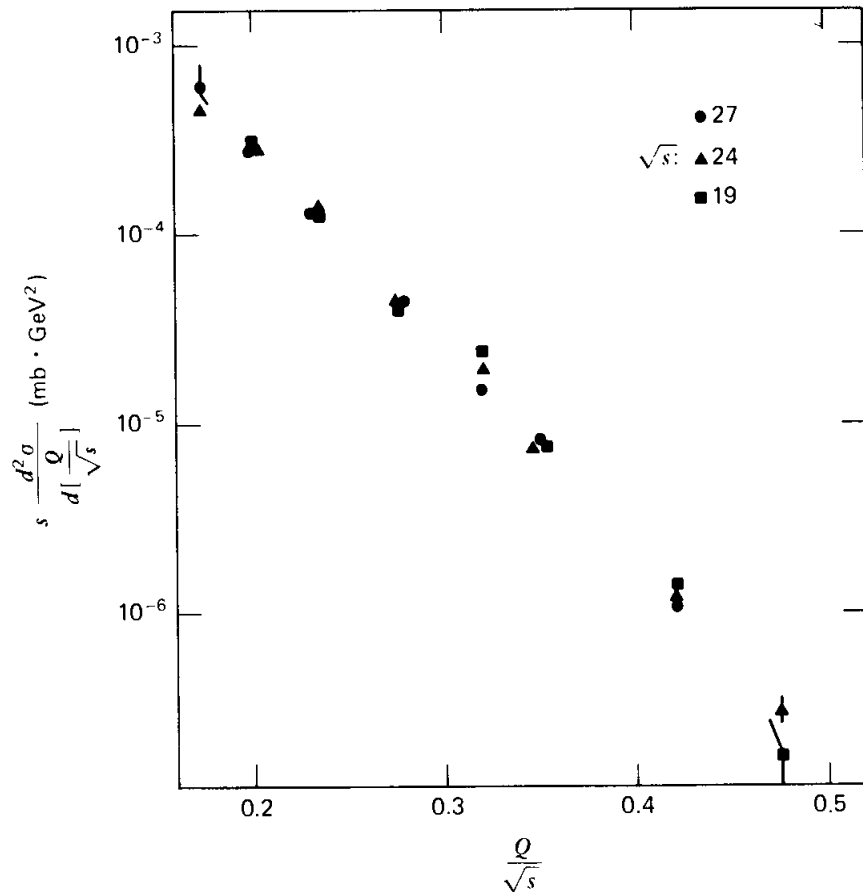


Fig. 11.15 Scaling in the Drell-Yan process. The scaling test shown is equivalent to (11.58); $\sqrt{s}$ in GeV. Data are from the Fermi laboratory.

EXERCISE 11.8 Express the sum in (11.57) in terms of the valence and sea quark structure functions introduced in Chapter 9. Do the same for lepton pair production in $\bar{p}p$ and $\pi^\pm p$ interactions.

EXERCISE 11.9 The data for an (isoscalar) carbon target indicate that the ratio

$$\frac{\sigma(\pi^+ C \rightarrow \mu^- \mu^+ X)}{\sigma(\pi^- C \rightarrow \mu^- \mu^+ X)}$$

is approximately unity for small values of Q^2/s but decreases toward $\frac{1}{4}$ as Q^2/s approaches 1. Explain why this is in agreement with the Drell-Yan prediction. Note that $xy = Q^2/s$.

EXERCISE 11.10 Including diagrams with gluons introduces logarithmic scaling violations in (11.58). Draw the diagrams which give an $O(\alpha^2\alpha_s)$ contribution to the lepton pair cross section. Compute the cross section for the diagram shown in Fig. 11.13e.